

賦予口語語言模型思考能力

Enabling Thinking for Spoken Language Models

姜成翰 ([Cheng-Han Chiang](#))

National Taiwan University

Outline

1. Background: Spoken Language Models
2. STITCH: Thinking while speaking
3. SHANKS: Thinking while listening
4. Conclusion

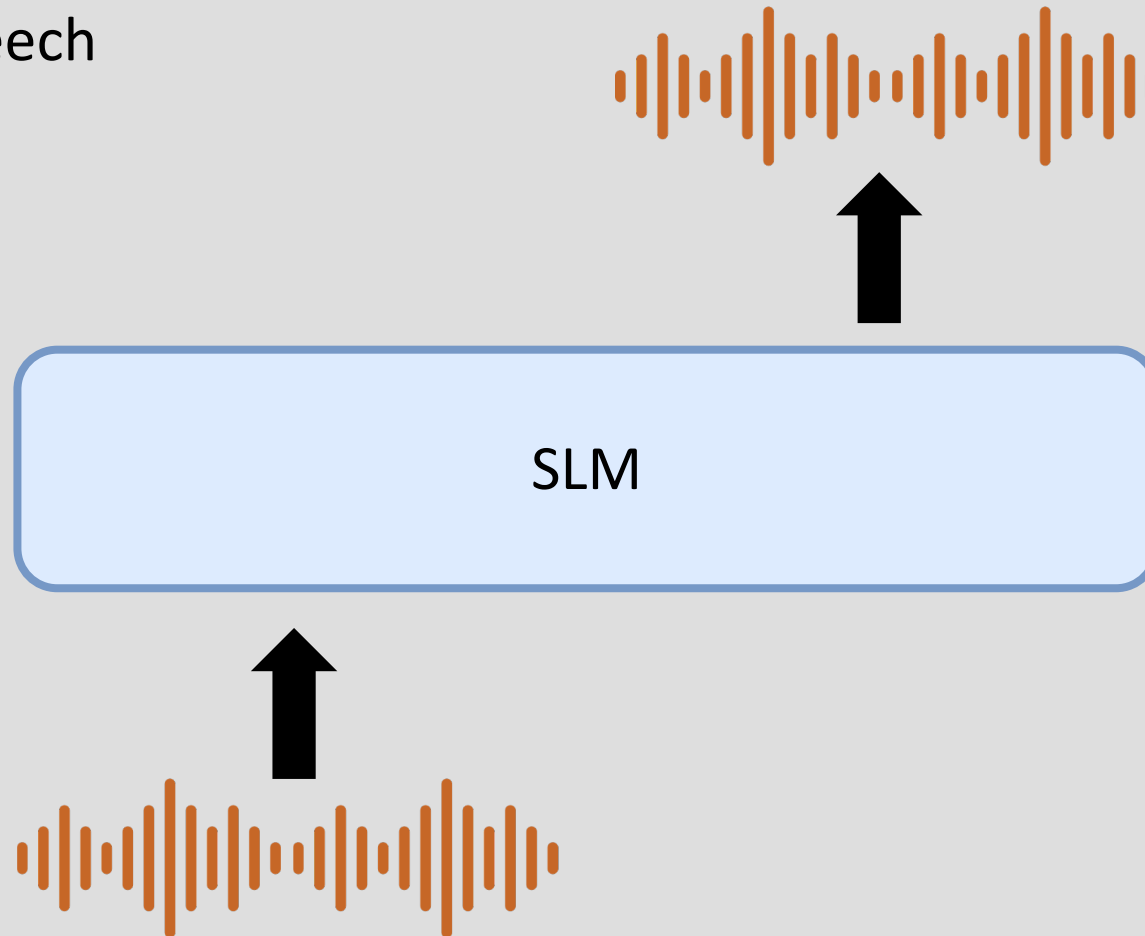
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Background: Spoken Language Models (SLMs)

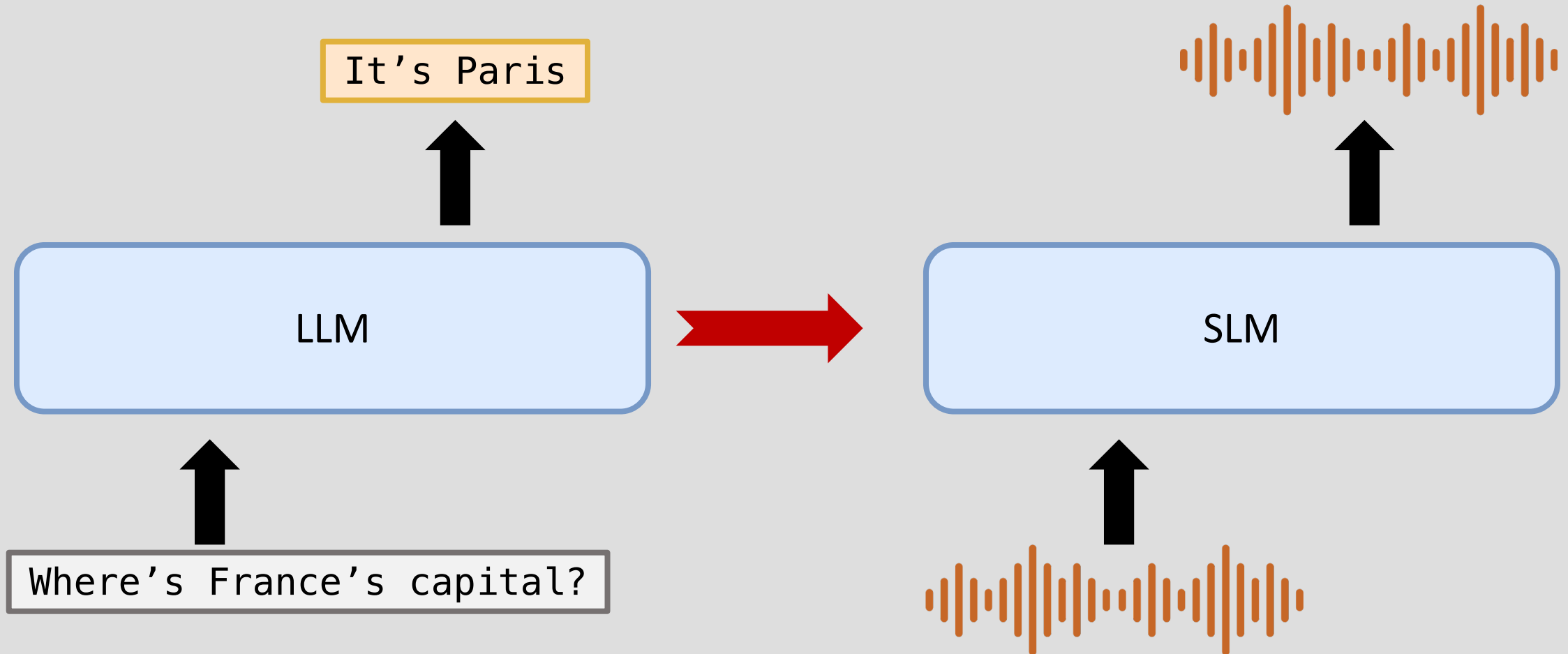
- Spoken language Models (SLMs)
 - Input: Speech
 - Output: Speech



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Background: Spoken Language Models (SLMs)

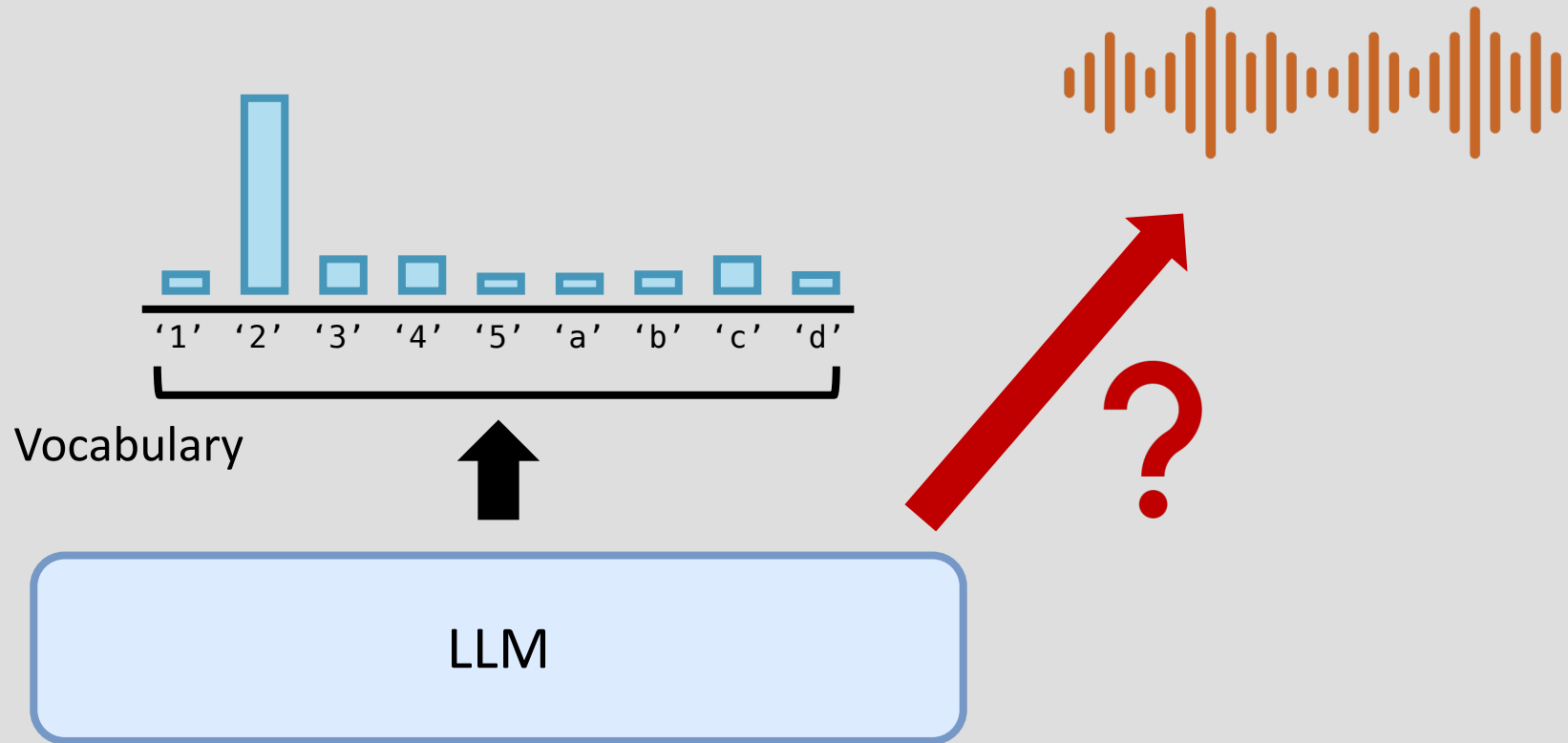
- SLMs are typically trained from text-only LLMs



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Background: Spoken Language Models (SLMs)

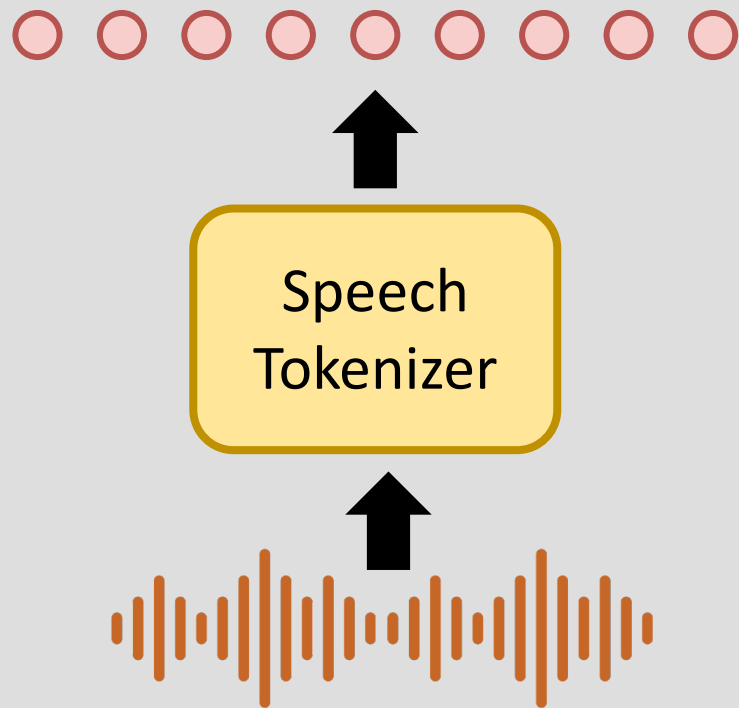
- LLMs can only output discrete text tokens, how to make it an SLM that can generate continuous speech waveform?



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Background: Speech Tokens

- We can convert continuous speech waveform into discrete tokens by quantization
- The quantization model (a neural network) is called the “*speech tokenizer*”

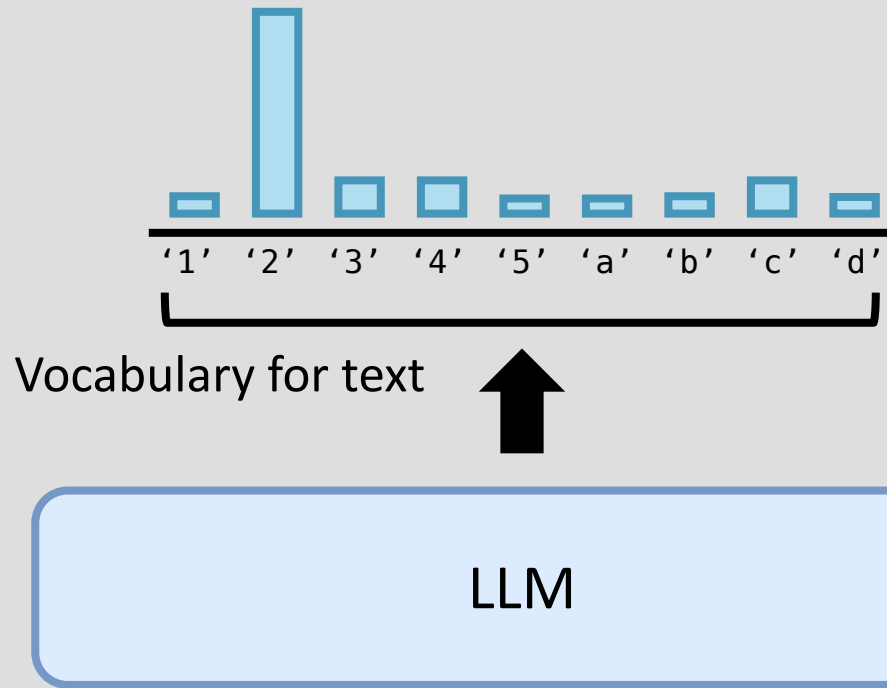


Waveform	Token
	< speech_token_0 >
	< speech_token_1 >
(...)	(...)
	< speech_token_N >

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Background: Spoken Language Models (SLMs)

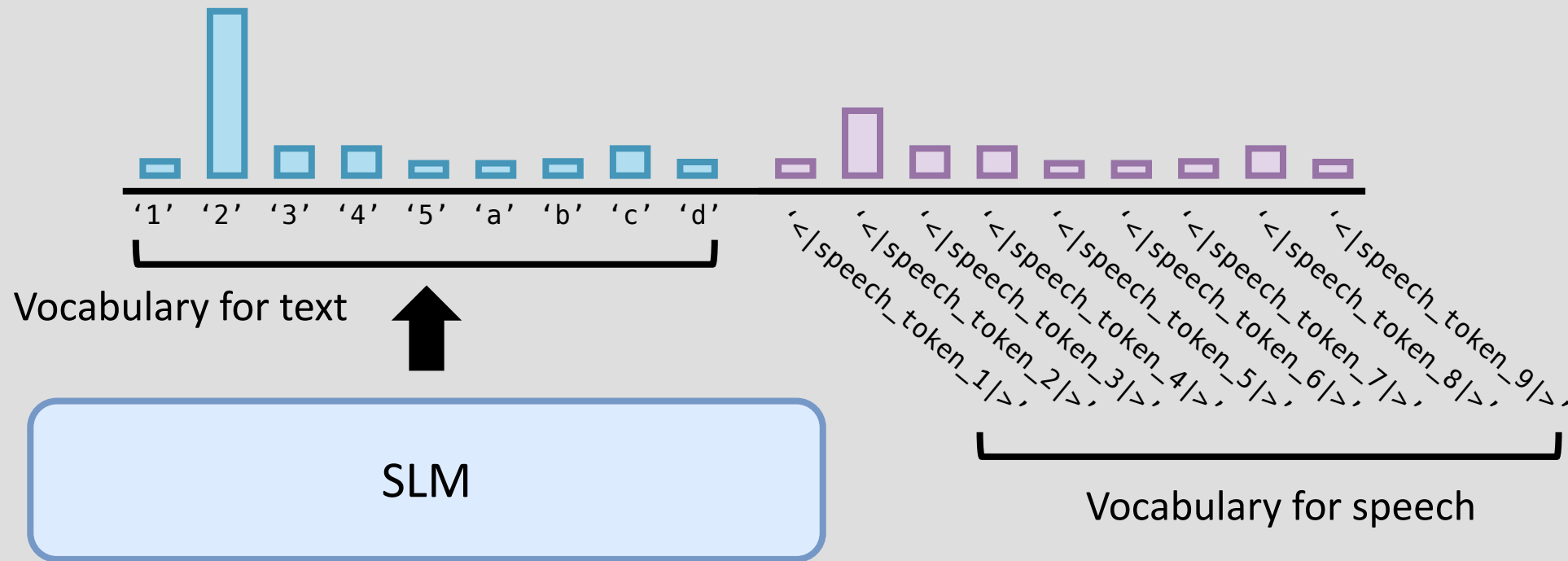
- SLMs can be obtained from LLMs by adding speech tokens in the vocabulary



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Background: Spoken Language Models (SLMs)

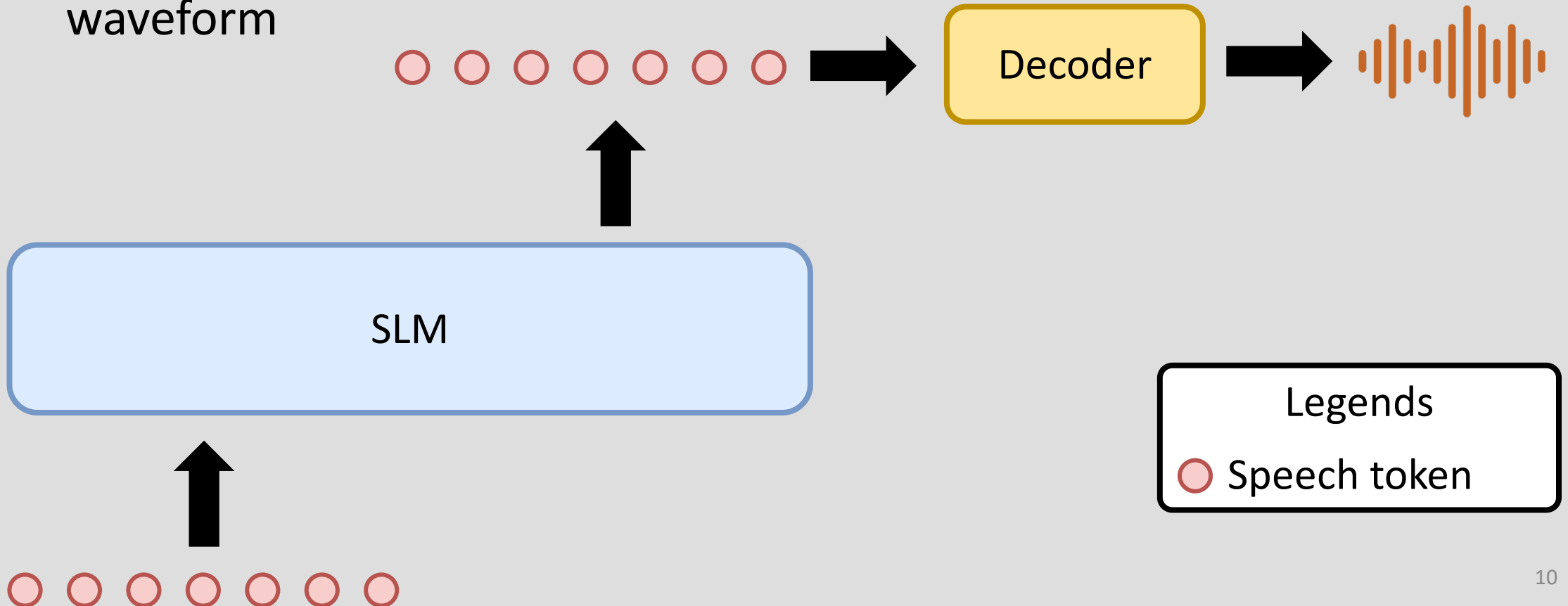
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Background: Spoken Language Models (SLMs)

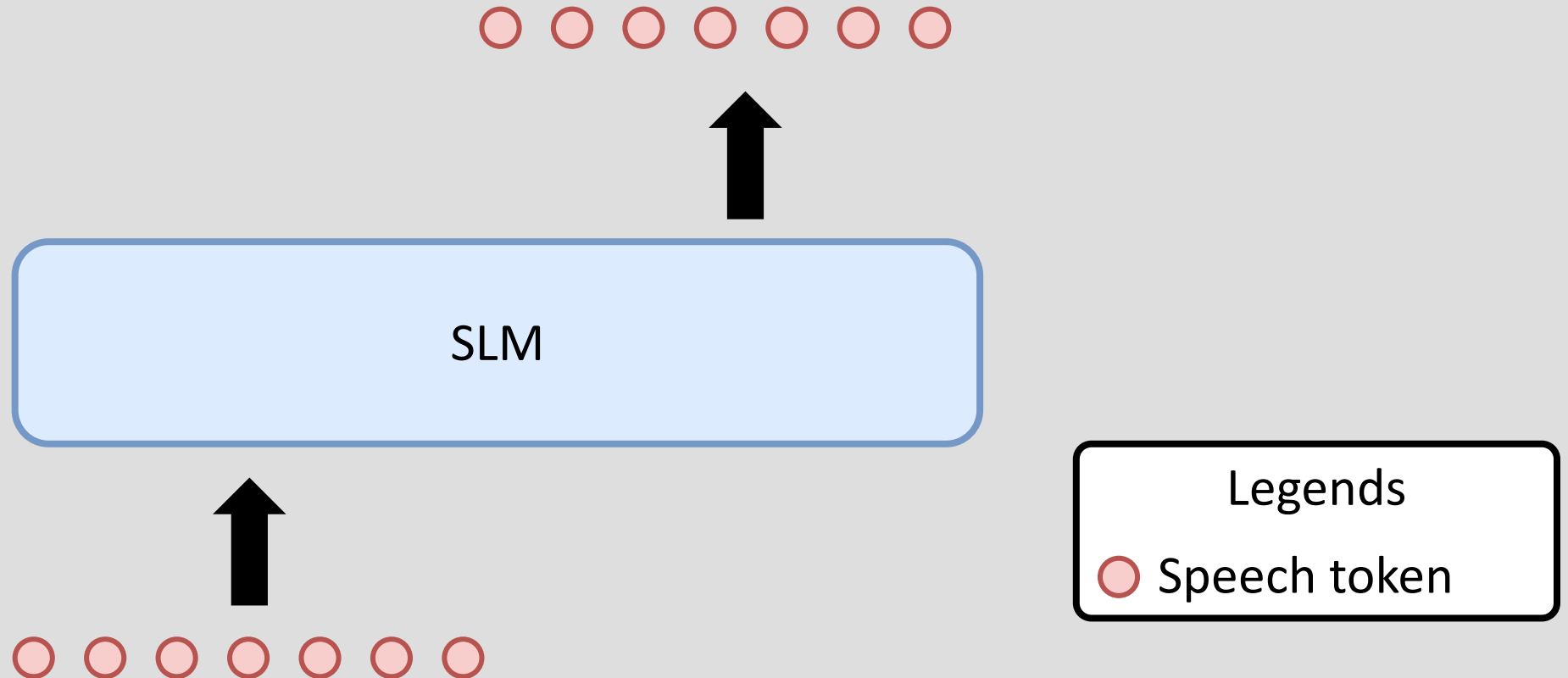
- SLMs are trained to take speech tokens as the input and generate audio tokens as the output
- A decoder (a neural network) converts the speech tokens into waveform



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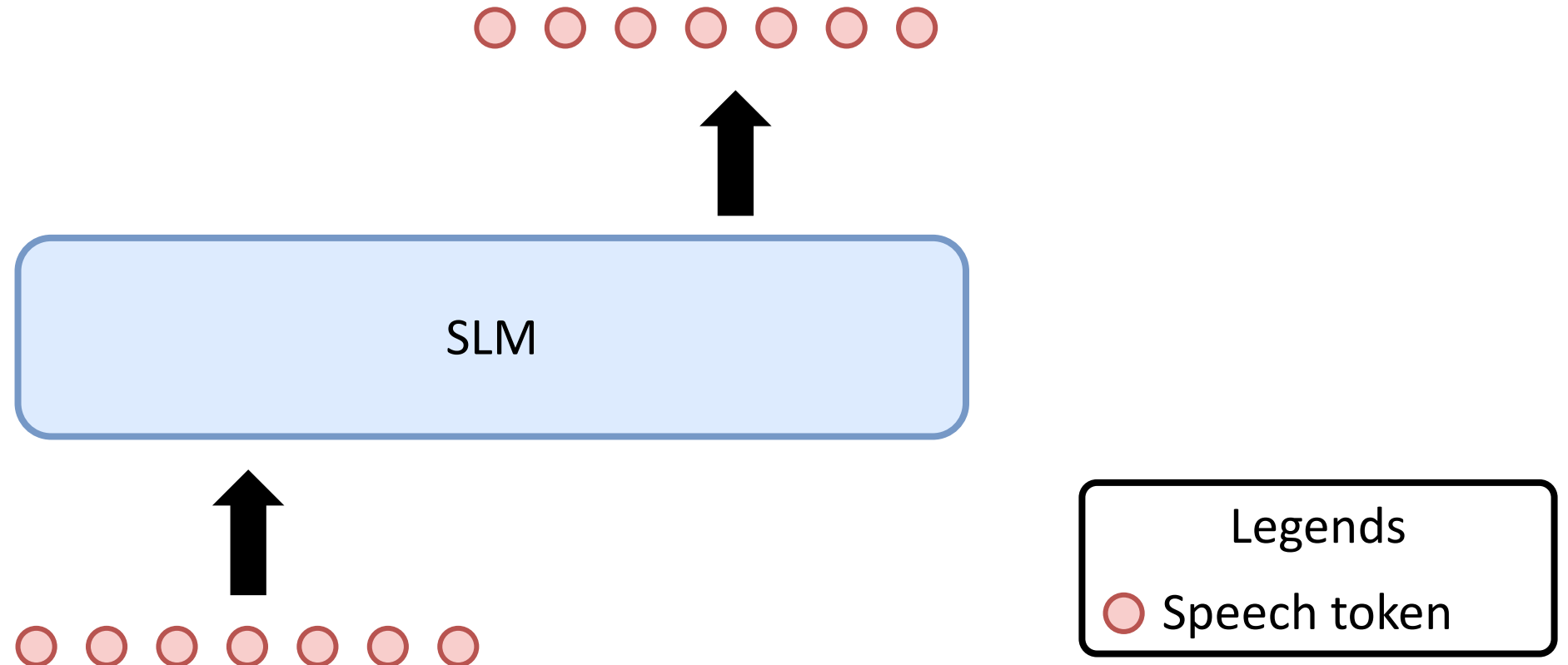
Background: Spoken Language Models (SLMs)

- We have an SLM that can already take speech input and generate speech output



Motivation: Reasoning for SLMs

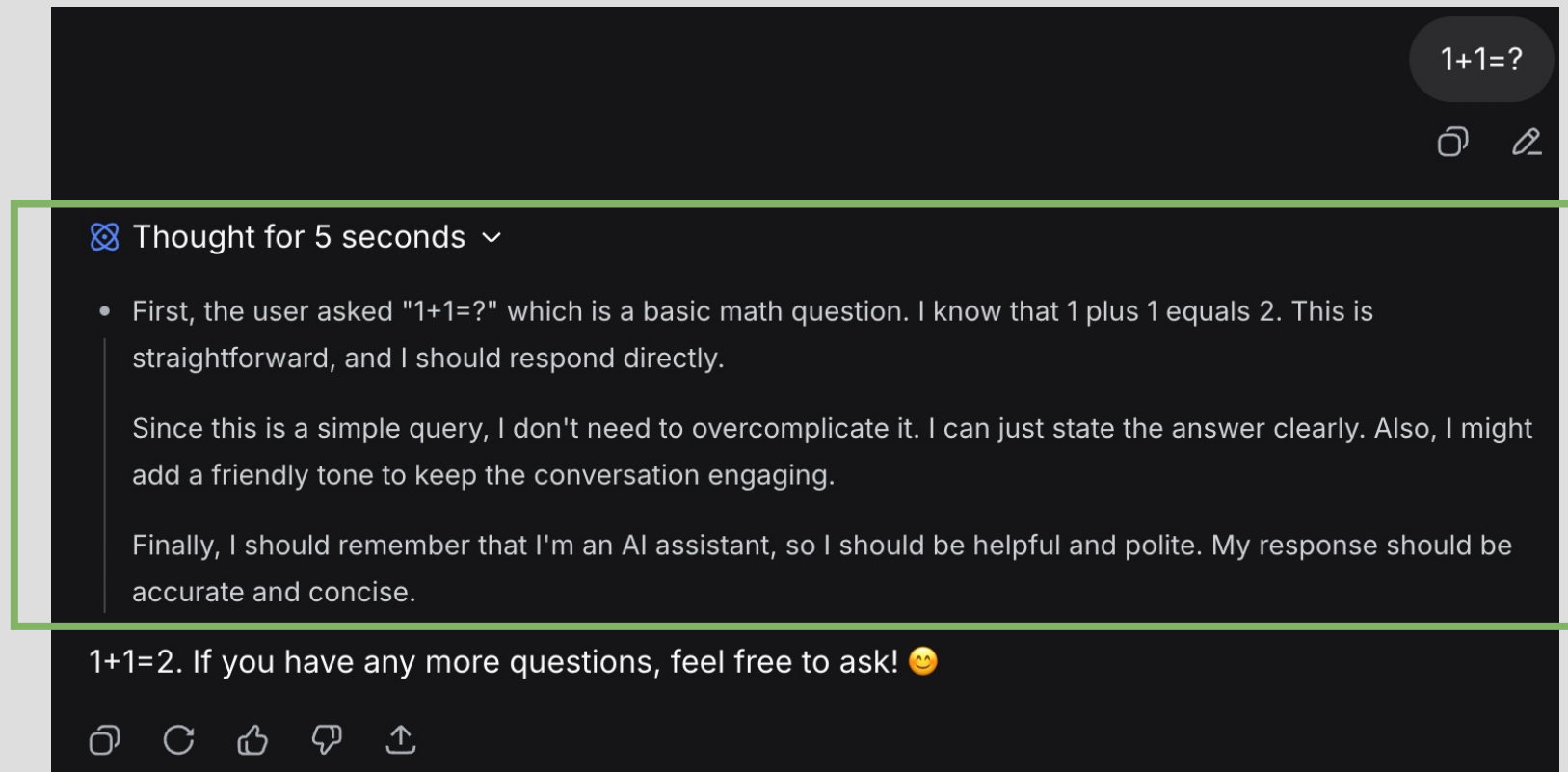
- SLMs generates speech tokens without any intermediate reasoning
- This is different from “reasoning LLMs”



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Motivation: Reasoning for SLMs

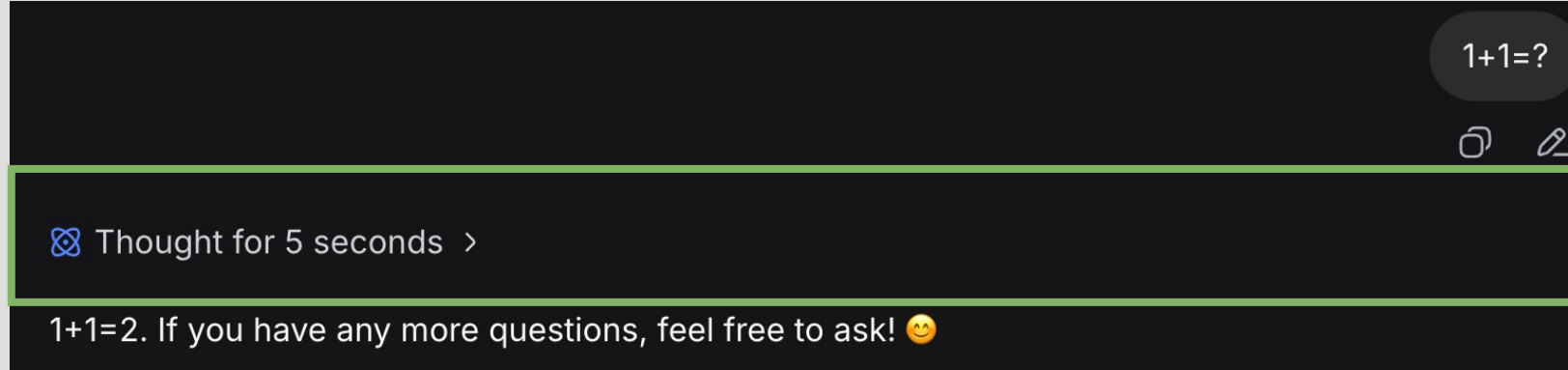
- Reasoning LLMs (e.g., Deepseek-r1, GPT o-series models) can generate reasoning before emitting the final response



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Motivation: Reasoning for SLMs

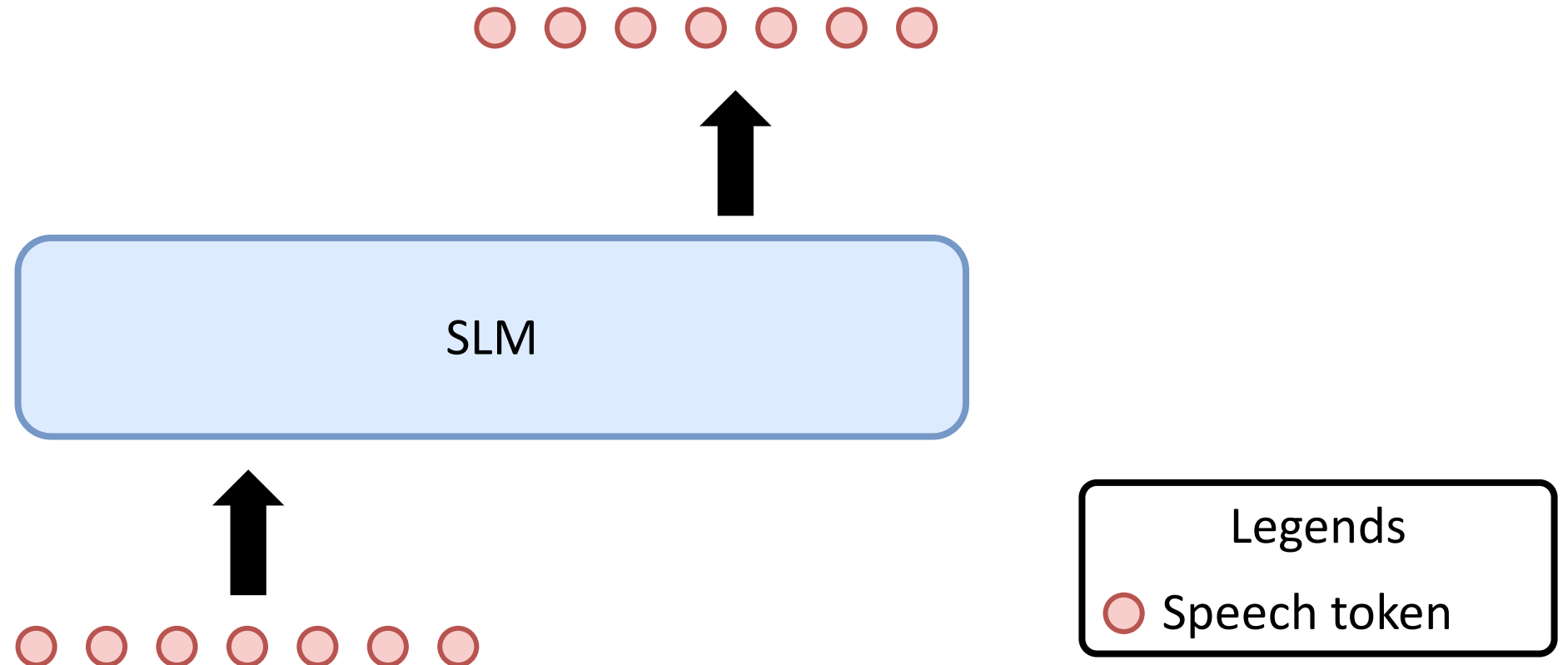
- The reasoning is designed to be hidden from the user



- We will use the term “thinking” to refer to the reasoning process hidden to the user
 - 免責聲明: We do not use this word to indicate that the model can really “*think*” like humans

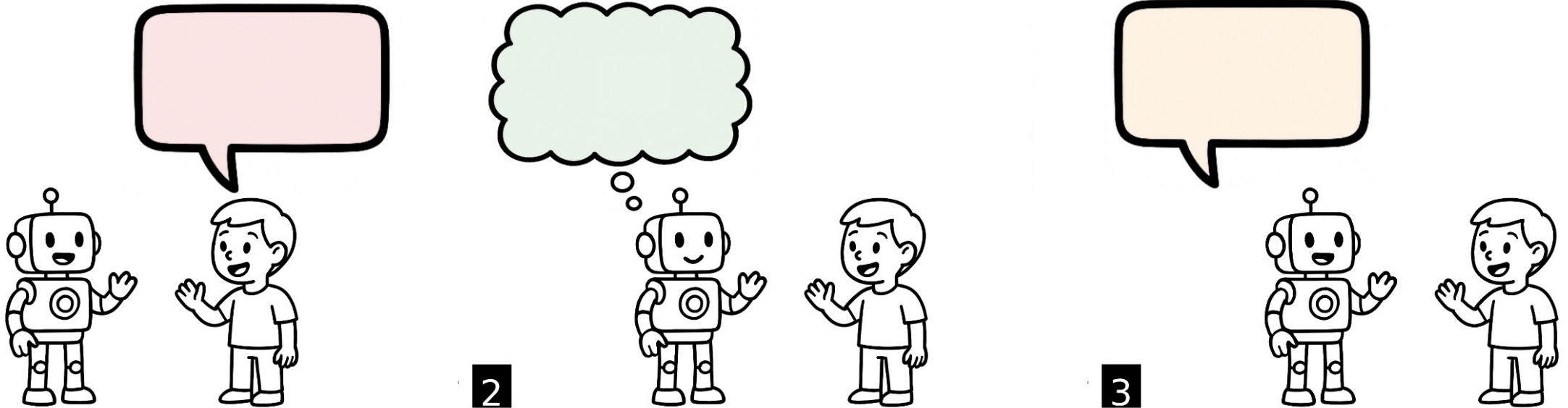
Motivation: Reasoning for SLMs

- SLMs generates speech tokens without any intermediate reasoning
- How can we make the SLMs reason to improve the final response?



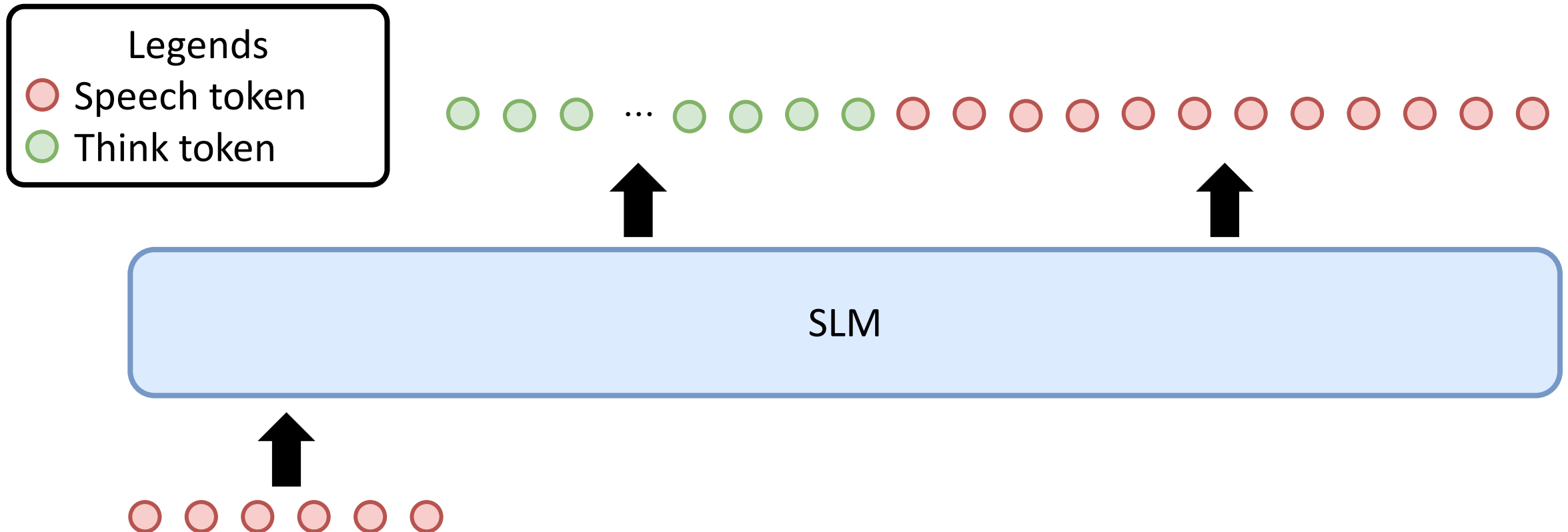
Setting 1: Listen → Think → Speak

- This directly borrows the idea from reasoning LLM: receive the user input, think silently (hidden from the user)



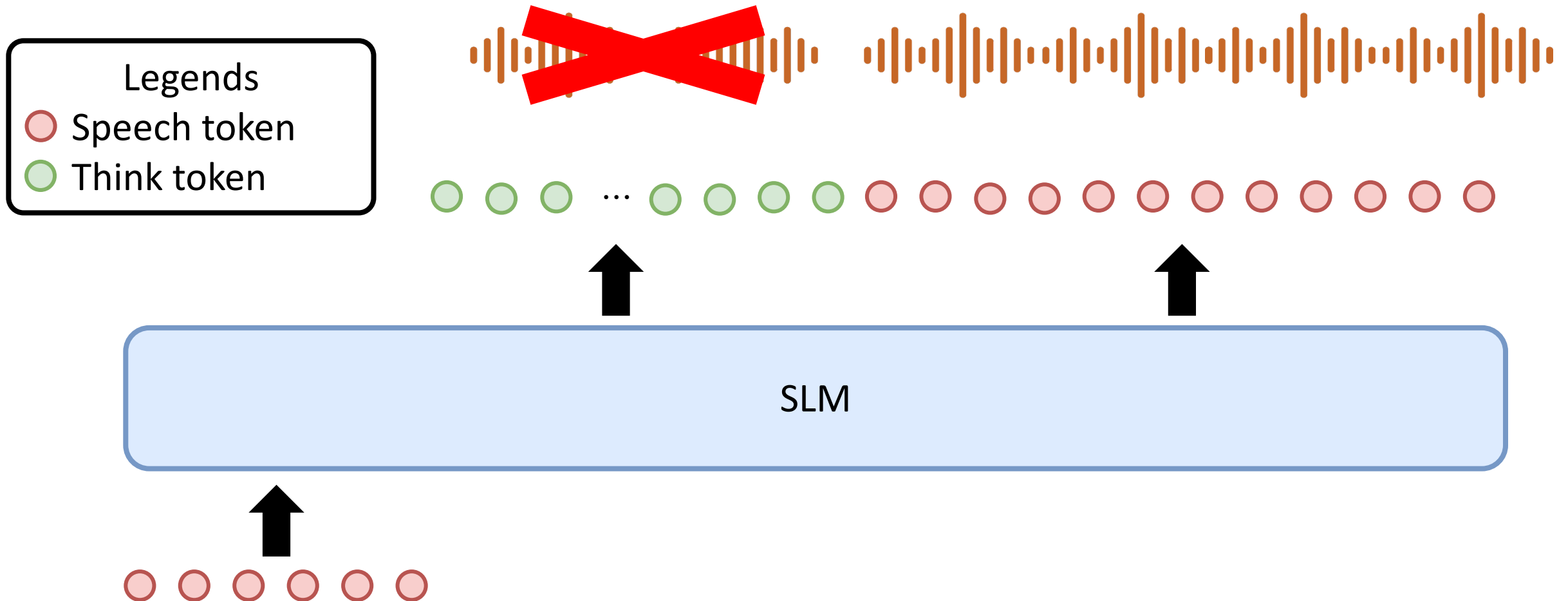
Thinking before Speaking

- Generate some think tokens before generating the speech tokens
 - The think tokens are **text tokens**; they are called think tokens because they are used for thinking



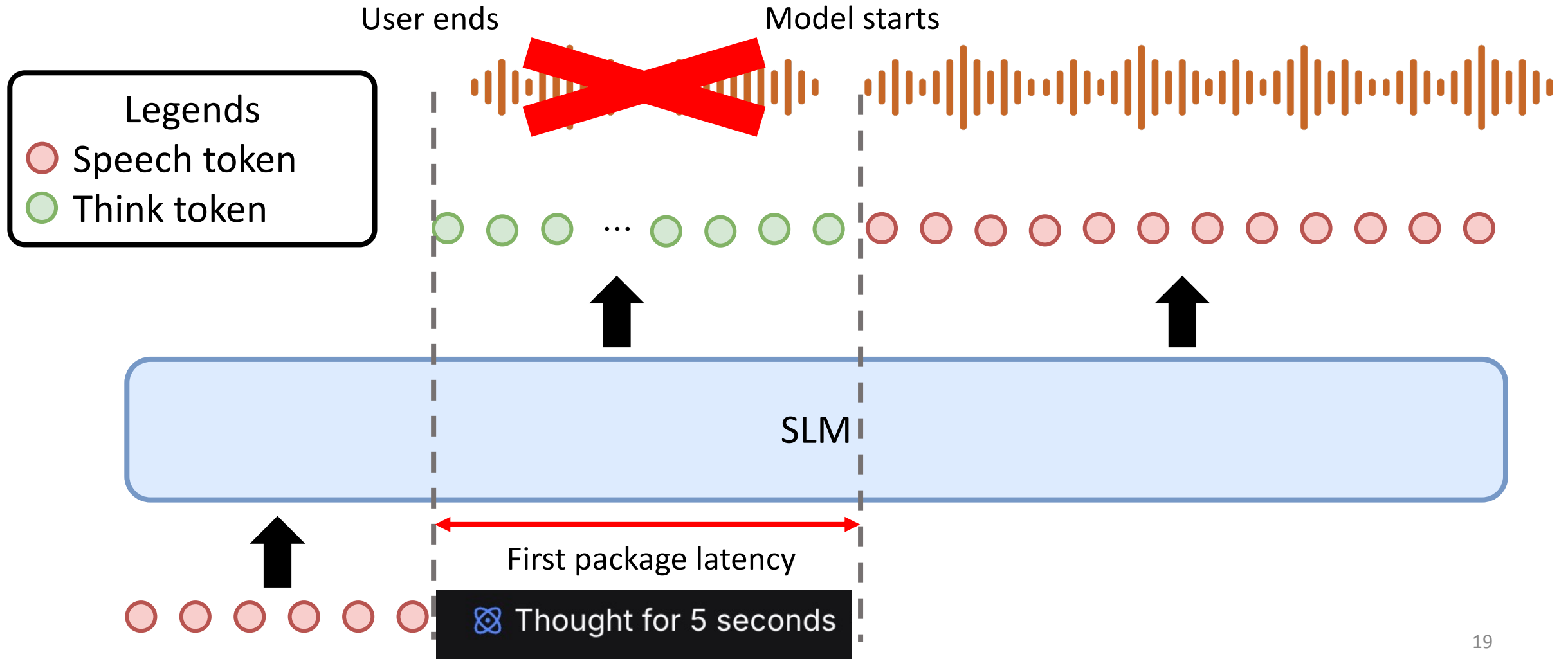
Thinking before Speaking

- Thinking tokens will not be spoken



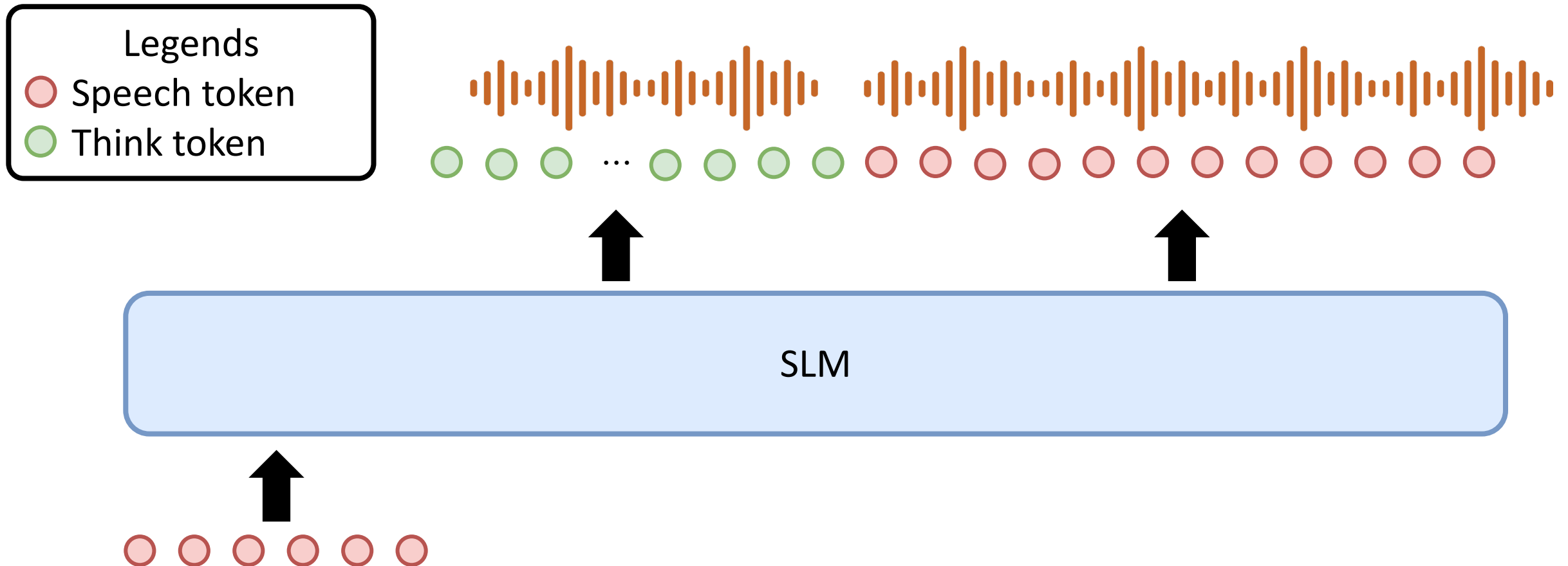
Thinking before Speaking

- Thinking before speaking creates a very high latency



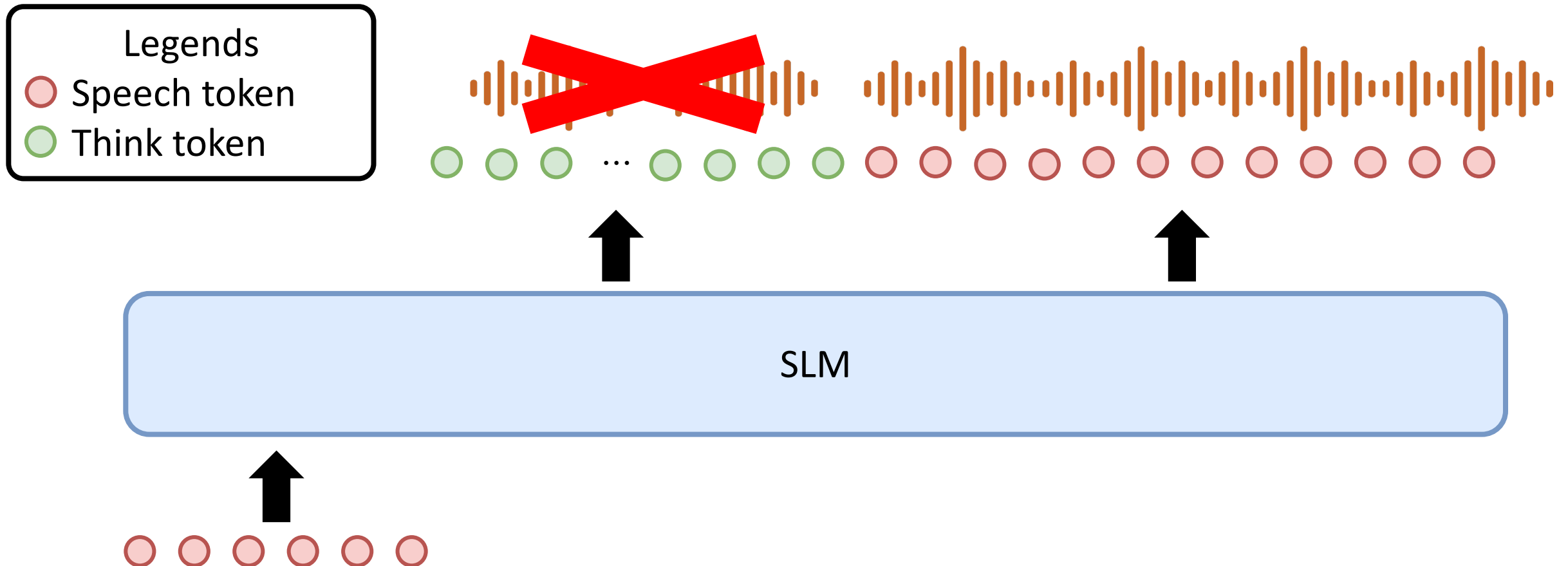
Thinking before Speaking

- Why not speak out the think tokens?
 - Thinking may be too long and hard to understand when spoken



Thinking before Speaking

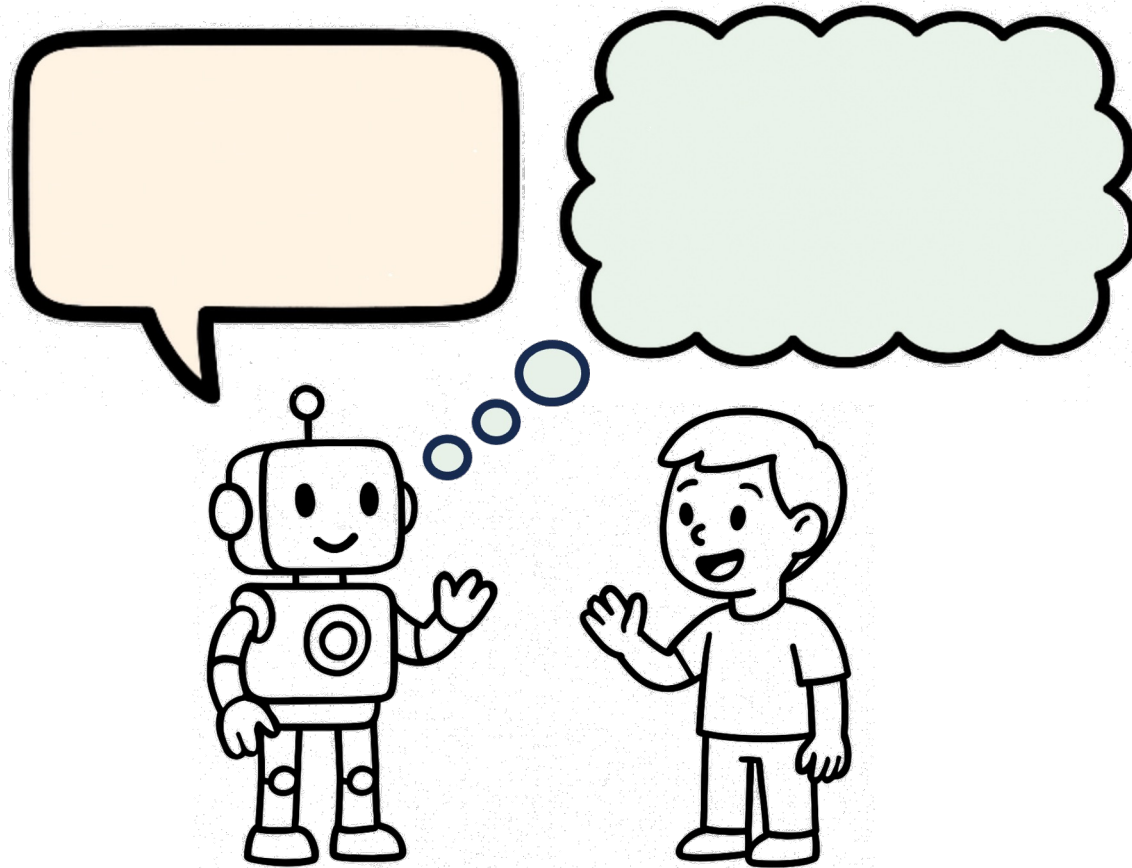
- Thinking *before* speaking is unlikely a good solution



“When should SLM think?”

Setting 2: Listen → Think + Speak

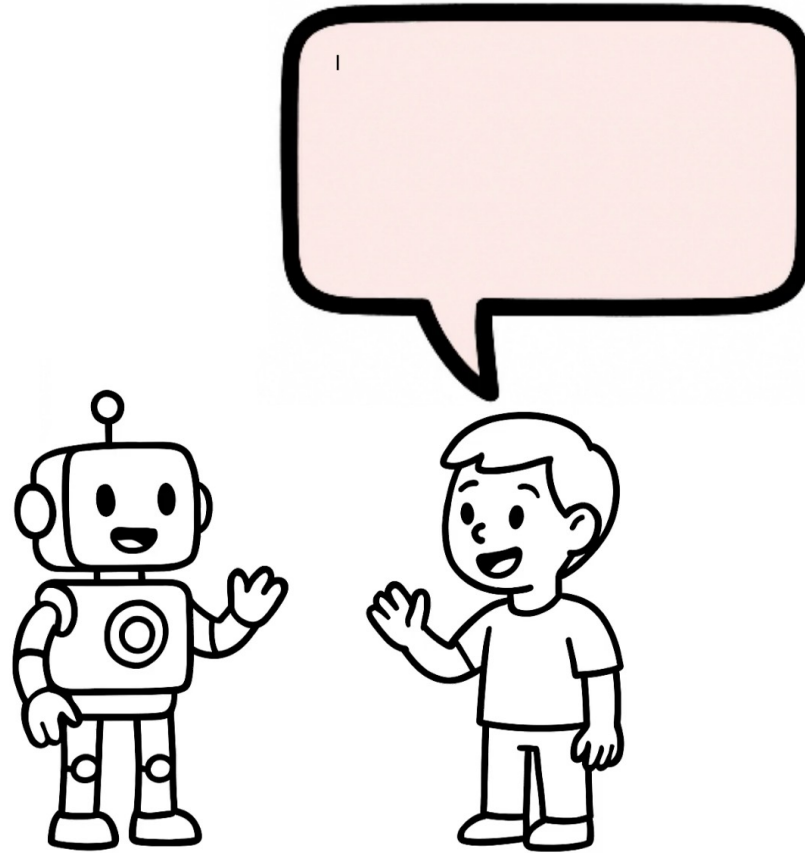
- Humans can think while speaking



STITCH: Simultaneous talking and thinking

Setting 3: Listen + Think → Speak

- Humans can also think while listening



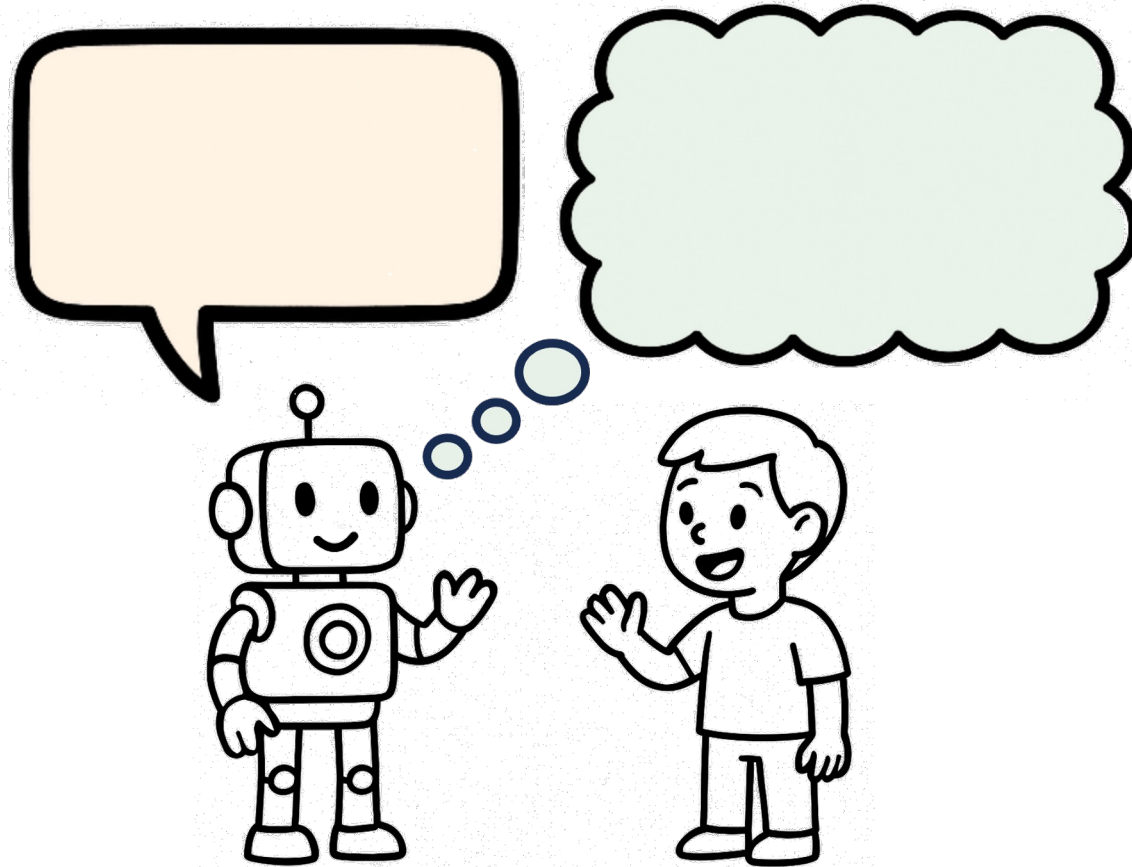
SHANKS: Simultaneous listening and thinking

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Setting 2: Listen → Think + Speak

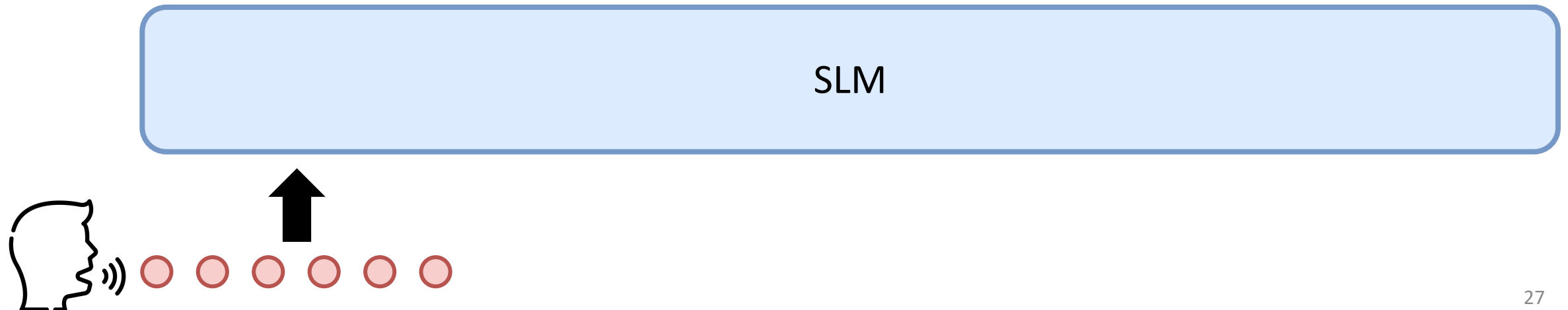
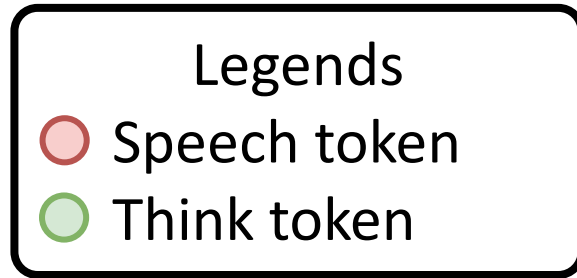
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STITCH: Simultaneous talking and thinking

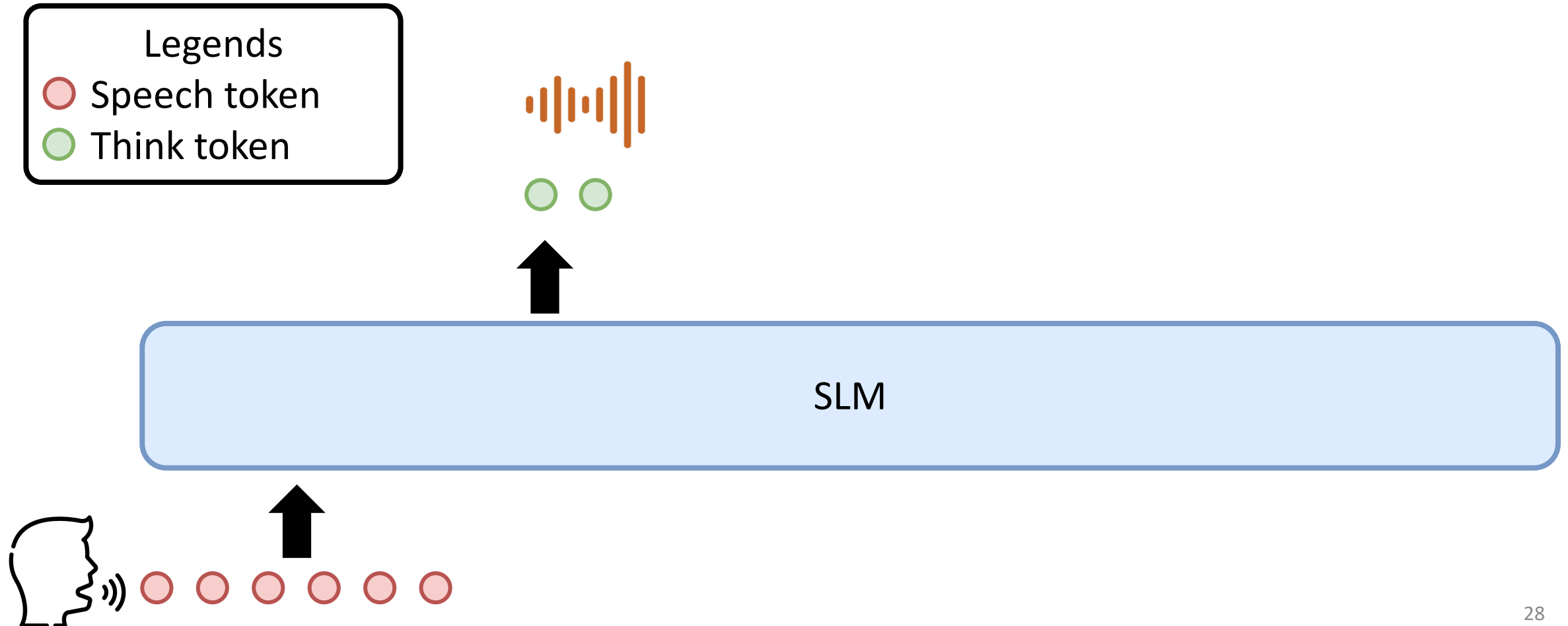
Goal: Thinking while Speaking

- We want to make SLMs think while speaking



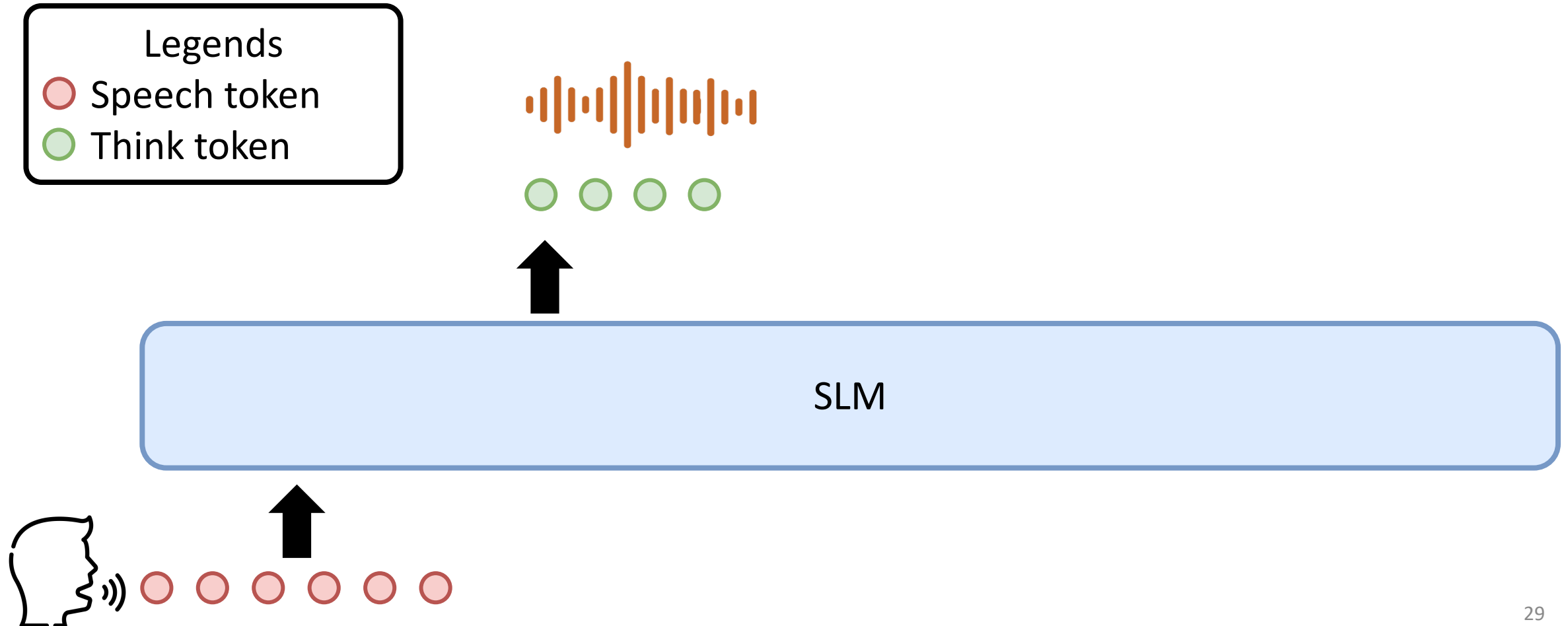
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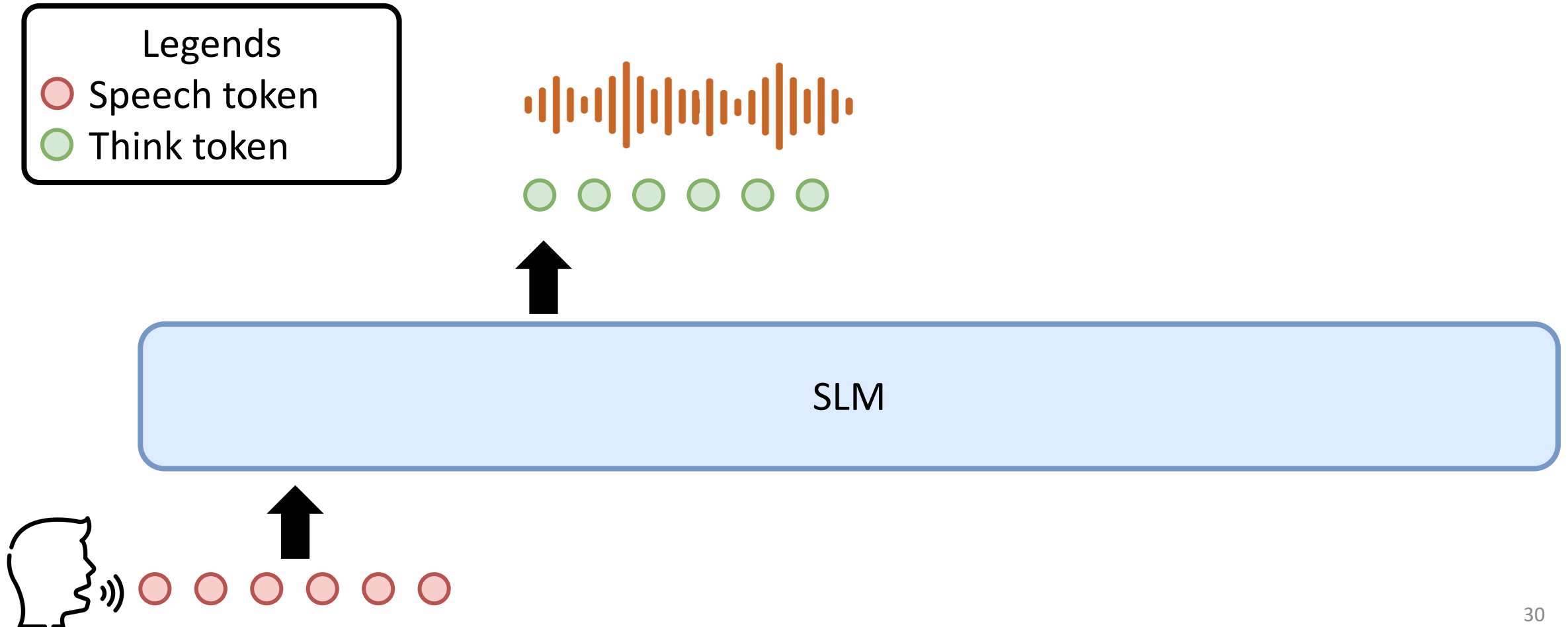
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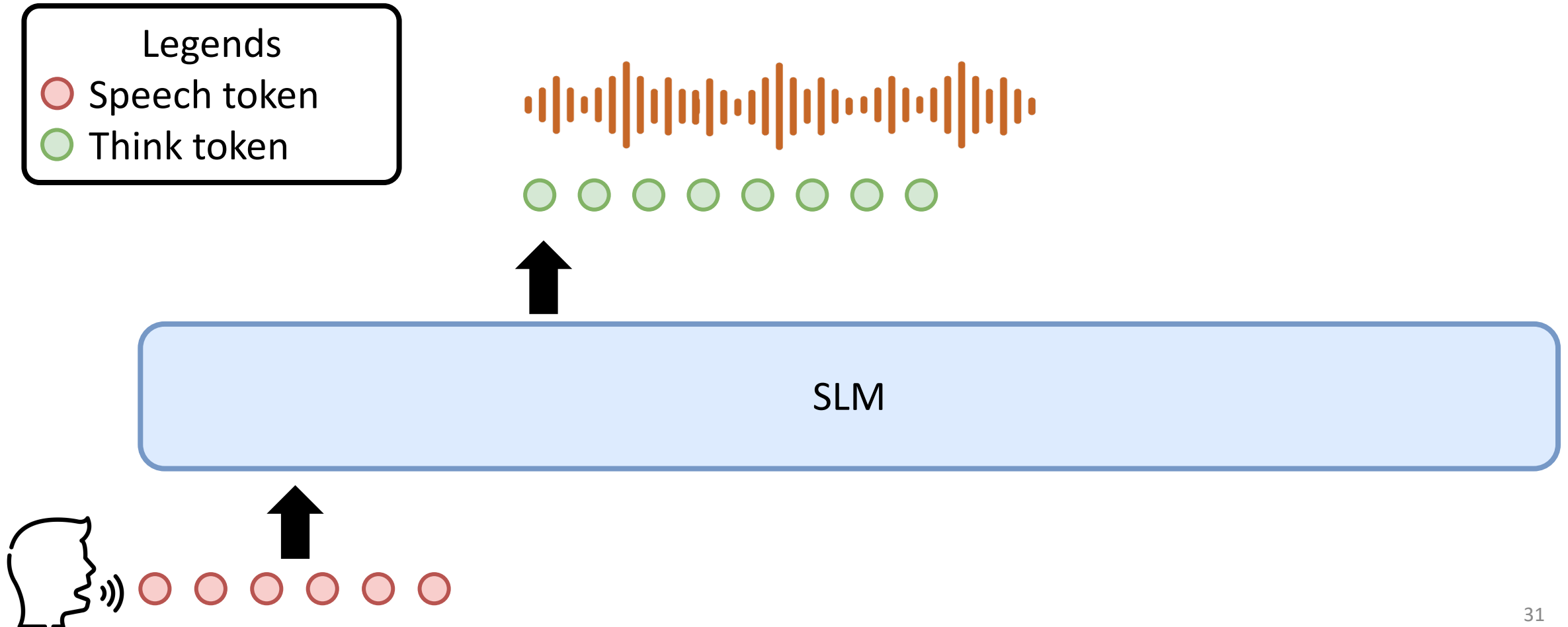
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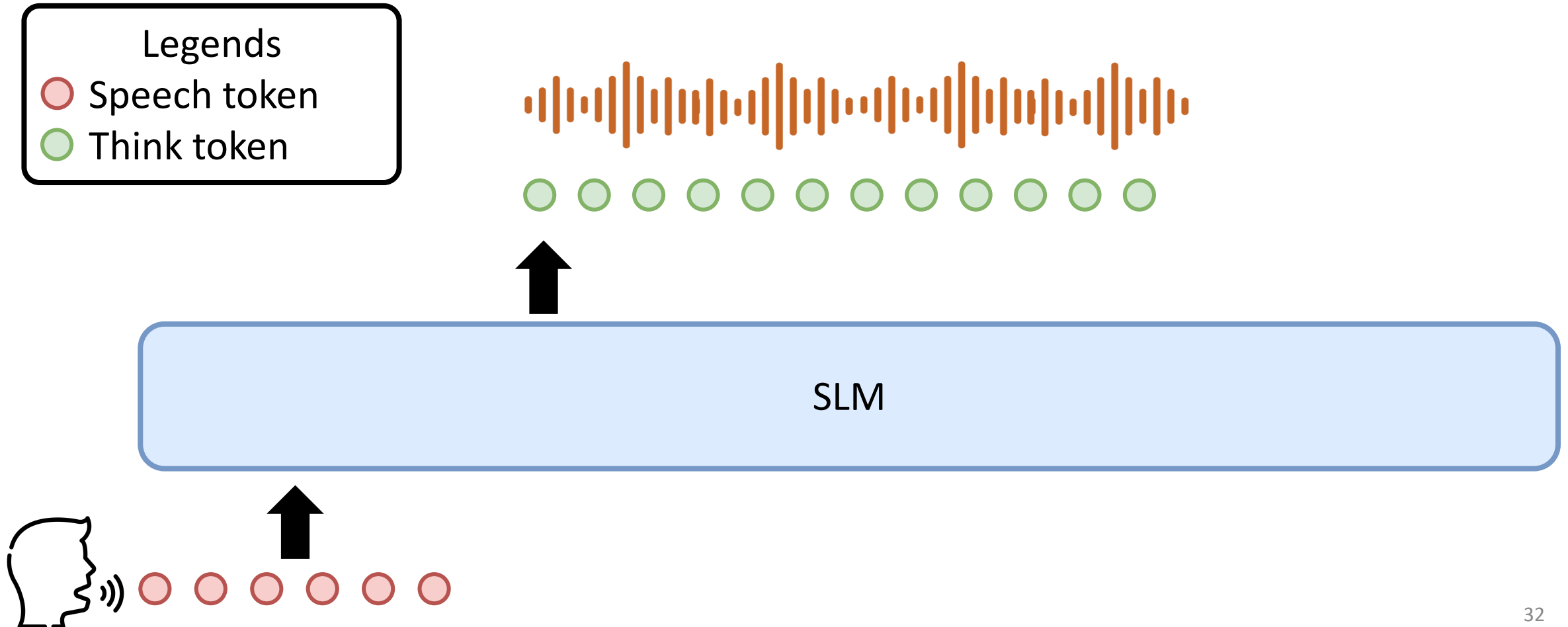
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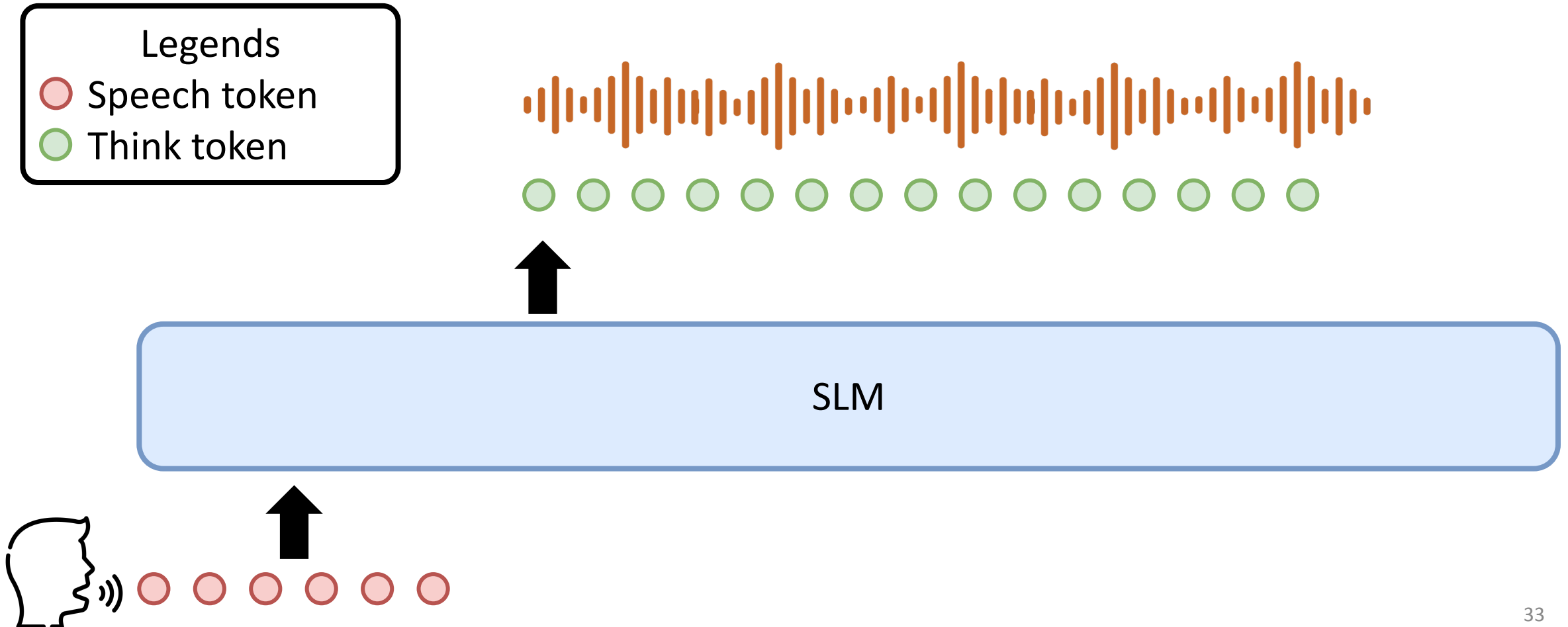
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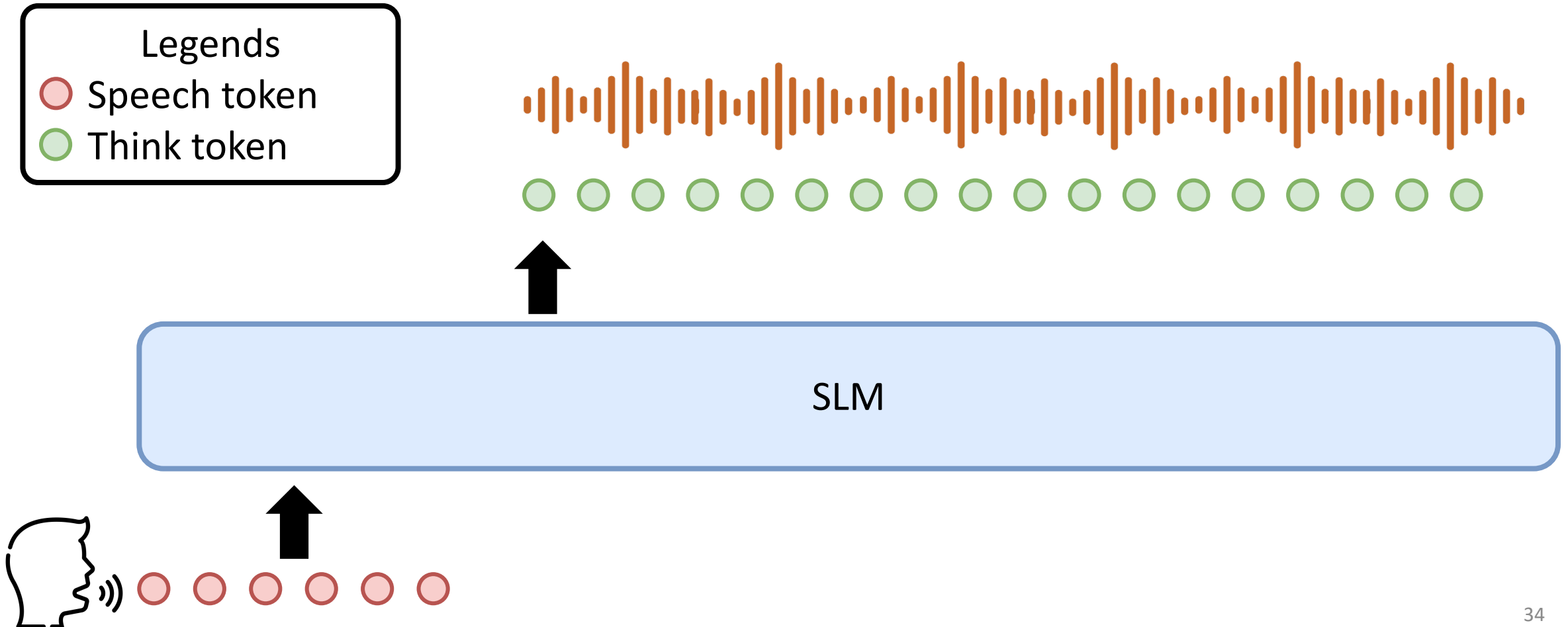
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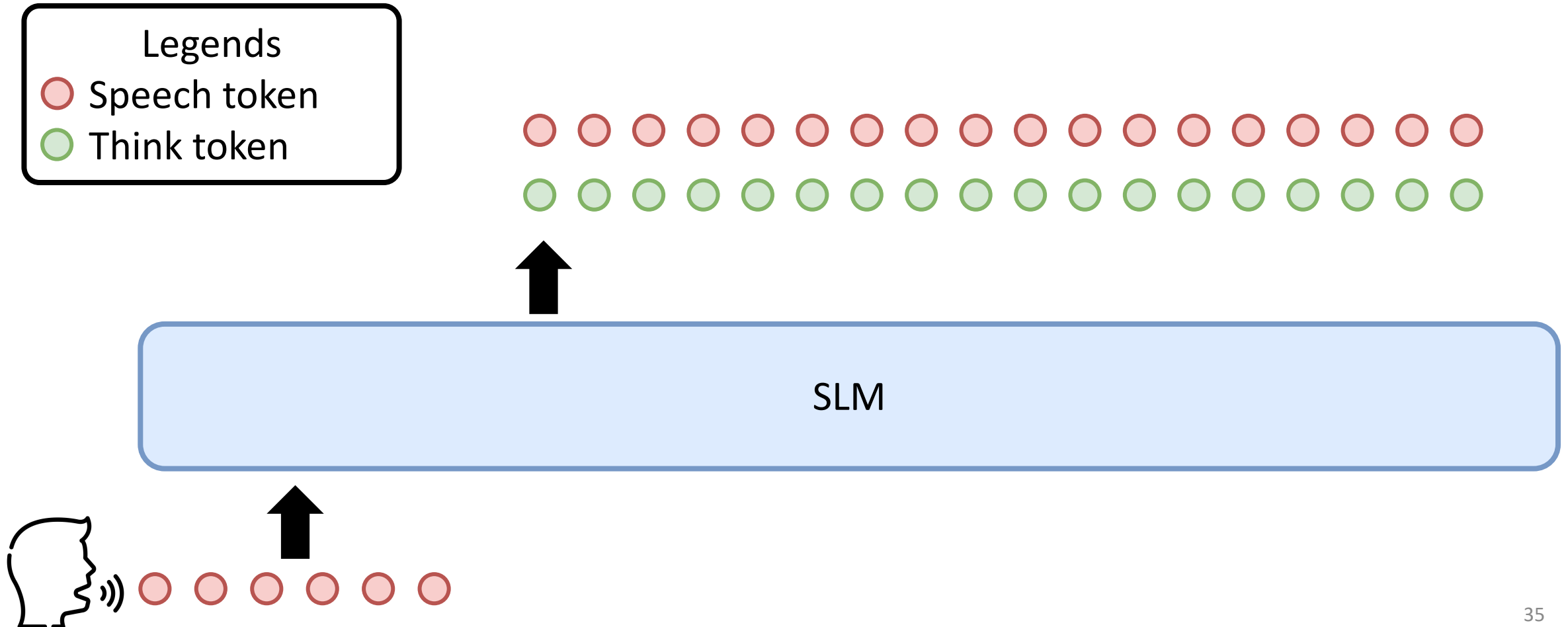
Goal: Thinking while Speaking

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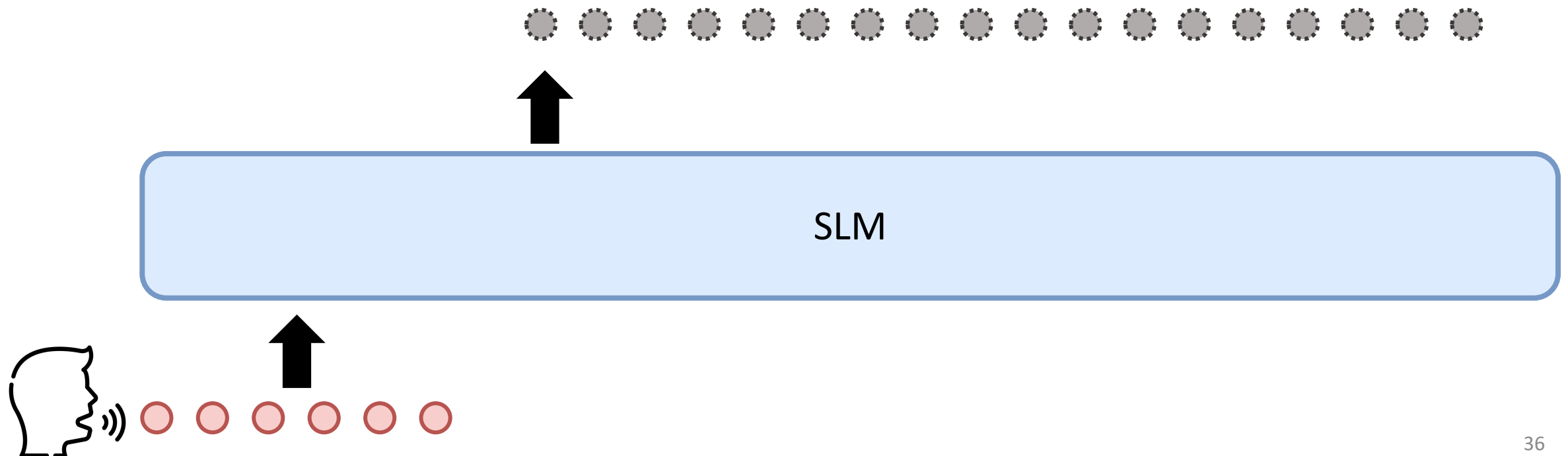
Goal: Thinking while Speaking

- SLMs generate speech token to form the speech responses



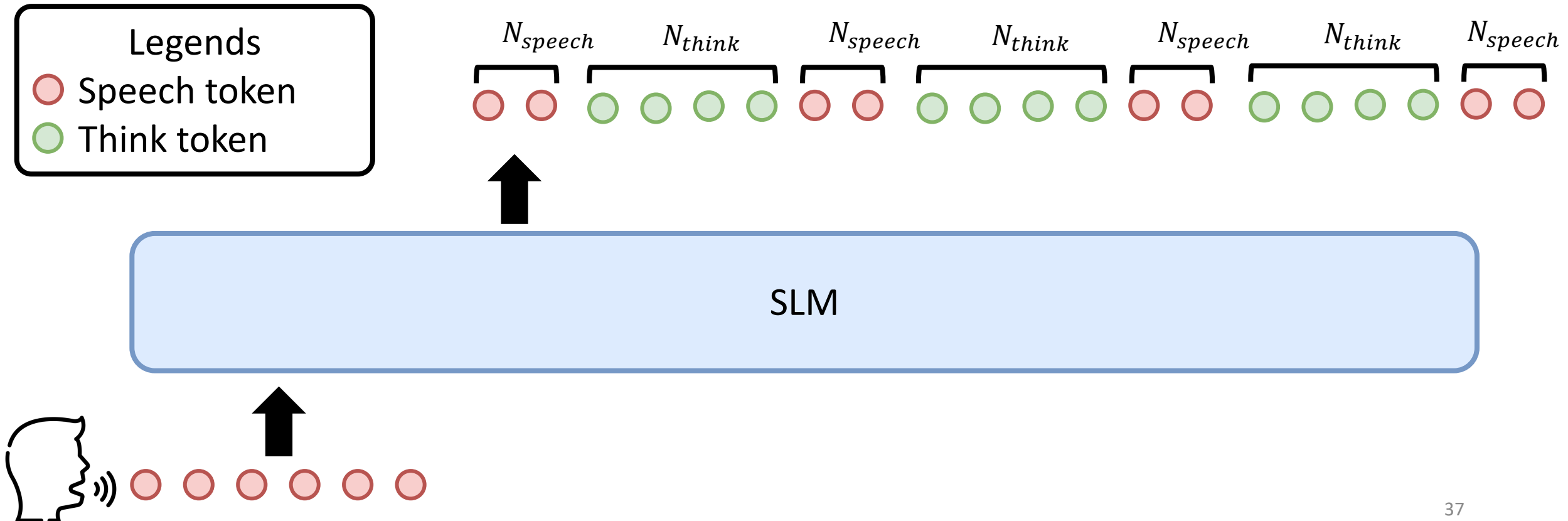
Goal: Thinking while Speaking

- Language model can only generate a single stream of token at a time
- Question: How do we use a single output stream of token but make it as if there are two types of tokens?



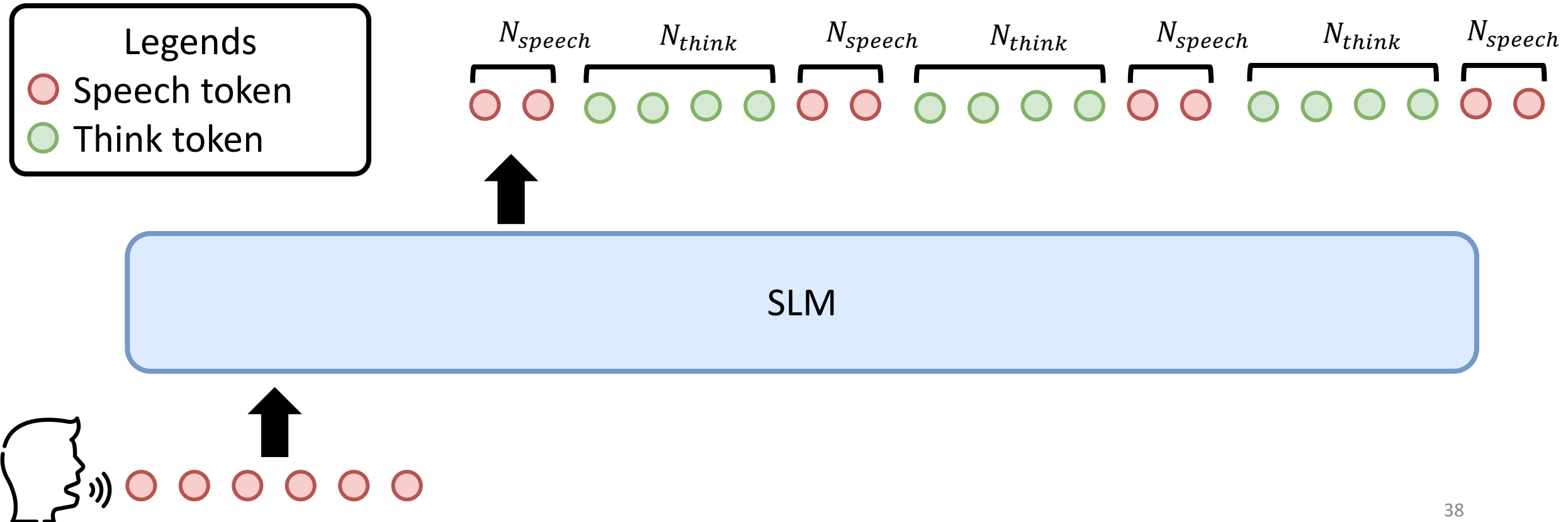
STITCH: Simultaneous Thinking and Talking

- Solution: Generate interleaved **chunks** of thinking and speech tokens
 - Thinking tokens: For **unspoken** internal thinking
 - Speech tokens: For the **spoken** responses



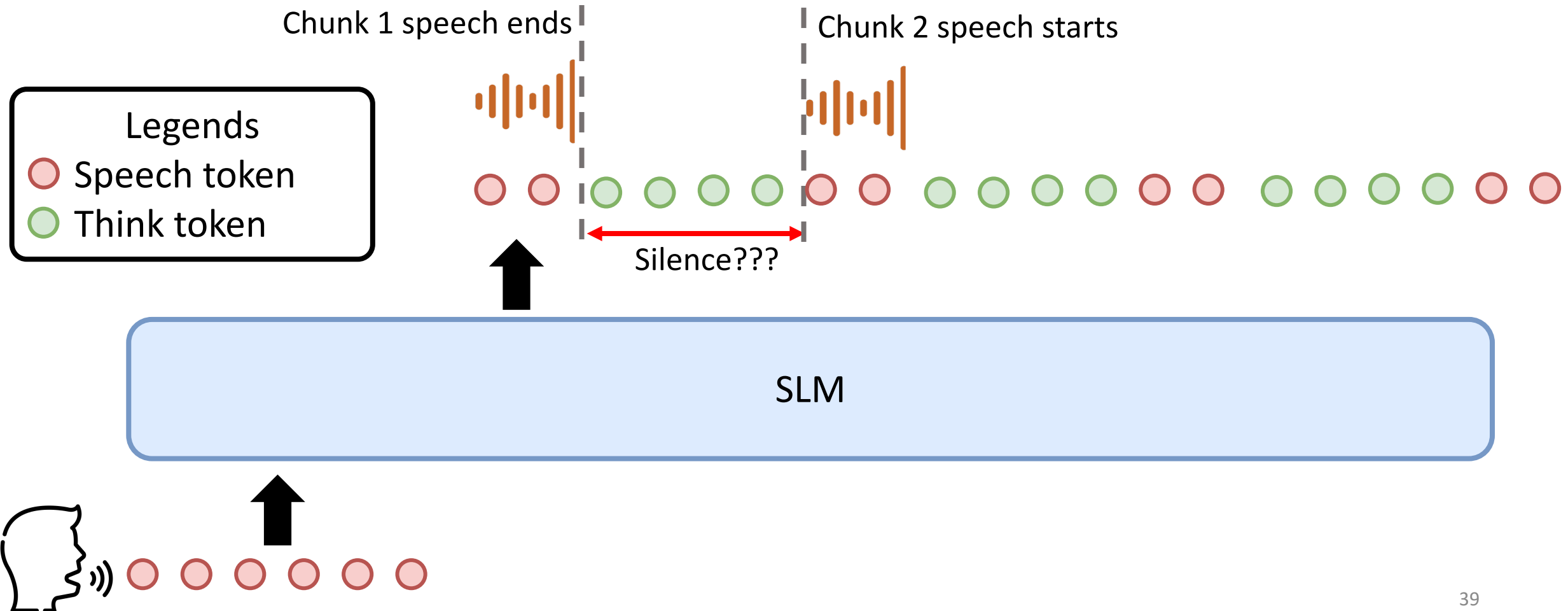
STITCH: Simultaneous Thinking and Talking

- The number of tokens in each chunk is fixed
 - N_{speech} : number of speech tokens in a speech chunk
 - N_{think} : number of thinking tokens in a thinking chunk



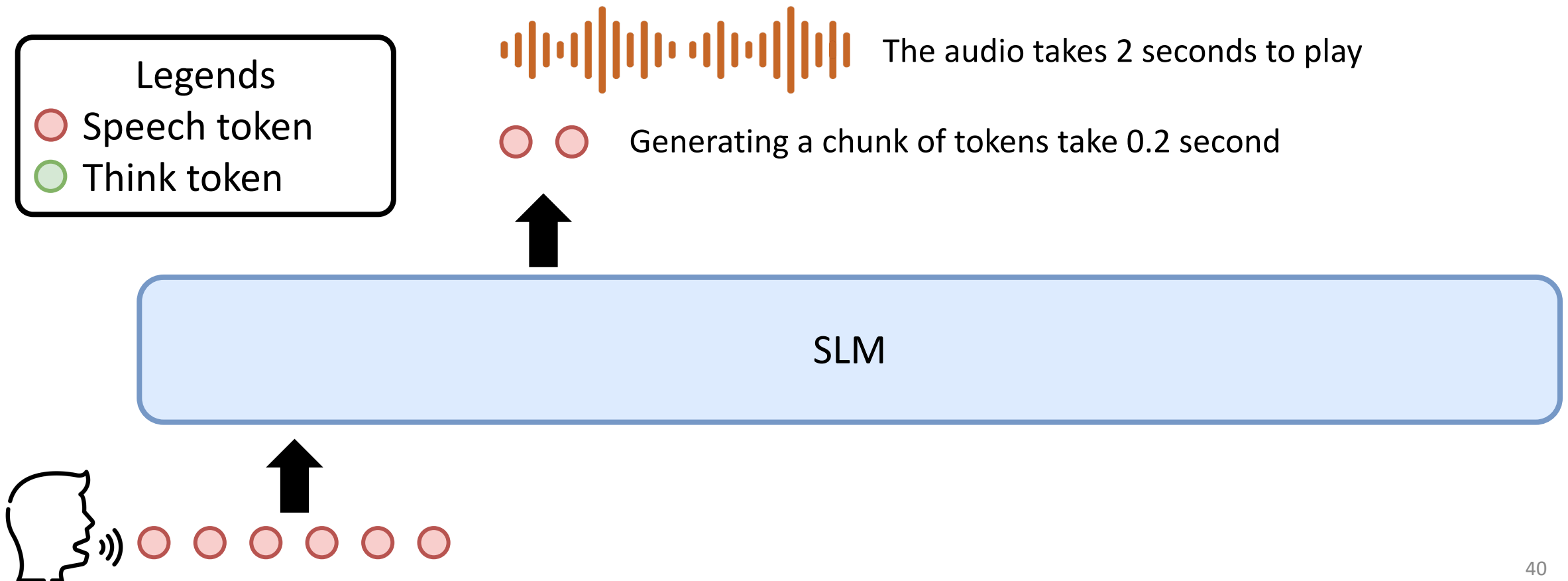
STITCH: Simultaneous Thinking and Talking

- Question: Does the unspoken thinking chunks break the speech response?



STITCH: Simultaneous Thinking and Talking

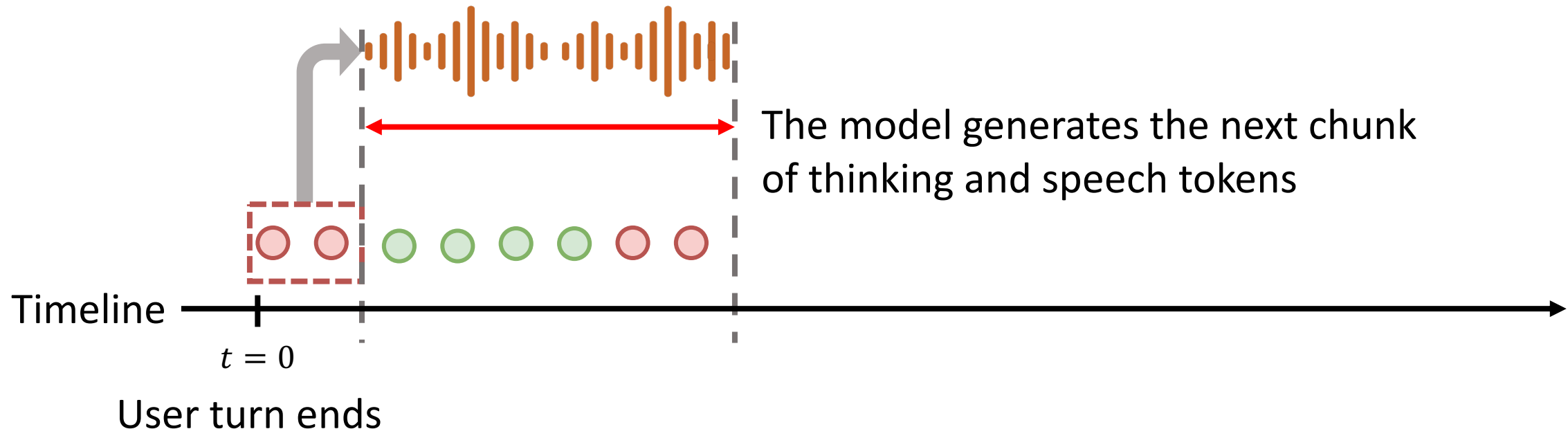
- Answer: No. Because the time for generating the speech tokens in a chunk is less than the time of the audio for that chunk



STITCH: Simultaneous Thinking and Talking

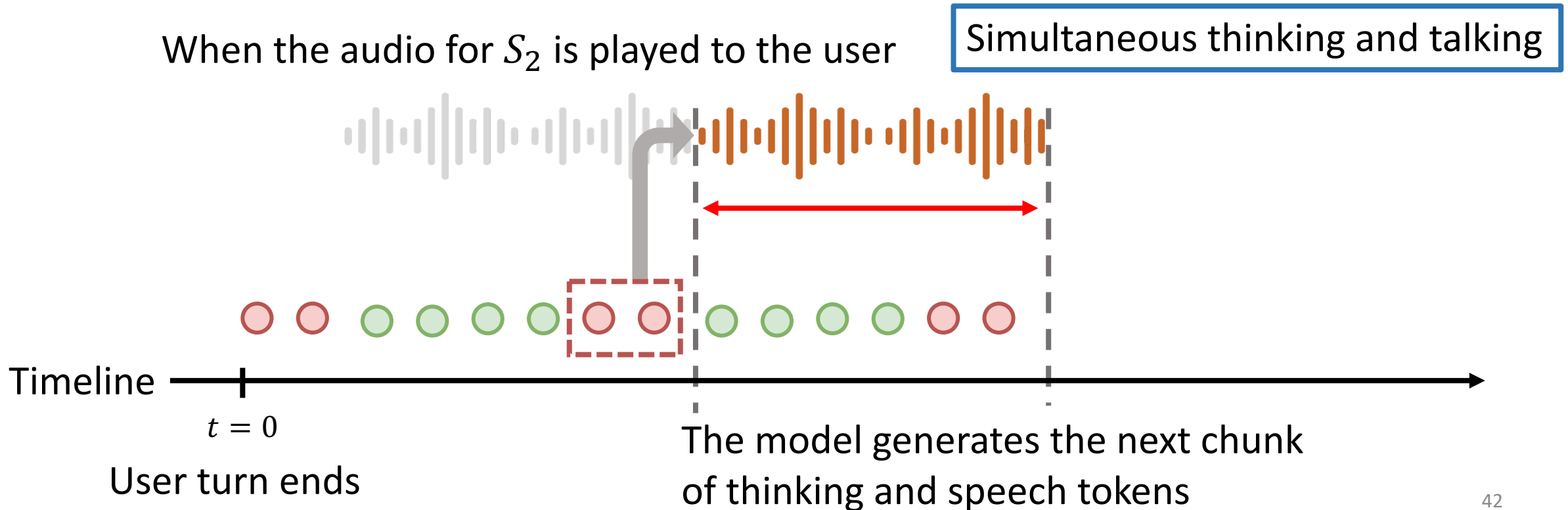
- By carefully selecting the ratio of N_{think} and N_{speech} , the duration of the audio for a speech token chunk can cover the time to generate $N_{think} + N_{speech}$ tokens

When the audio for S_1 is played to the user



STITCH: Simultaneous Thinking and Talking

- By carefully selecting the ratio of the : N_{think}, N_{speech} , the time of the speech of N_{speech} speech token can cover the time to generate $N_{think} + N_{speech}$ tokens



STITCH: Simultaneous Thinking and Talking

- Example of interleaved speech-thinking tokens
- Merging the thinking token chunks forms a complete CoT reasoning

Claire makes a 3-egg omelet every morning for breakfast. How many dozens of eggs will she eat in 4 weeks?

<|audio_0|> <|audio_17|> (...) <|audio_24|>[SOPR] ## Inner Monologue This is a math question. I need to think step by step. To solve this problem, we need to determine how many dozens of eggs Claire will eat in 4 weeks if she makes a 3-egg omelette every morning. 1. **Calculate the number of days in 4 weeks:** - There are 7 days in a week. - Therefore, in 4 weeks, there are $(4 \times 7 = 28)$ [EOPR] <|audio_3|> <|audio_3|> (...) <|audio_12|>[SOPR] days. 2. **Determine the number of eggs Claire eats per day:** - Claire makes a 3-egg omelette every morning. - Therefore, she eats 3 eggs per day. 3. **Calculate the total number of eggs Claire eats in 28 days:** - Since she eats 3 eggs per day, in 28 days, she will eat $(3 \times 28 = 84)$ eggs. 4. **Convert the total number [EOPR] <|audio_2|> <|audio_51|> (...) <|audio_32|>[SOPR] of eggs from whole eggs to dozens:** - There are 12 eggs in a dozen. - To find out how many dozens of eggs she eats, divide the total number of eggs by 12. - $(\frac{84}{12} = 7)$ dozens. Thus, Claire will eat 7 dozens of eggs in 4 weeks. ##### 7 [EOPR][EOR] <|audio_16|><|audio_127|>(...) <|audio_224|>

STITCH: Simultaneous Thinking and Talking

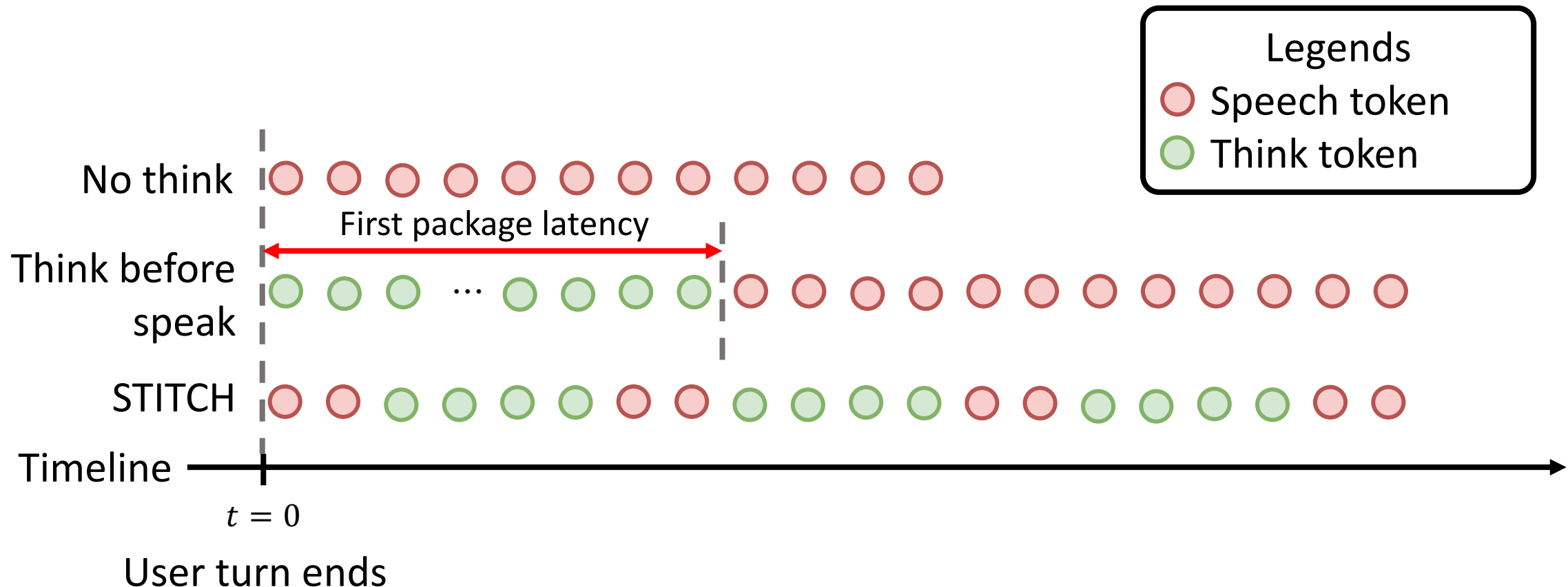
- Example of interleaved speech-think tokens
 - We convert the speech tokens into the corresponding text for better understanding

Claire makes a 3-egg omelet every morning for breakfast. How many dozens of eggs will she eat in 4 weeks?

Claire makes a 3-egg omelette every morning[SOPR] ## Inner Monologue This is a math question. I need to think step by step. To solve this problem, we need to determine how many dozens of eggs Claire will eat in 4 weeks if she makes a 3-egg omelette every morning. 1. **Calculate the number of days in 4 weeks:** - There are 7 days in a week. - Therefore, in 4 weeks, there are $(4 \times 7 = 28)$ [EOPR] , and there are 28 days in 4 weeks. That [SOPR]) days. 2. **Determine the number of eggs Claire eats per day:** - Claire makes a 3-egg omelette every morning. - Therefore, she eats 3 eggs per day. 3. **Calculate the total number of eggs Claire eats in 28 days:** - Since she eats 3 eggs per day, in 28 days, she will eat $(3 \times 28 = 84)$ eggs. 4. **Convert the total number[EOPR] 's 84 eggs in total. Since there are 12 eggs [SOPR] of eggs from whole eggs to dozens:** - There are 12 eggs in a dozen. - To find out how many dozens of eggs she eats, divide the total number of eggs by 12. - $(\frac{84}{12} = 7)$ dozens. Thus, Claire will eat 7 dozens of eggs in 4 weeks. #### 7 [EOPR][EOR] in a dozen, she'll eat 7 dozens of eggs in 4 weeks. The answer is 7 dozens.

STITCH: Simultaneous Thinking and Talking

- First package latency (the time user needs to wait) comparison
 - Think before speak $\gg$ No think = STITCH



STITCH: Training

- Given a question

Claire makes a 3-egg omelet every morning for breakfast. How many dozens of eggs will she eat in 4 weeks?

- Generate the CoT reasoning



and the spoken response



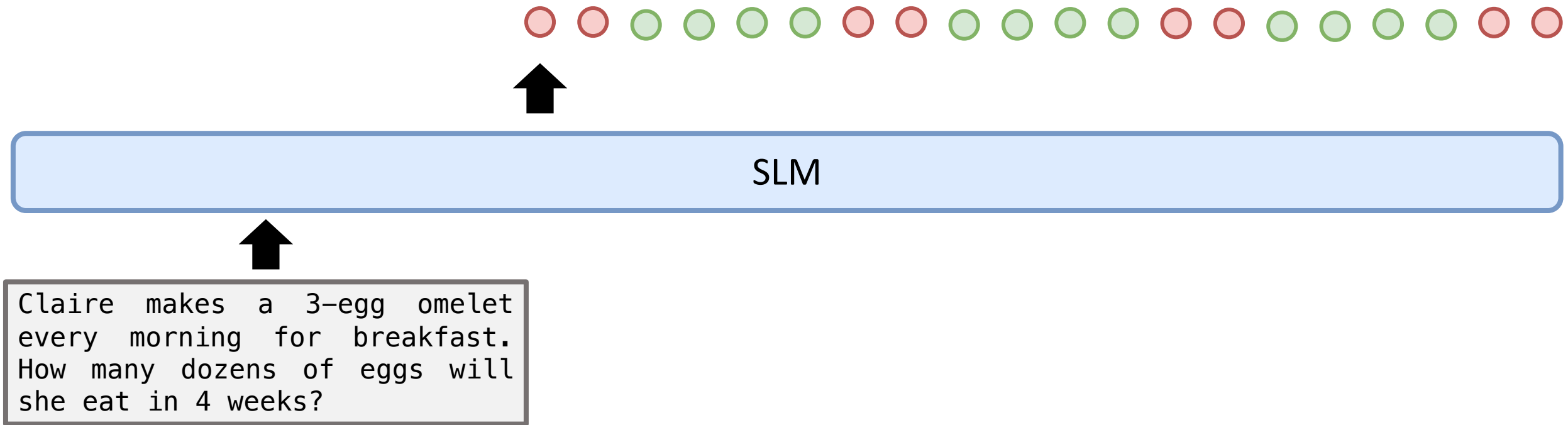
STITCH: Training

- Interleave the N_{speech} speech tokens and N_{think} thinking tokens



STITCH: Training

- Train the model to predict the interleaved speech/thinking tokens



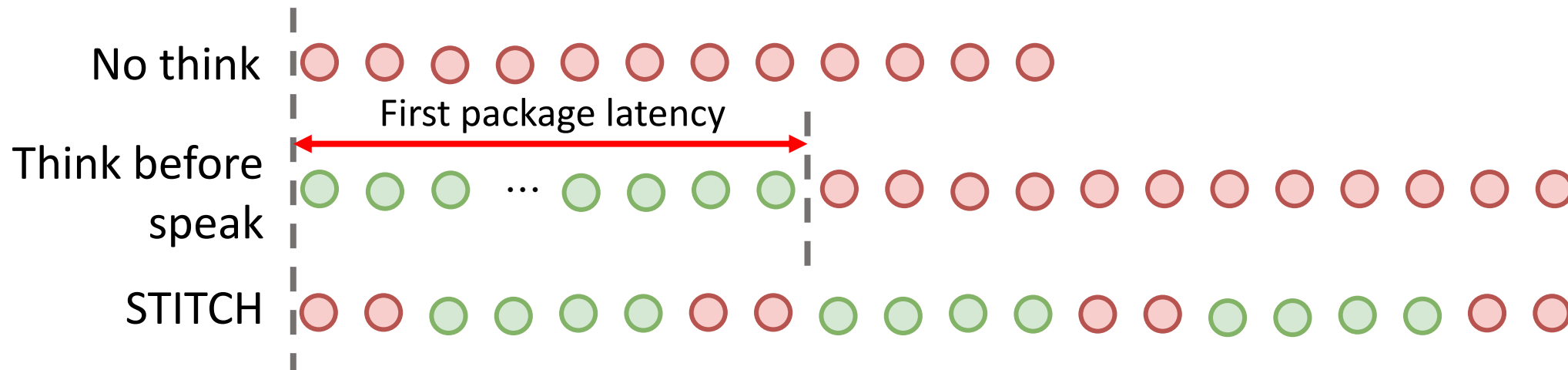
STITCH: Experiment Setting

- Dataset: Four mathematic reasoning datasets
- Evaluation metric
 - Accuracy
 - First package latency (seconds)

STITCH: Experiment Results

Compared methods

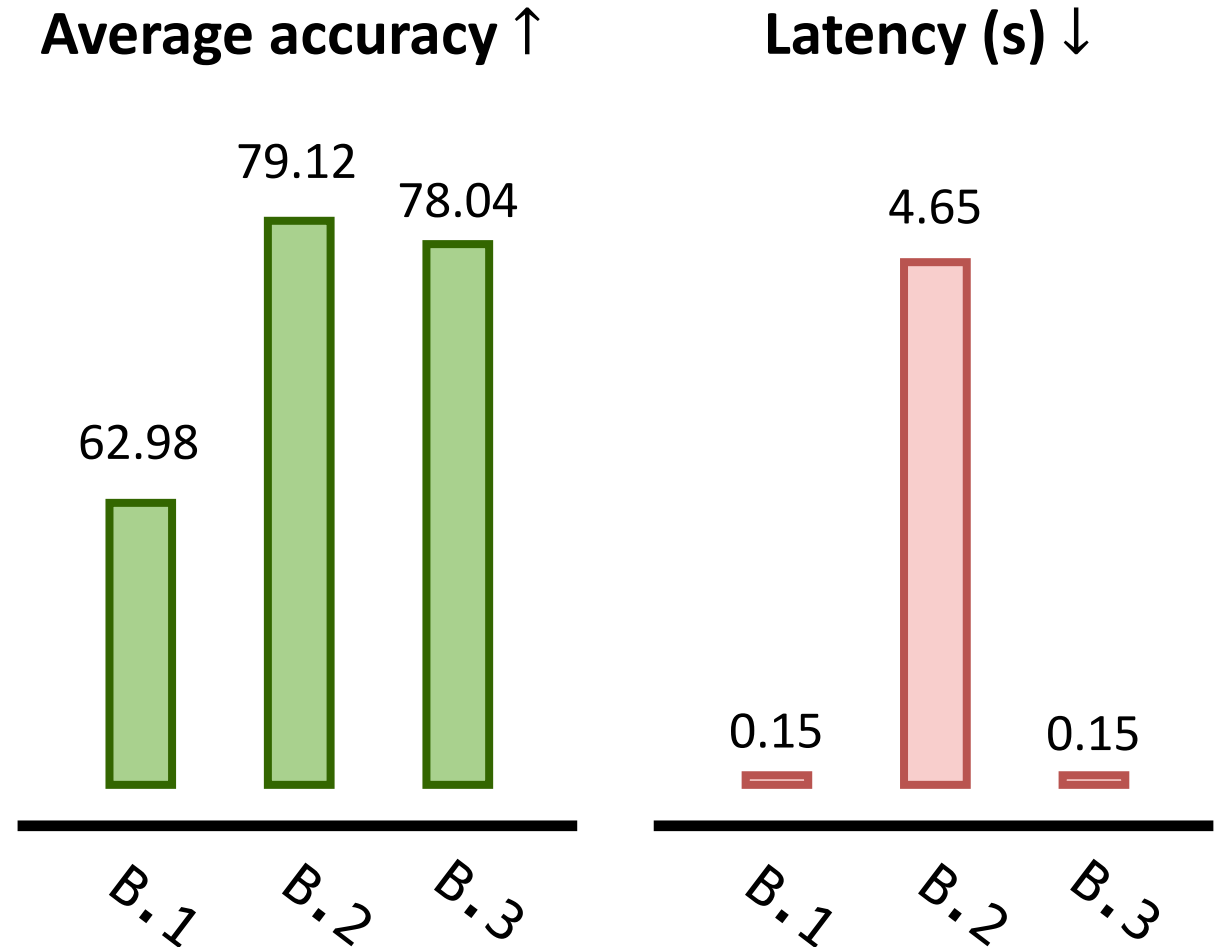
- B.1: No think
- B.2: Think before speaking
- B.3: STITCH (thinking while speaking)



STITCH: Experiment Results

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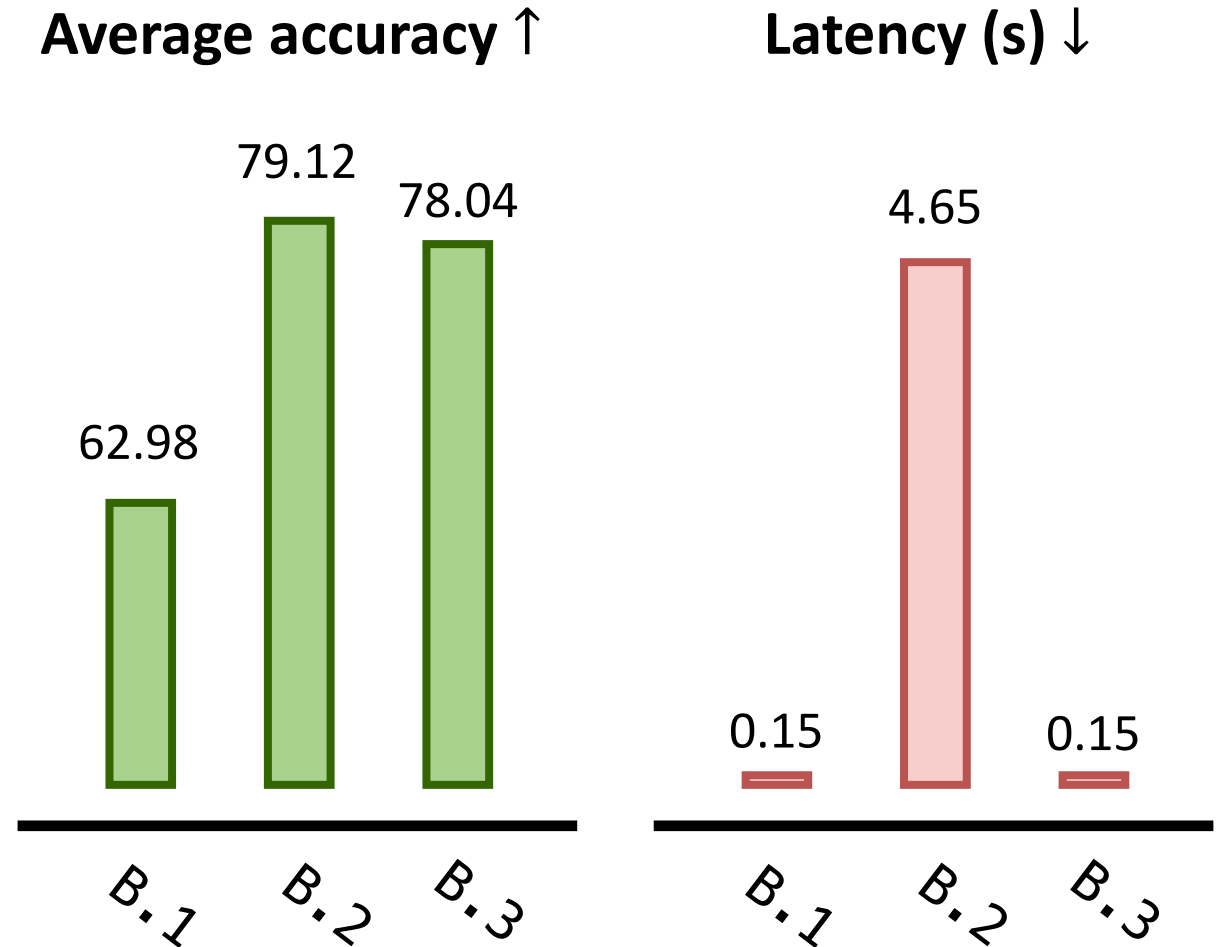


STITCH: Experiment Results

Compared methods

- B.1: No think
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- B.3: STITCH (thinking while speaking)

- No thinking has a low latency and low accuracy

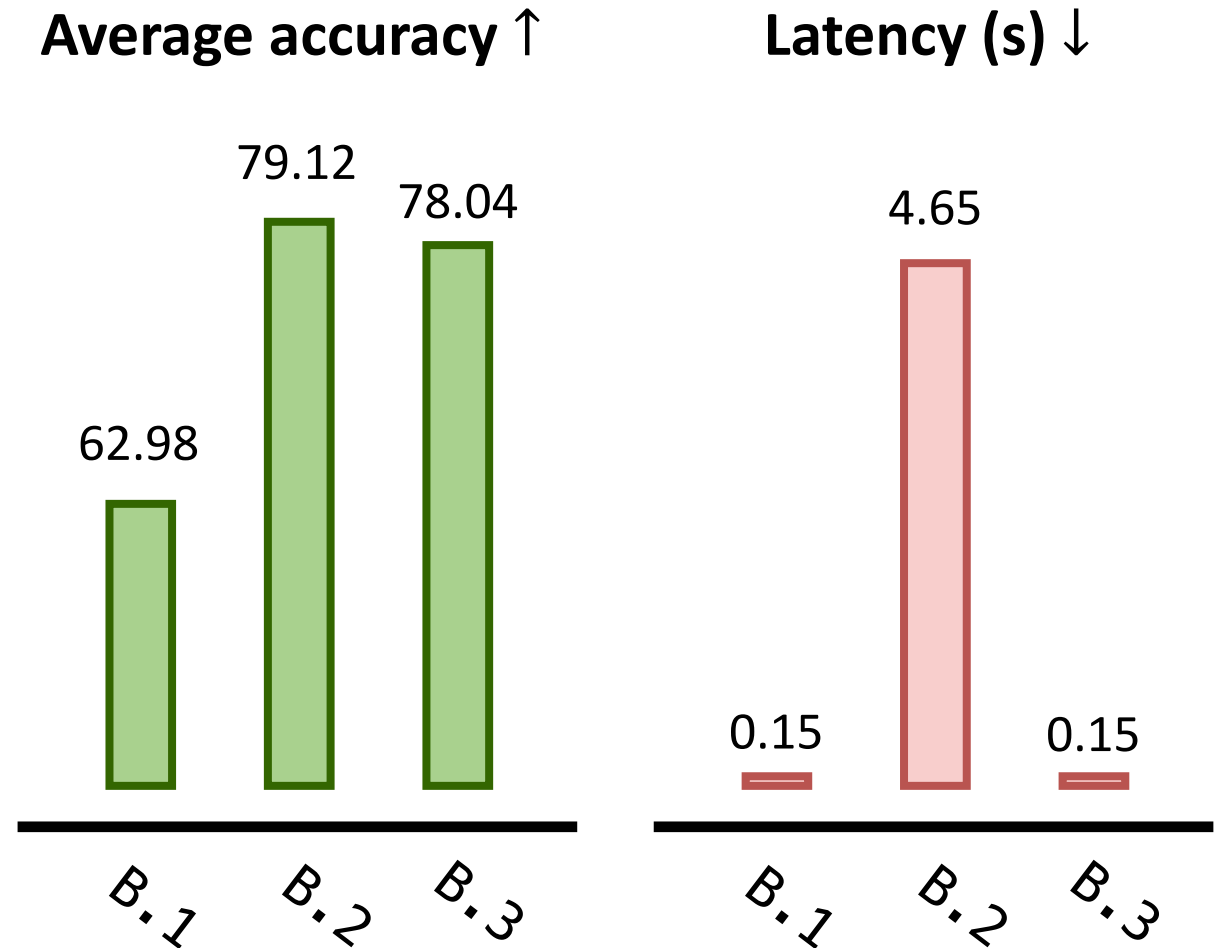


STITCH: Experiment Results

Compared methods

- B.1: No think
- B.2: Think before speaking
- B.3: STITCH (thinking while speaking)

- Thinking before speaking has a high accuracy but induces high latency

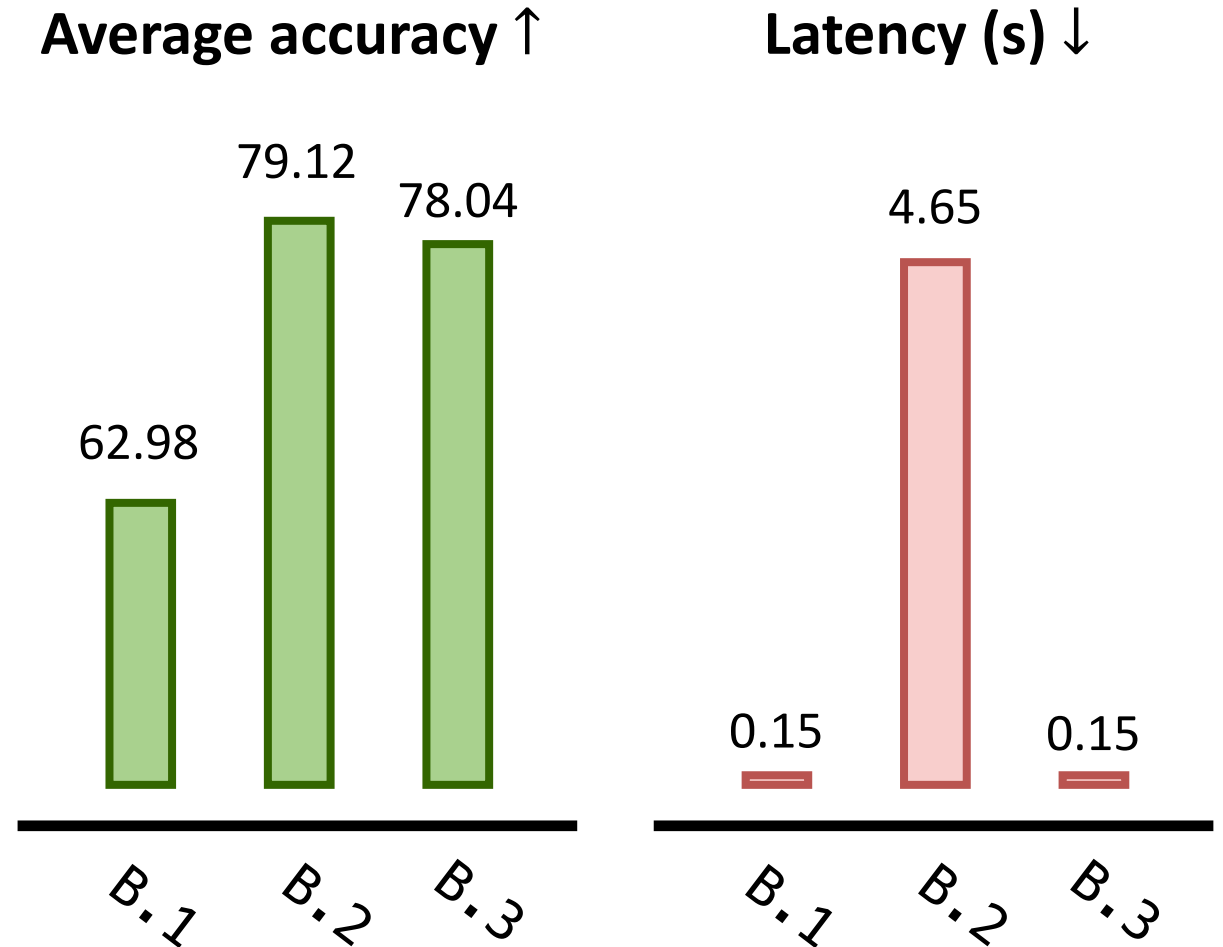


STITCH: Experiment Results

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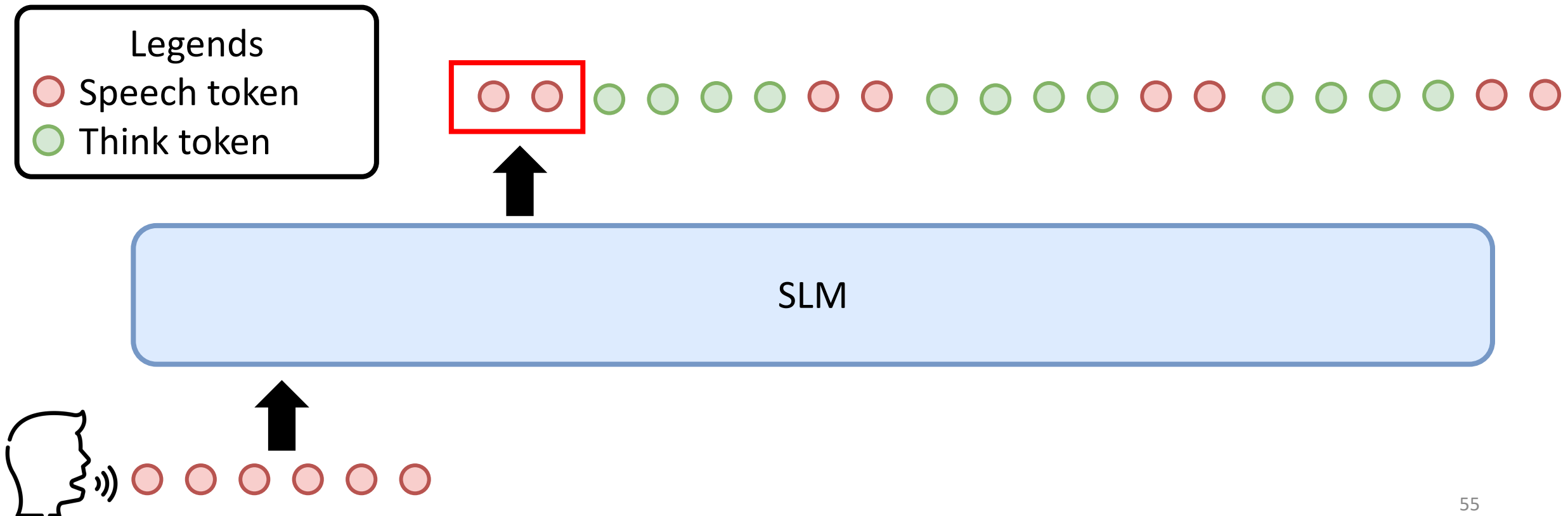
- B.1: No think
- B.2: Think before speaking
- B.3: STITCH (thinking while speaking)

- STITCH has low latency but high accuracy



Analysis

- The first speech chunk is not generated based on any thinking
- But the performance almost does not drop
- Why?



STITCH: Simultaneous Thinking and Talking

- The first chunk of speech output is mostly a rephrase of the question, so it can be generated without any reasoning

Claire makes a 3-egg omelet every morning for breakfast. How many dozens of eggs will she eat in 4 weeks?

Claire makes a 3-egg omelette every morning[SOPR] ## Inner Monologue This is a math question. I need to think step by step. To solve this problem, we need to determine how many dozens of eggs Claire will eat in 4 weeks if she makes a 3-egg omelette every morning. 1. **Calculate the number of days in 4 weeks:** - There are 7 days in a week. - Therefore, in 4 weeks, there are $(4 \times 7 = 28)$ [EOPR] , and there are 28 days in 4 weeks. That [SOPR]) days. 2. **Determine the number of eggs Claire eats per day:** - Claire makes a 3-egg omelette every morning. - Therefore, she eats 3 eggs per day. 3. **Calculate the total number of eggs Claire eats in 28 days:** - Since she eats 3 eggs per day, in 28 days, she will eat $(3 \times 28 = 84)$ eggs. 4. **Convert the total number[EOPR] 's 84 eggs in total. Since there are 12 eggs [SOPR] of eggs from whole eggs to dozens:** - There are 12 eggs in a dozen. - To find out how many dozens of eggs she eats, divide the total number of eggs by 12. - $(\frac{84}{12} = 7)$ dozens. Thus, Claire will eat 7 dozens of eggs in 4 weeks. #### 7 [EOPR][EOR] in a dozen, she'll eat 7 dozens of eggs in 4 weeks. The answer is 7 dozens.

How Is the Perceptual Quality?

- Does interleaving reasoning and speech harms the perceptual quality?
- B.1: No think
- B.2: Think before speaking
- B.3: STITCH (thinking while speaking)

	MOS (1-5)
B.1	3.15
B.2	3.14
B.3	3.17

- All methods have a similar MOS score

Demo

User: Claire makes a 3 egg omelet every morning for breakfast. How many dozens of eggs will she eat in 4 weeks?

Summary

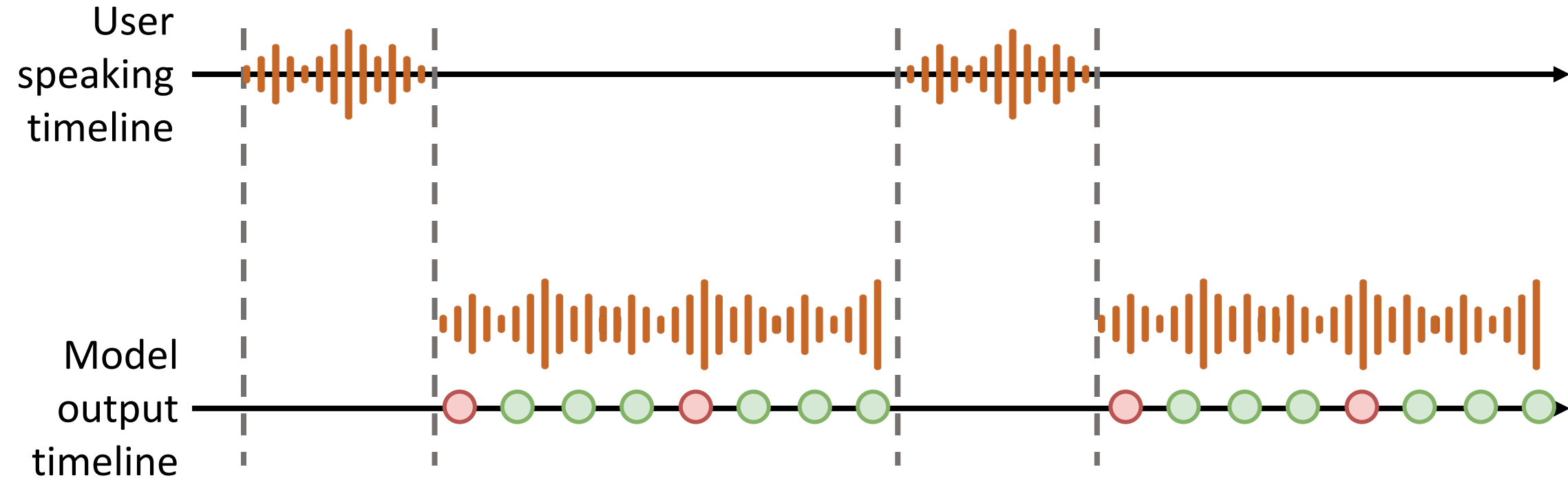
- Thinking while speaking can have a low latency as the no thinking model but has a much better performance on reasoning tasks

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3. **SHANKS: Thinking while listening**
4. Conclusion

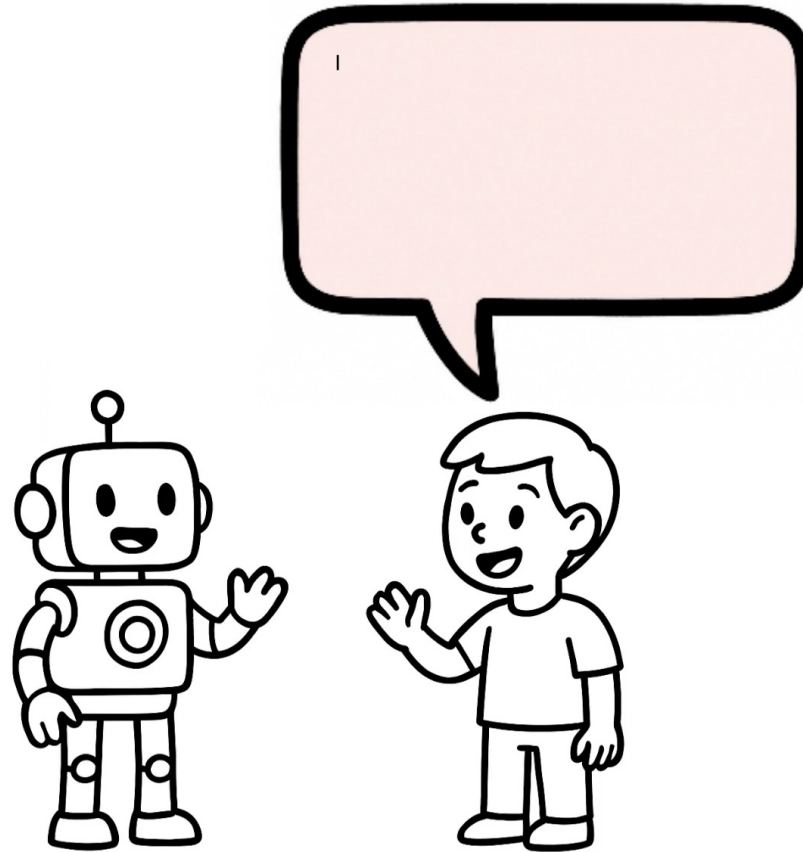
SHANKS: Simultaneous Hearing and Thinking

- Our previous model: Turn taking
 - User speaks → Model thinks and speak → User speaks → Model thinks and speak



Setting 3: Listen + Think → Speak

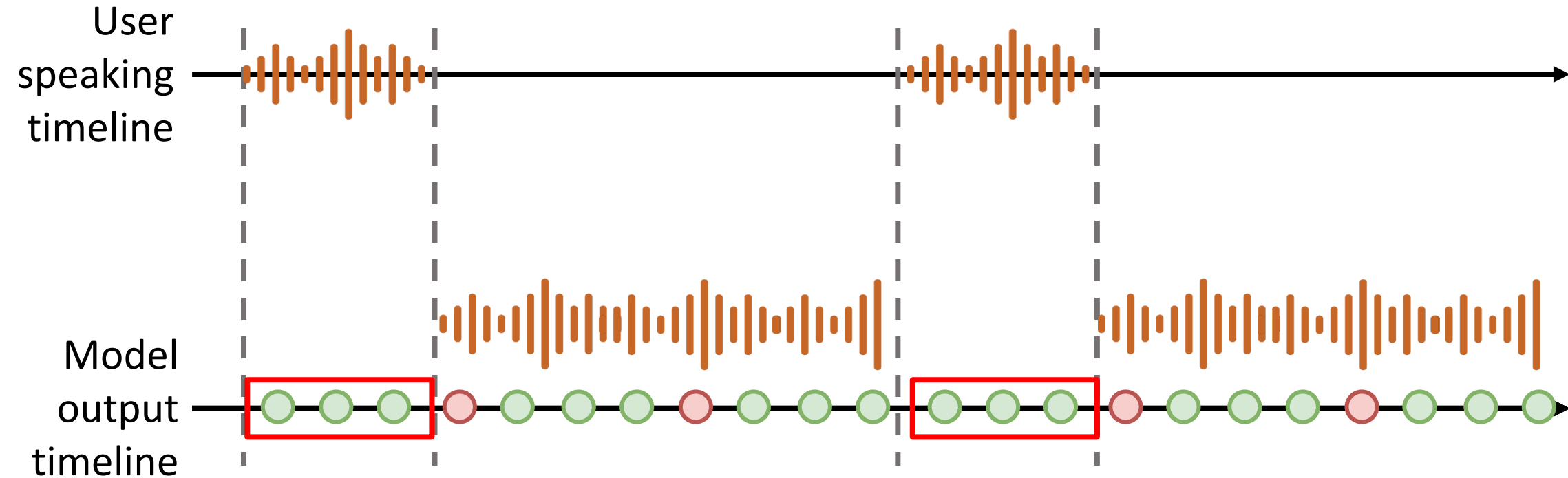
- Humans can also think while listening



SHANKS: Simultaneous listening and thinking

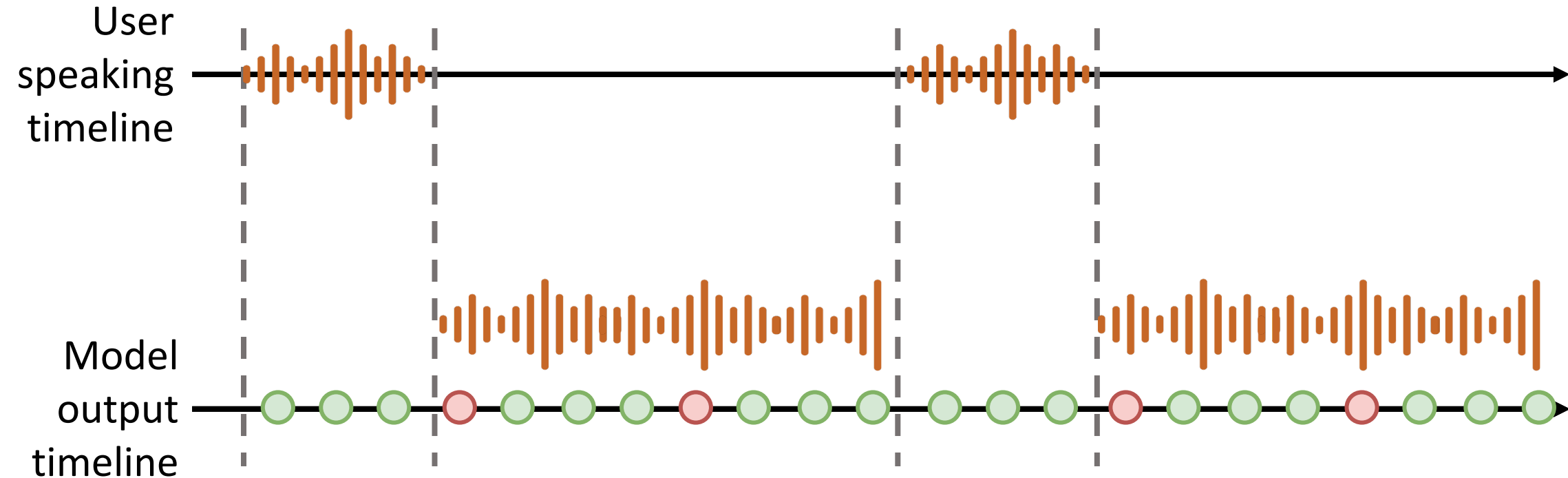
SHANKS: Simultaneous Hearing and Thinking

- Humans think when we are hearing



SHANKS: Simultaneous Hearing and Thinking

- What benefits can we get if we think when listening?
 1. We can interact with the speaker in real-time, e.g., interrupting the speaker
 2. We can prepare for the response while we are still listening



Goal: Thinking while Listening

- We want to make the model think while listening

Legends

- Speech token
- Think token

SLM



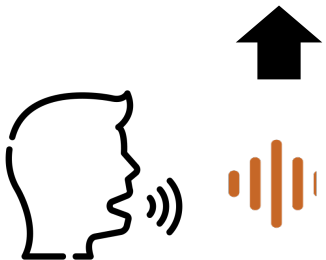
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Legends

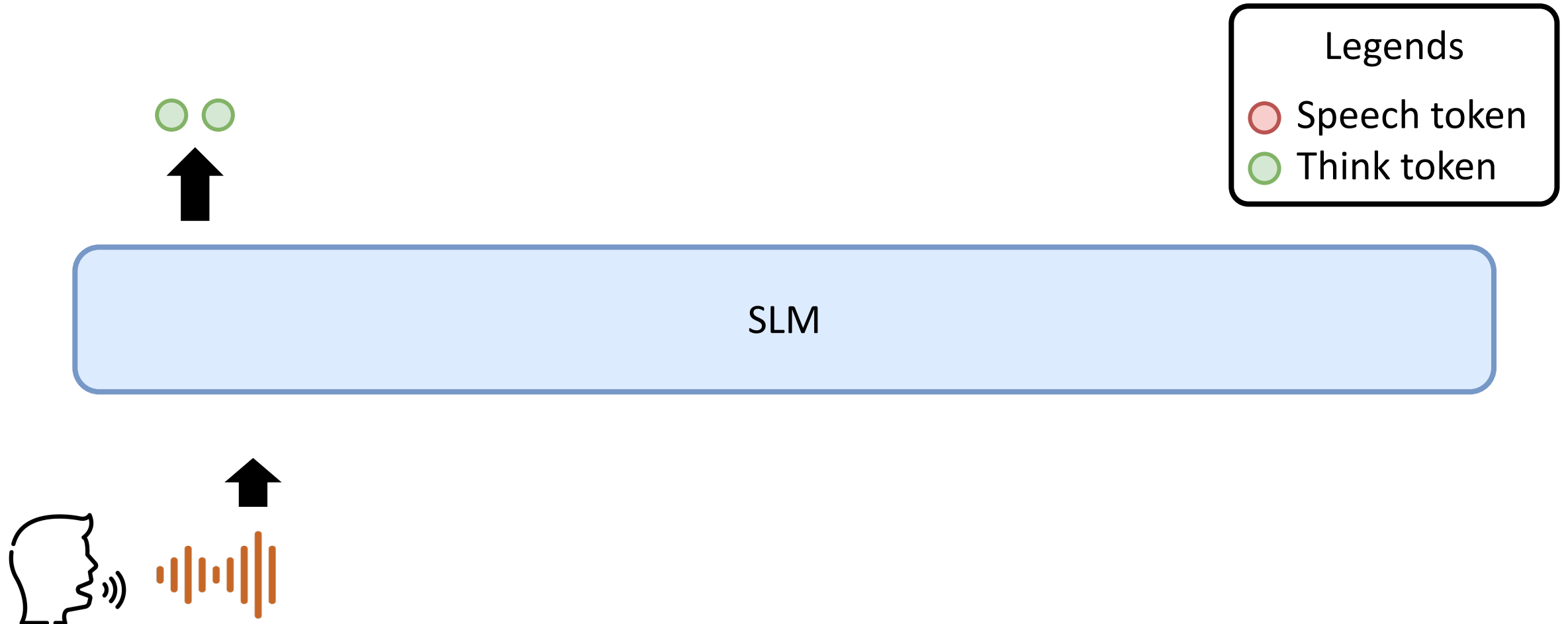
- Speech token
- Think token

SLM



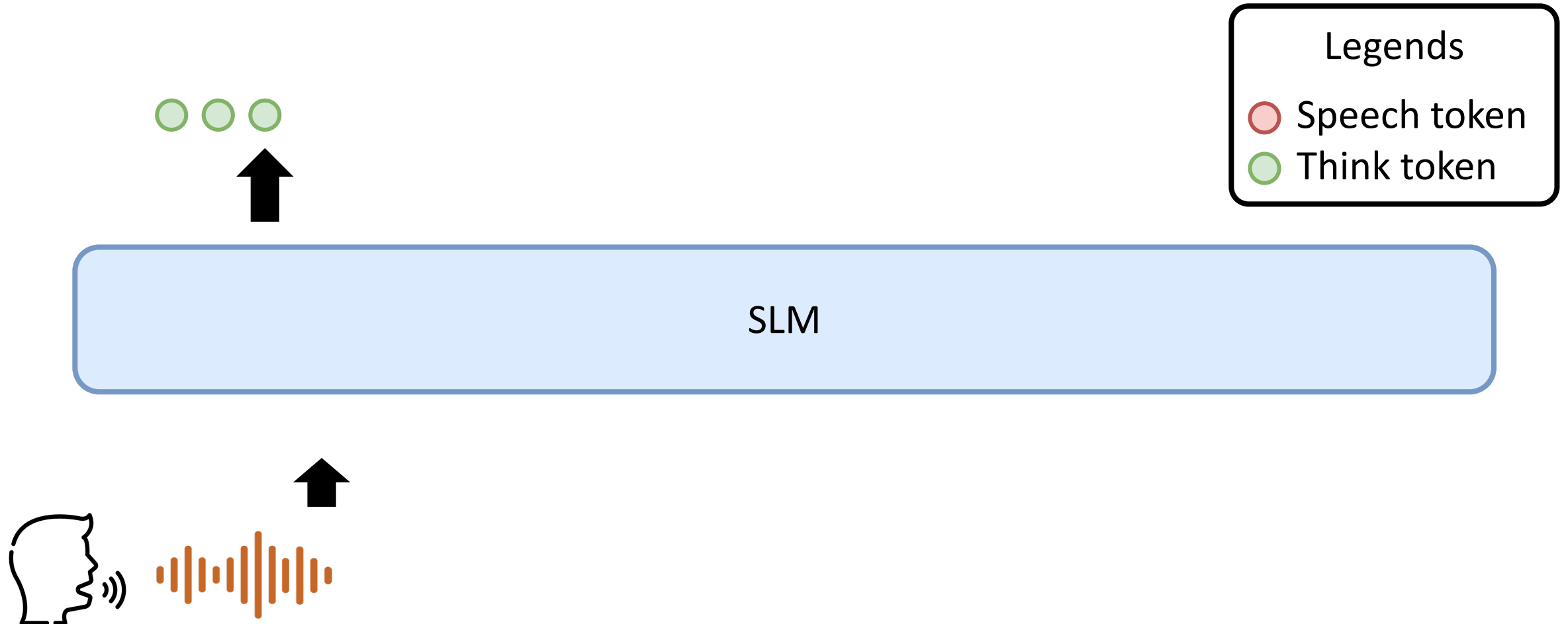
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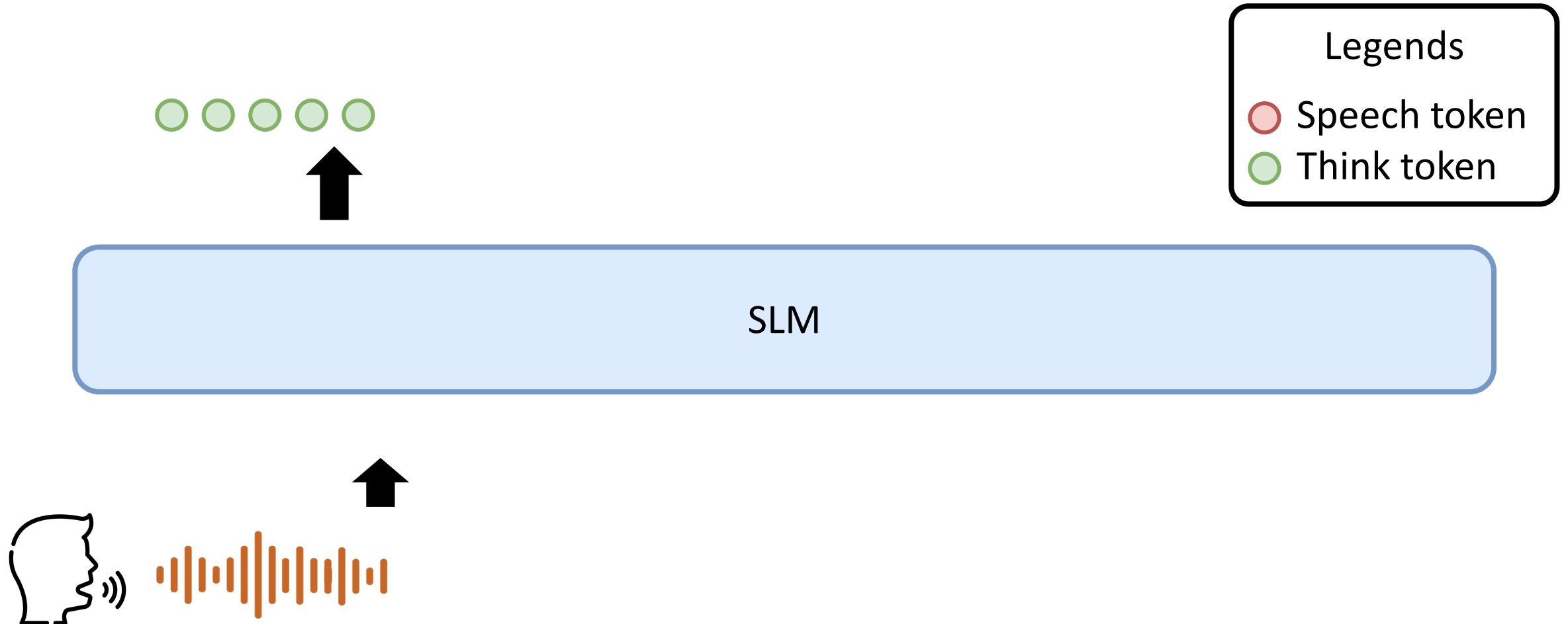
Goal: Thinking while Listening

- We want to make the model think while listening



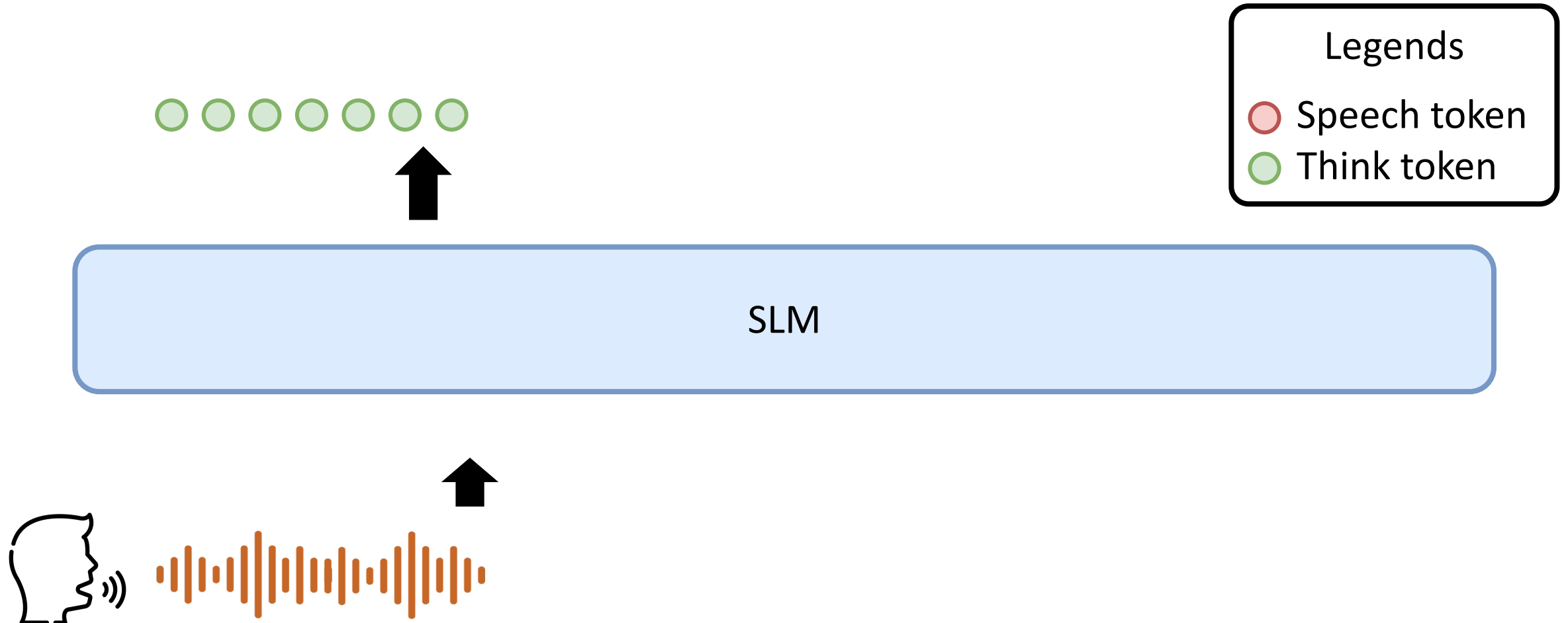
Goal: Thinking while Listening

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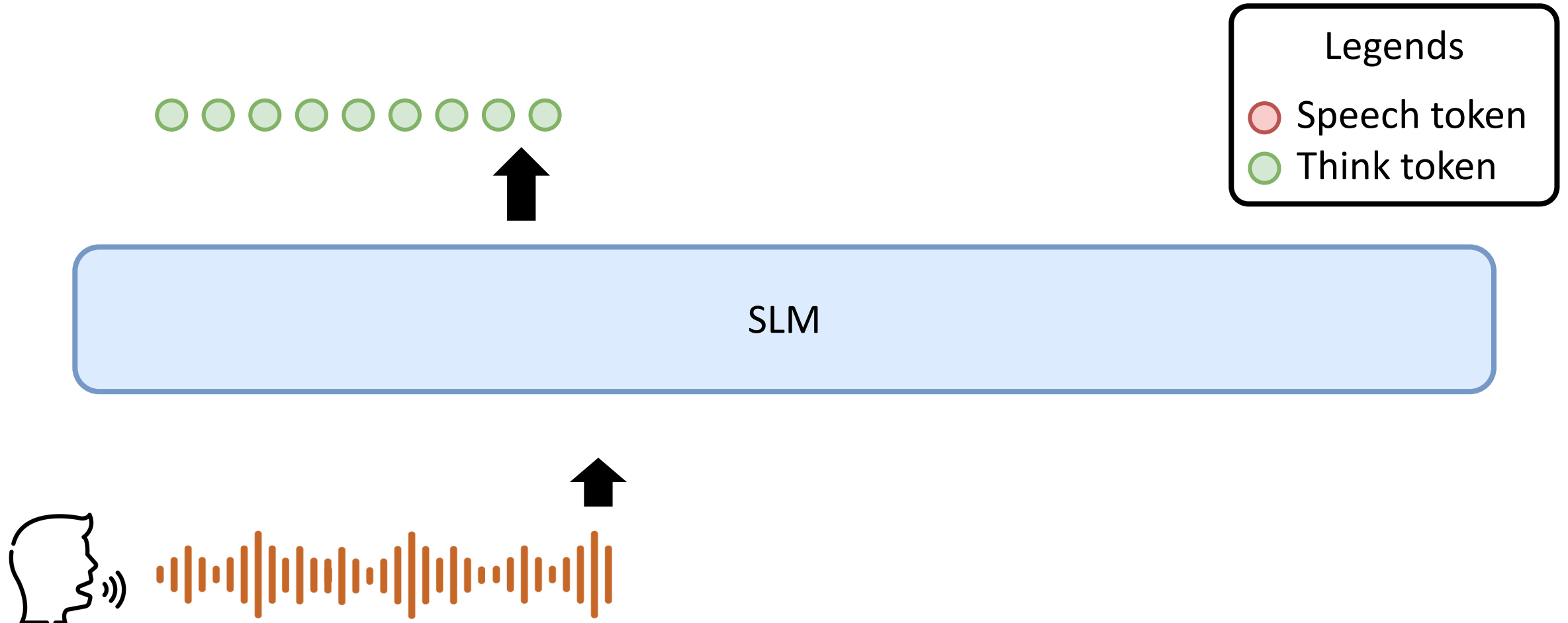
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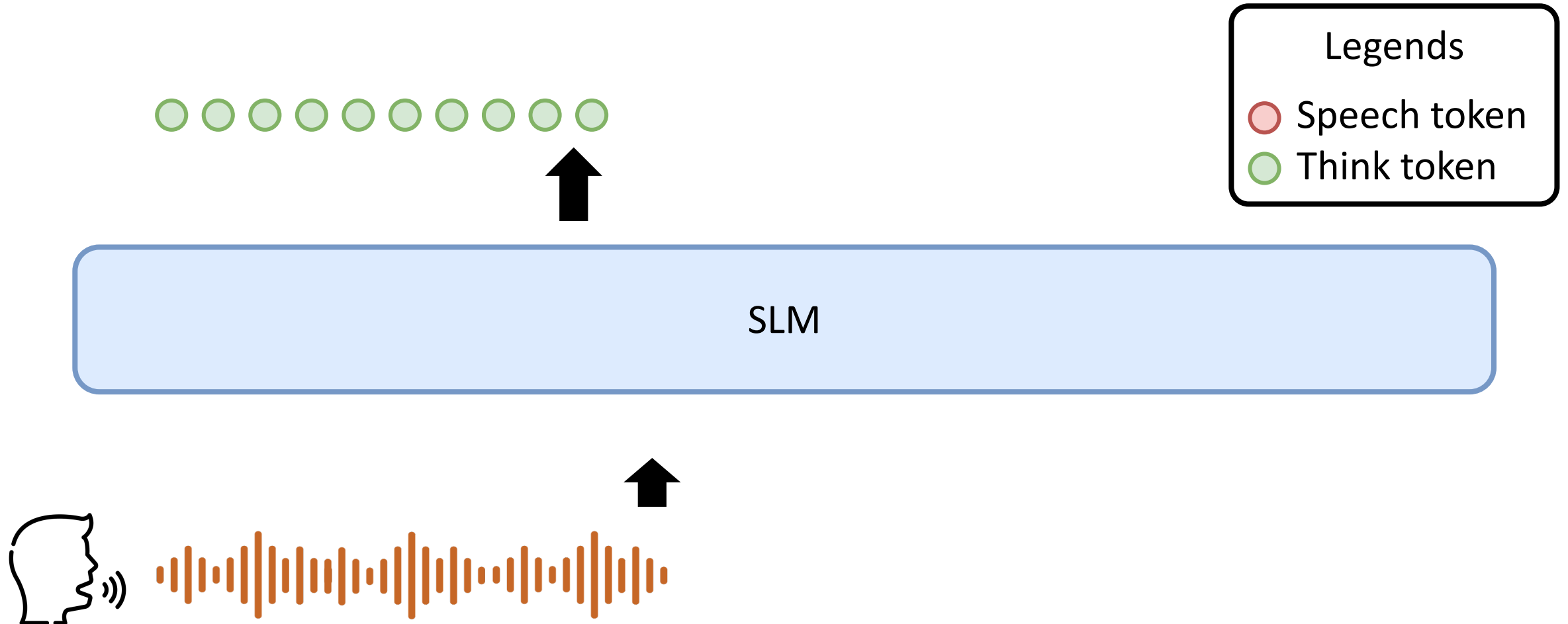
Goal: Thinking while Listening

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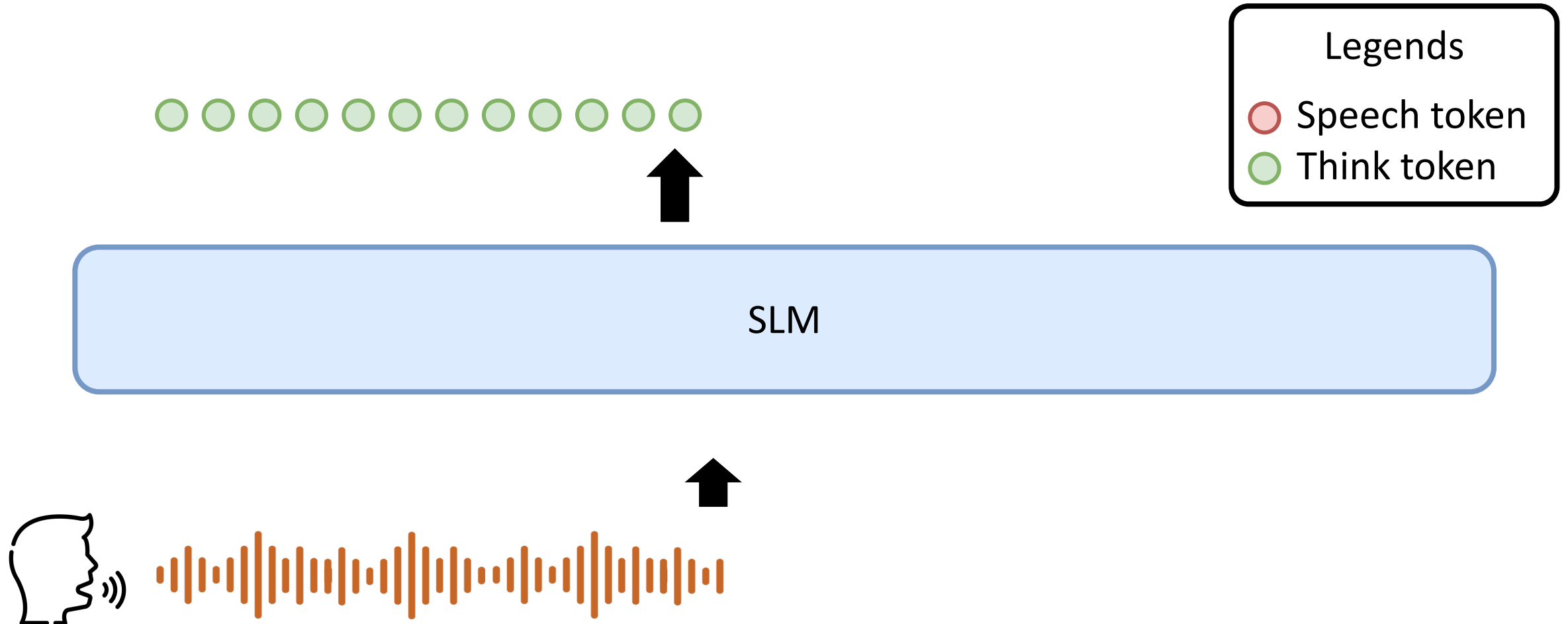
Goal: Thinking while Listening

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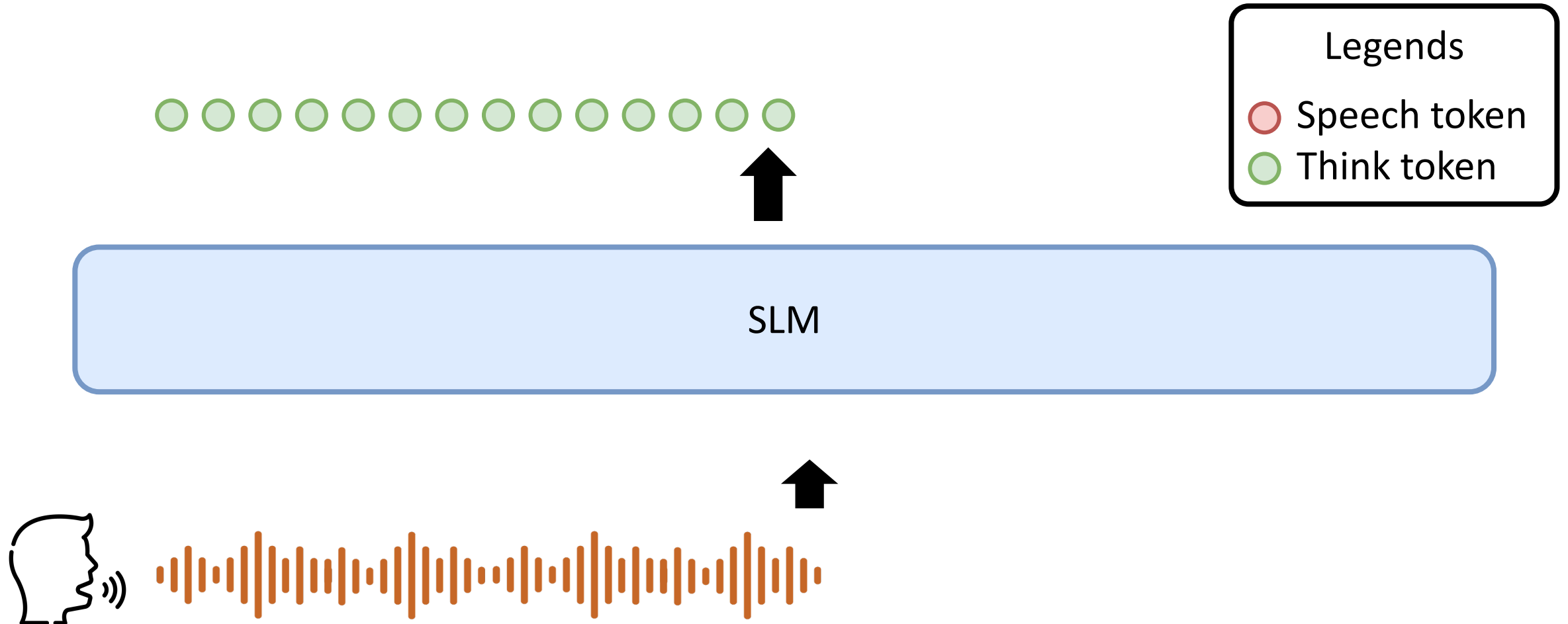
Goal: Thinking while Listening

- We want to make the model think while listening



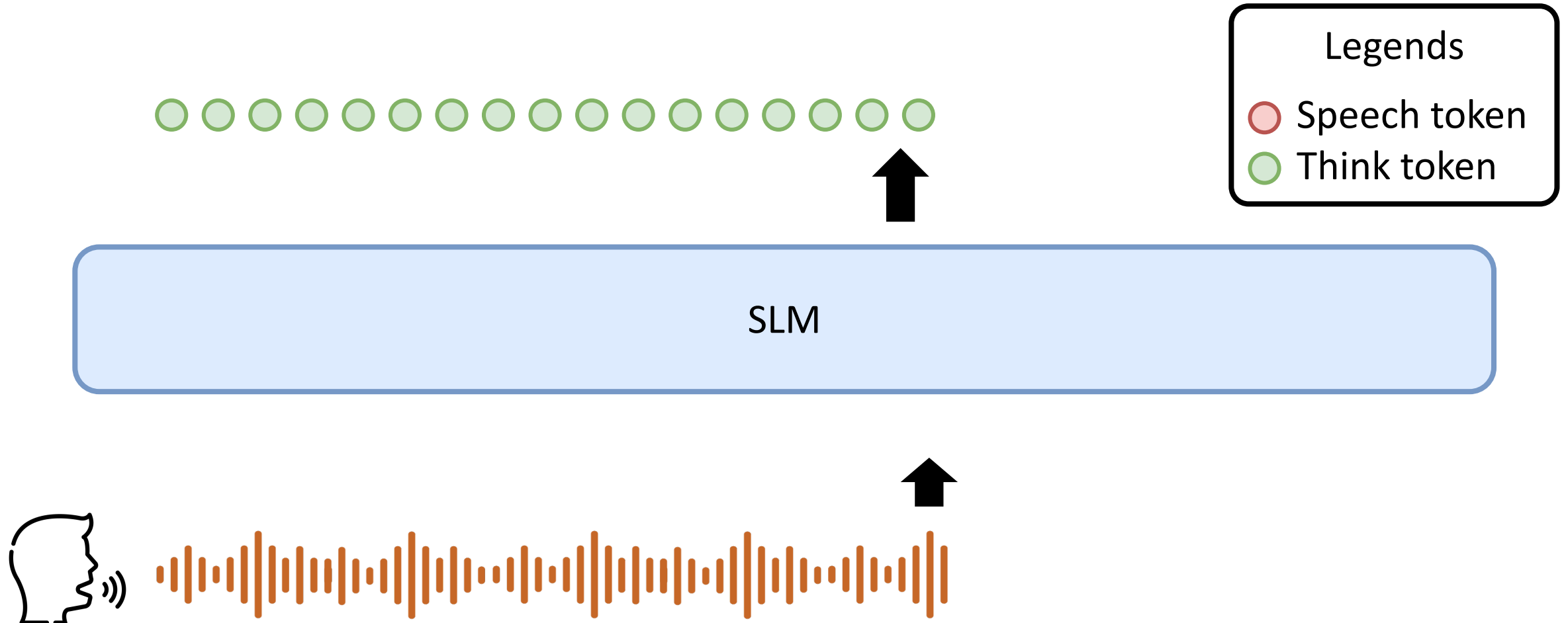
Goal: Thinking while Listening

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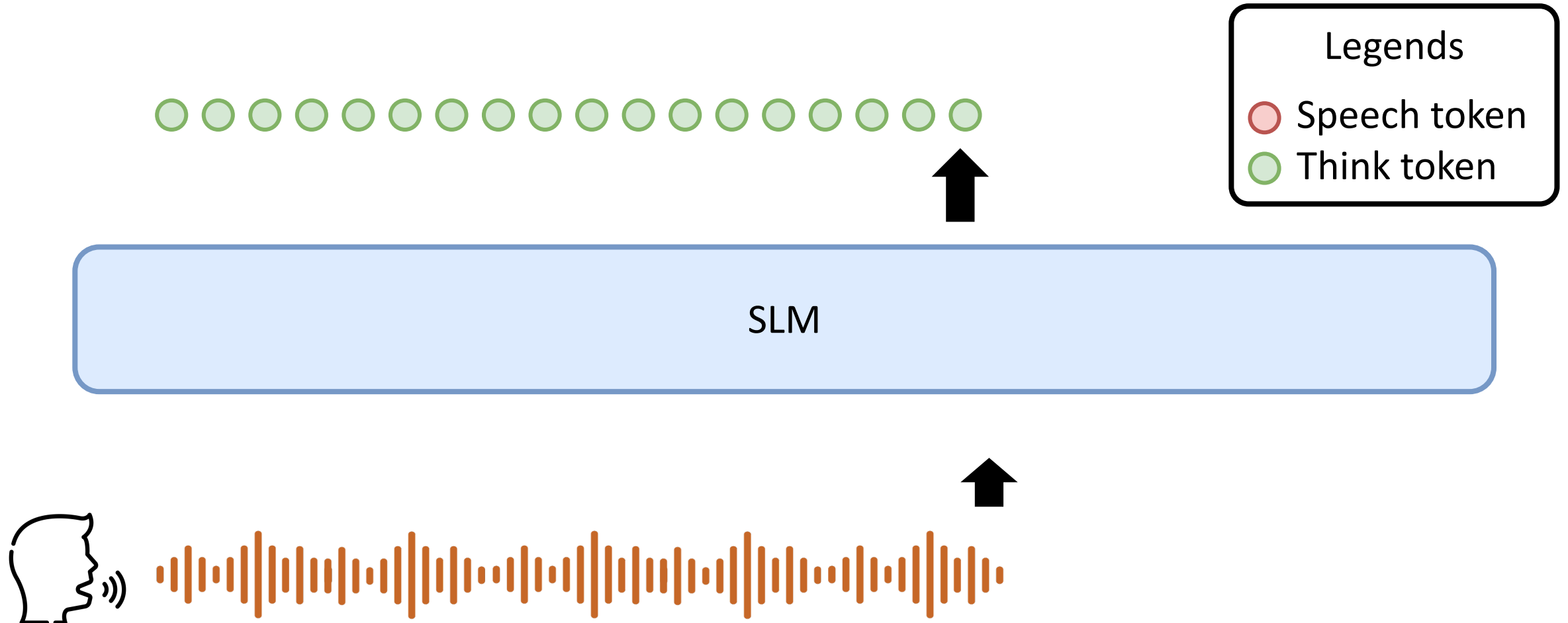
Goal: Thinking while Listening

- We want to make the model think while listening



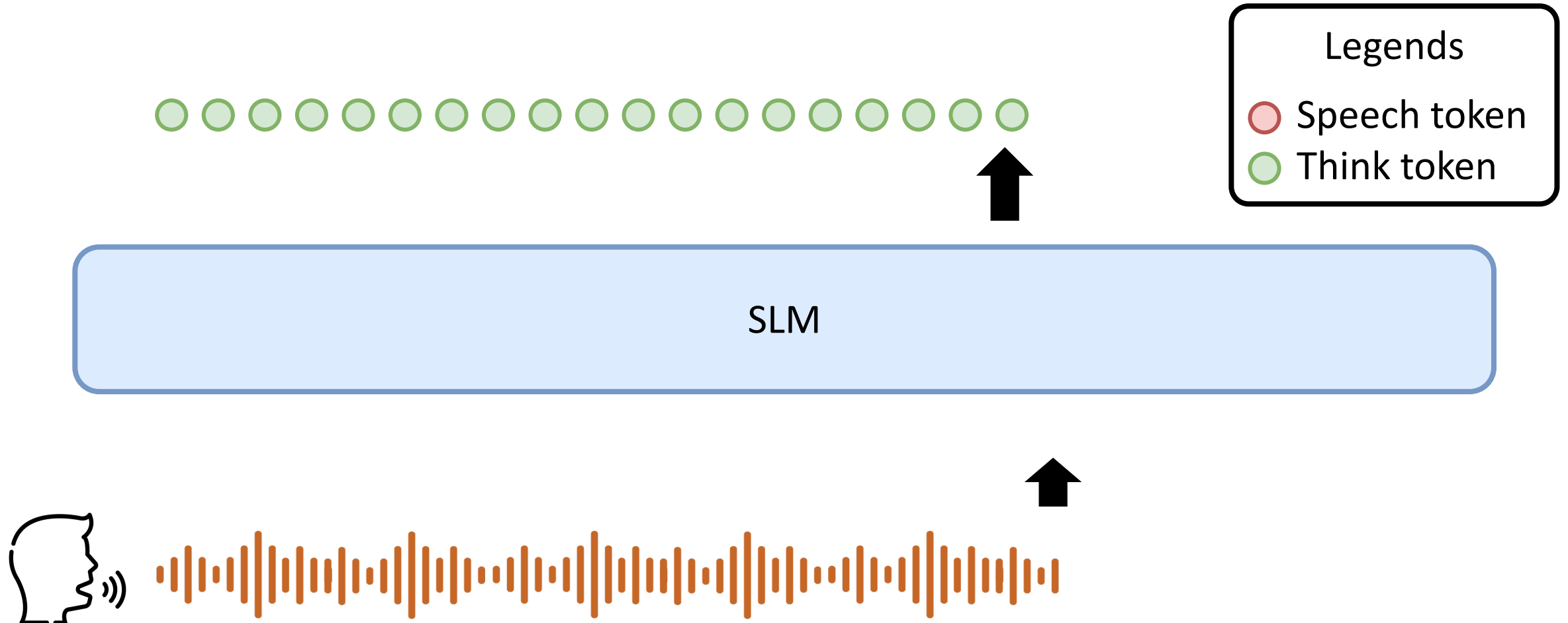
Goal: Thinking while Listening

- We want to make the model think while listening



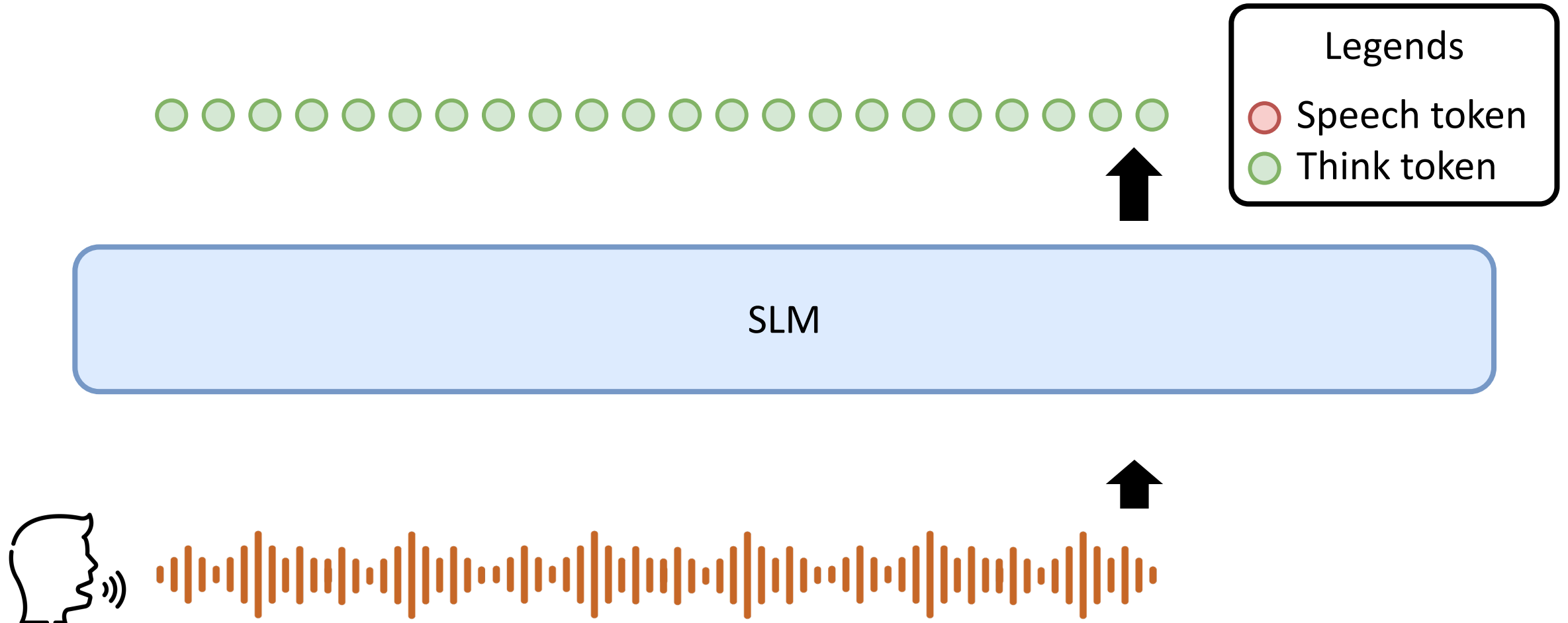
Goal: Thinking while Listening

- We want to make the model think while listening



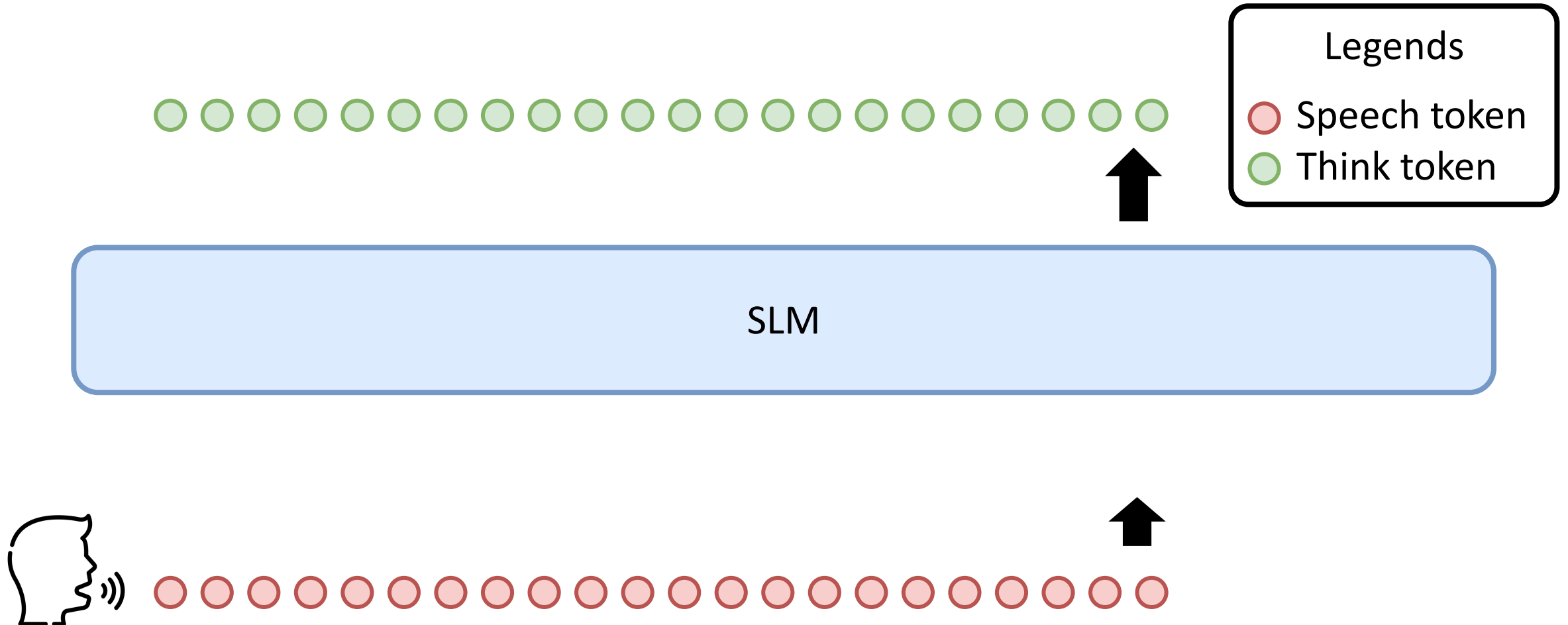
Goal: Thinking while Listening

- We want to make the model think while listening



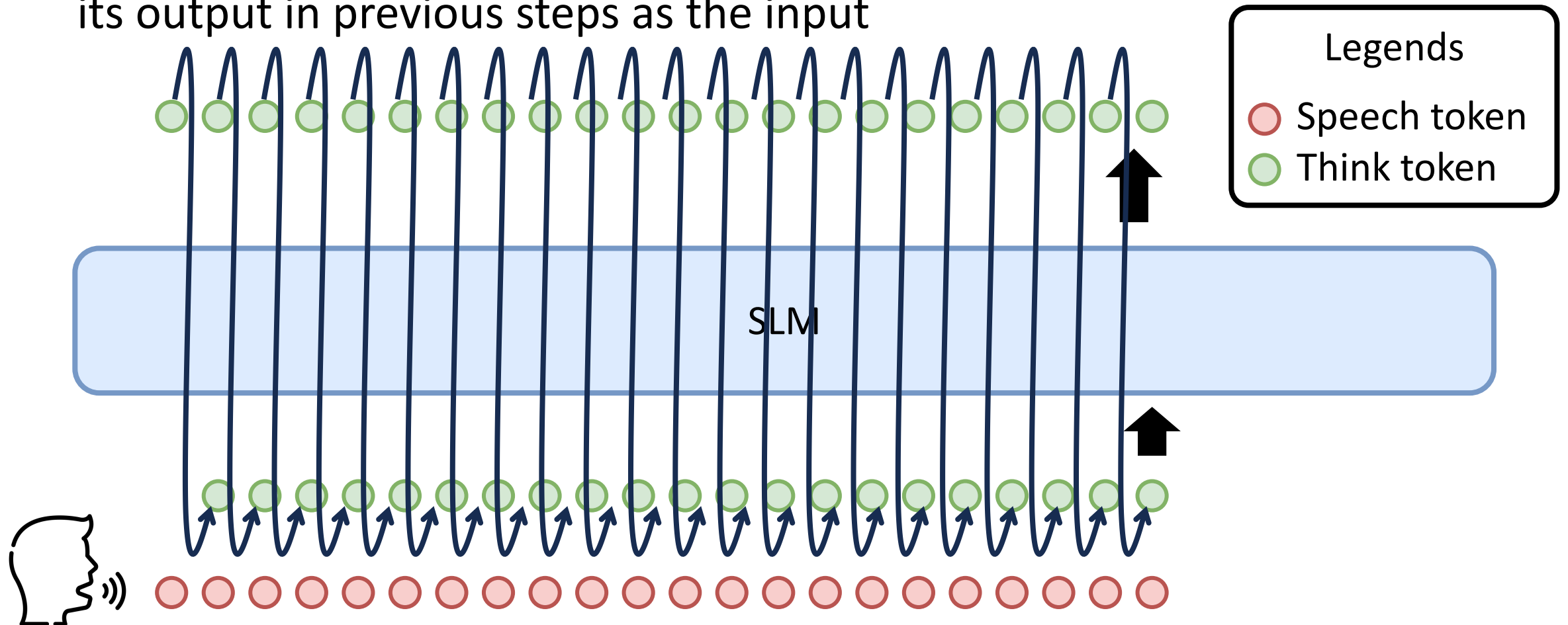
Goal: Thinking while Listening

- The speech input to the model is the speech tokens



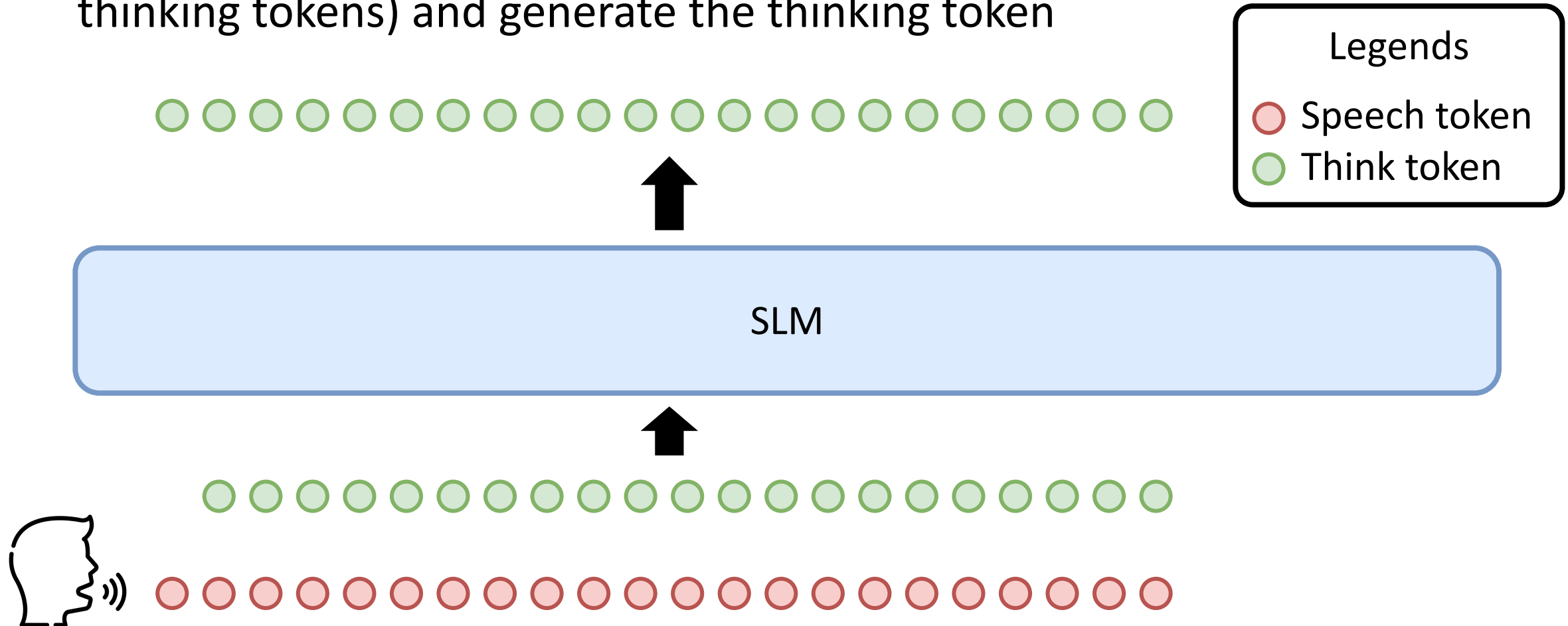
Goal: Thinking while Listening

- The model is an auto-regressive model, meaning that it should take its output in previous steps as the input



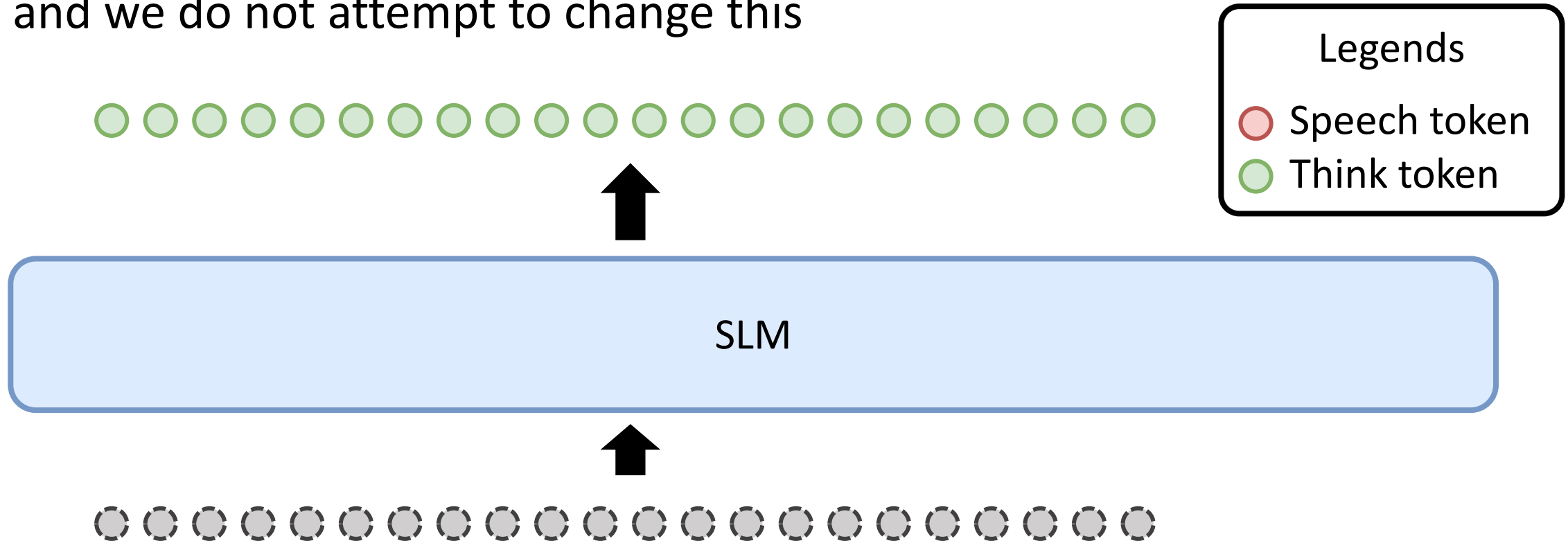
Goal: Thinking while Listening

- We want the model to take in two streams of tokens (speech and thinking tokens) and generate the thinking token



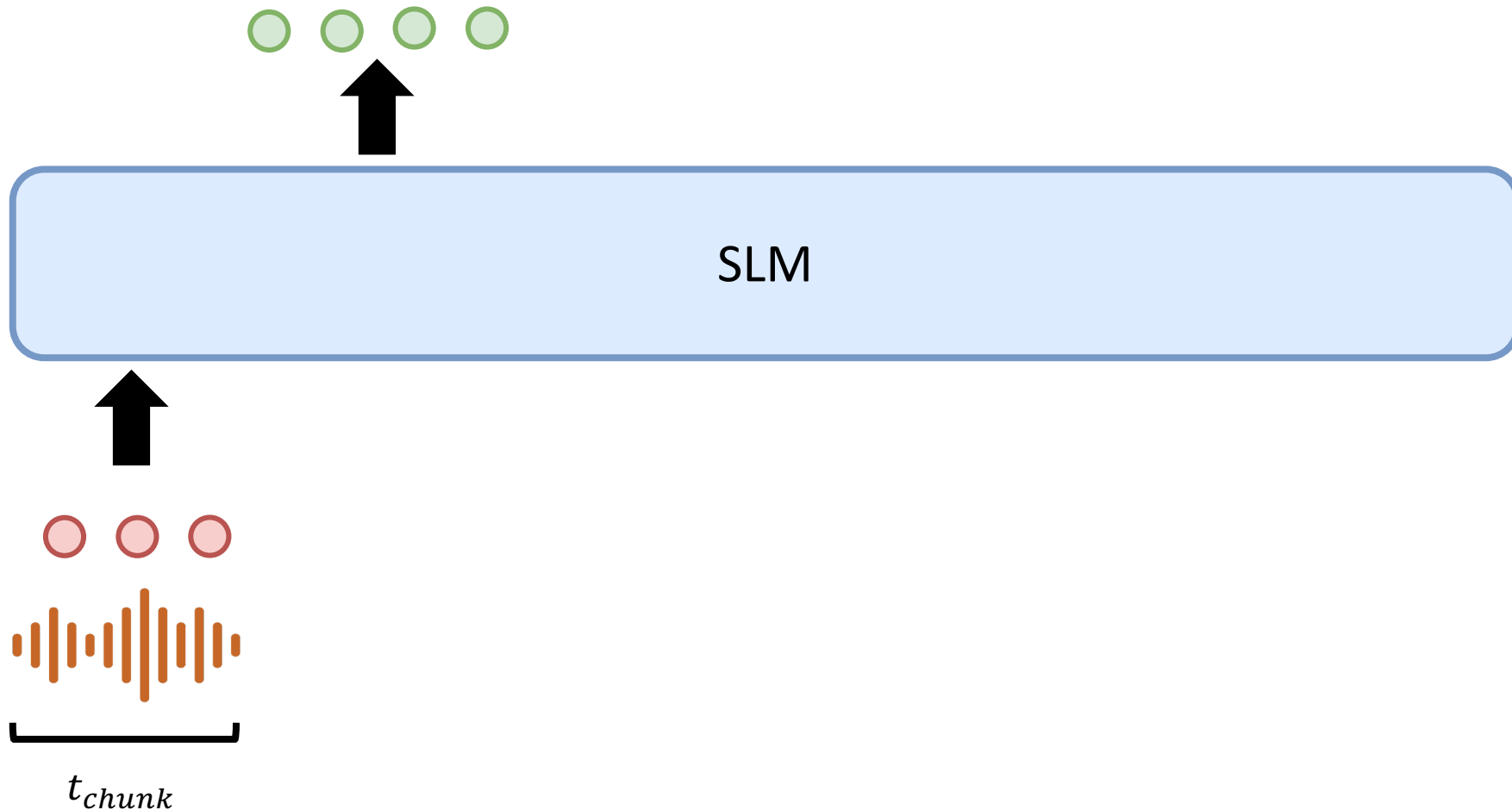
Goal: Thinking while Listening

- Again, the model can only take a single stream of tokens at a time, and we do not attempt to change this



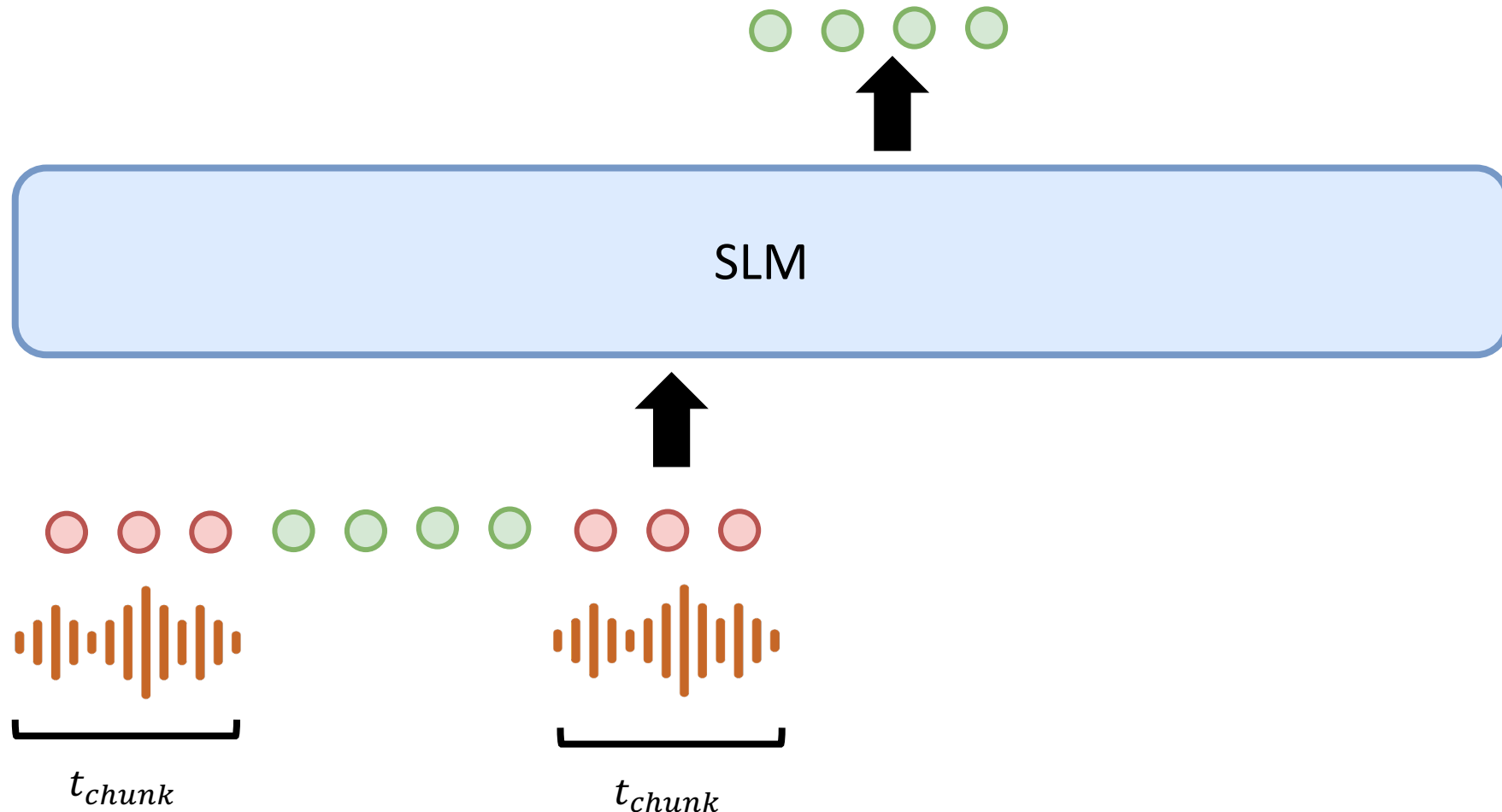
SHANKS: Method

- Chunk the user input speech into fixed duration t_{chunk} segments, and generate unspoken internal thinking based on each input chunk



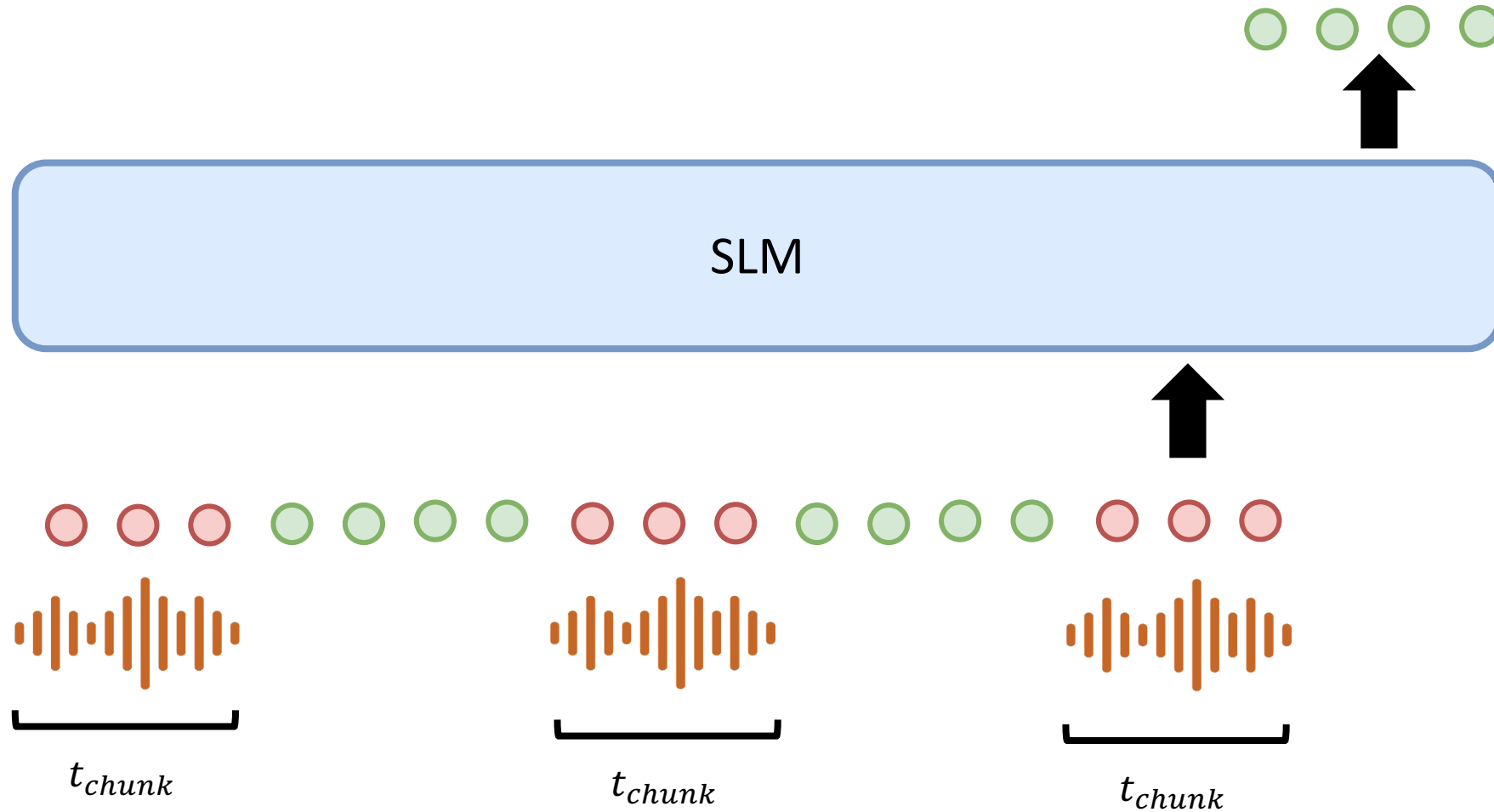
SHANKS: Method

- Given a new partial input speech and the previous thinking, generate a new partial thinking



SHANKS: Method

- Given a new partial input speech and the previous thinking, generate a new partial thinking



SHANKS: How It Works at Inference Time

- Core idea: Chunk the user input speech into fixed duration segments, and generate unspoken internal thinking based on each input chunk
 - In the following illustration, we will use $t_{chunk} = 4$

Full input
speech



Legends

- Speech token
- Think token

User starts to speak

SHANKS



$t = 0$

Legends

- Speech token
- Think token

User continue to speak

SHANKS



$t = 1$

Legends

- Speech token
- Think token

User continue to speak

SHANKS



$t = 2$

Legends

- Speech token
- Think token

User continue to speak

SHANKS



$t = 3$

Legends

- Speech token
- Think token

First chunk ($t = 0$ to $t = 4$) completed

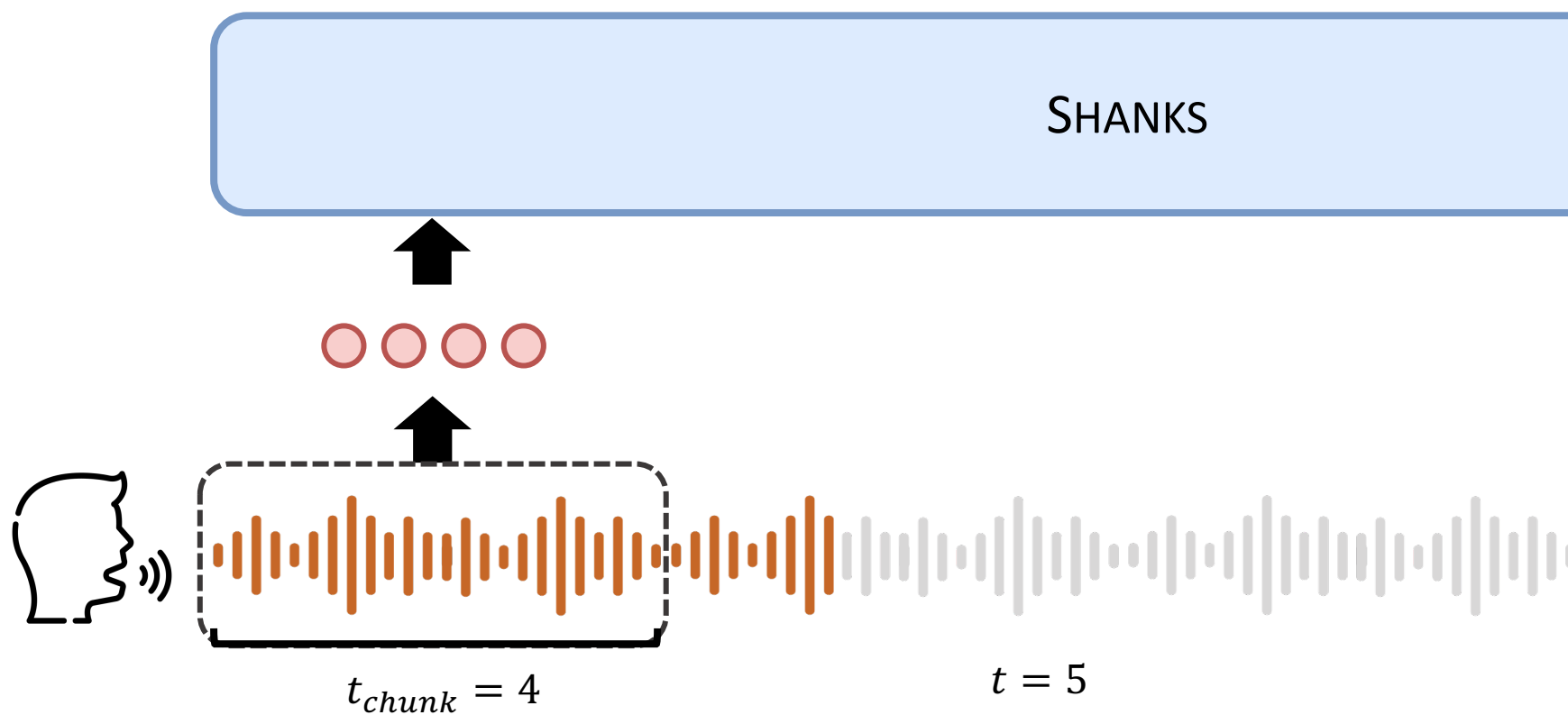
SHANKS



Legends

- Speech token
- Think token

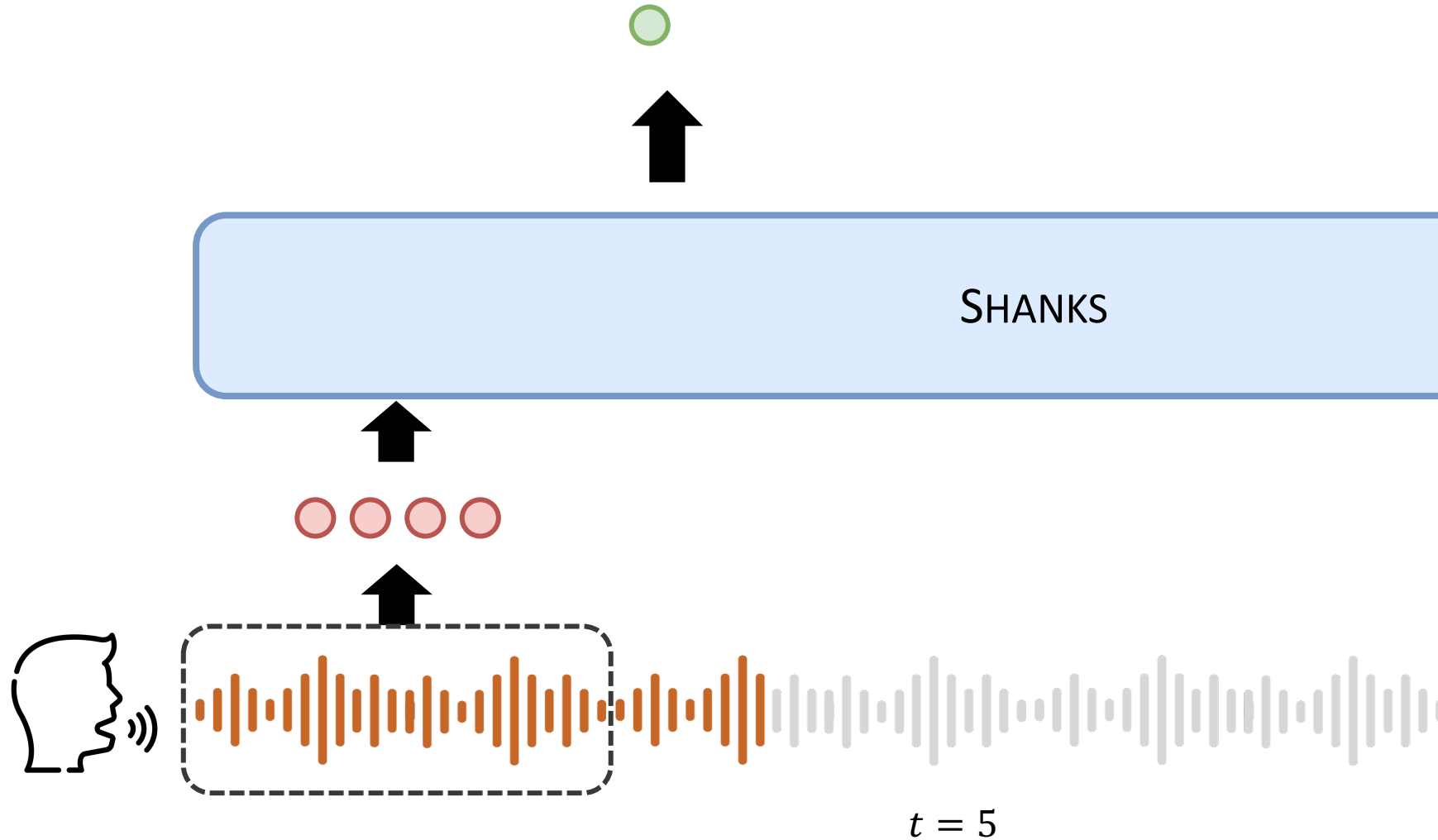
First chunk ($t = 0$ to $t = 4$) completed



Legends

- Speech token
- Think token

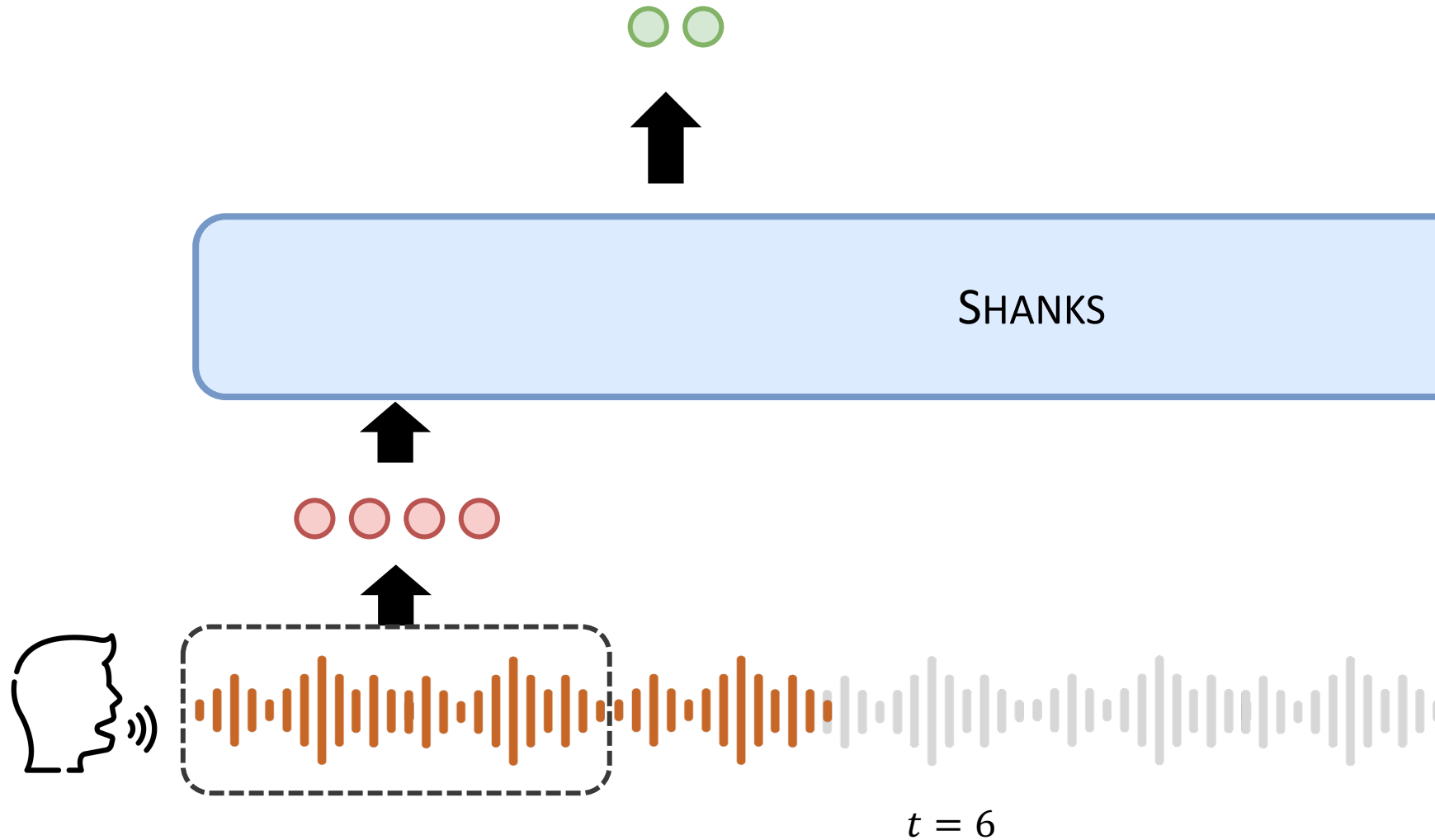
Model continue thinking based on the first chunk ($t = 0$ to $t = 4$)
The user keeps talking



Legends

- Speech token
- Think token

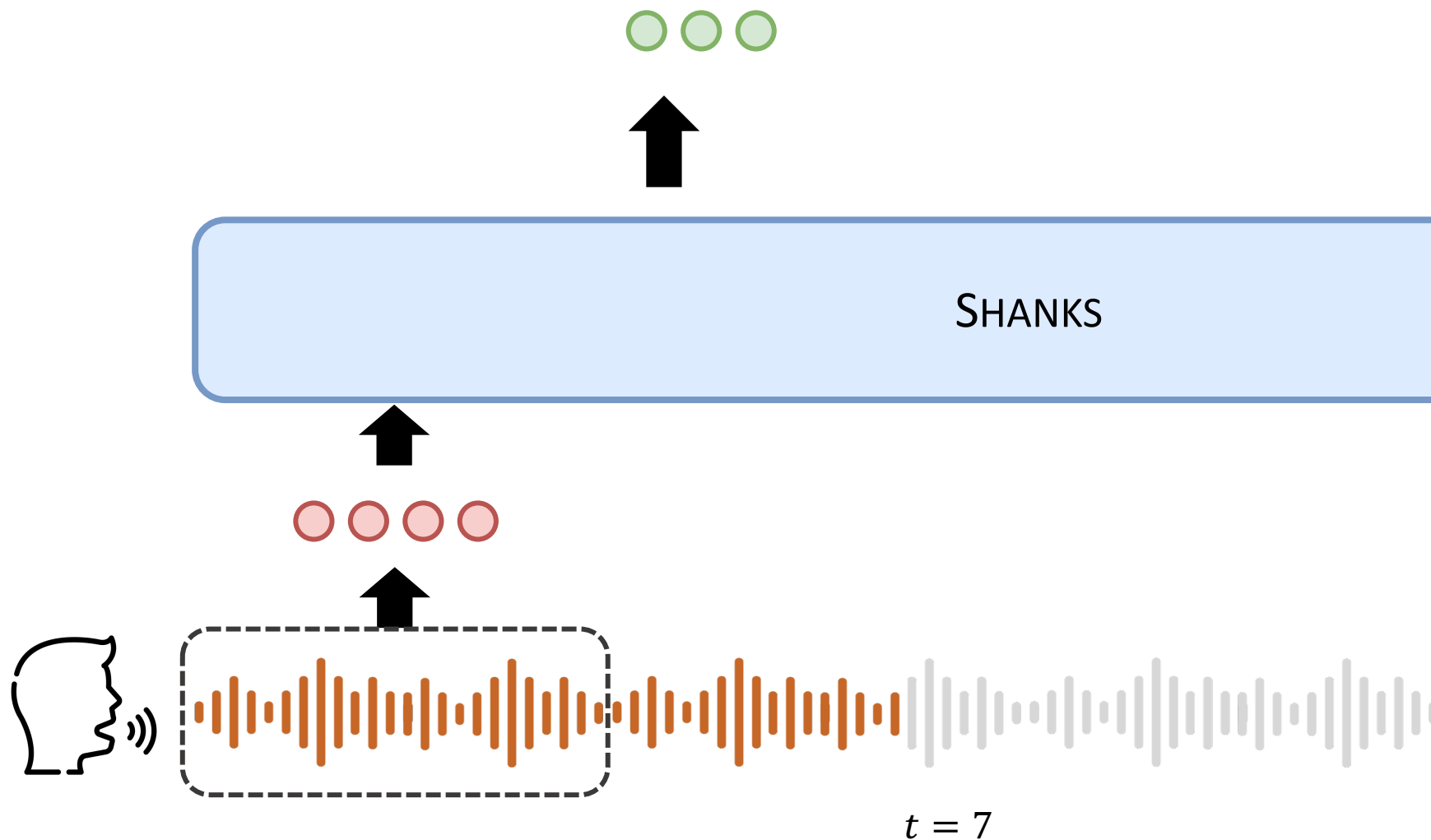
Model continue thinking based on the first chunk ($t = 0$ to $t = 4$)
The user keeps talking



Legends

- Speech token
- Think token

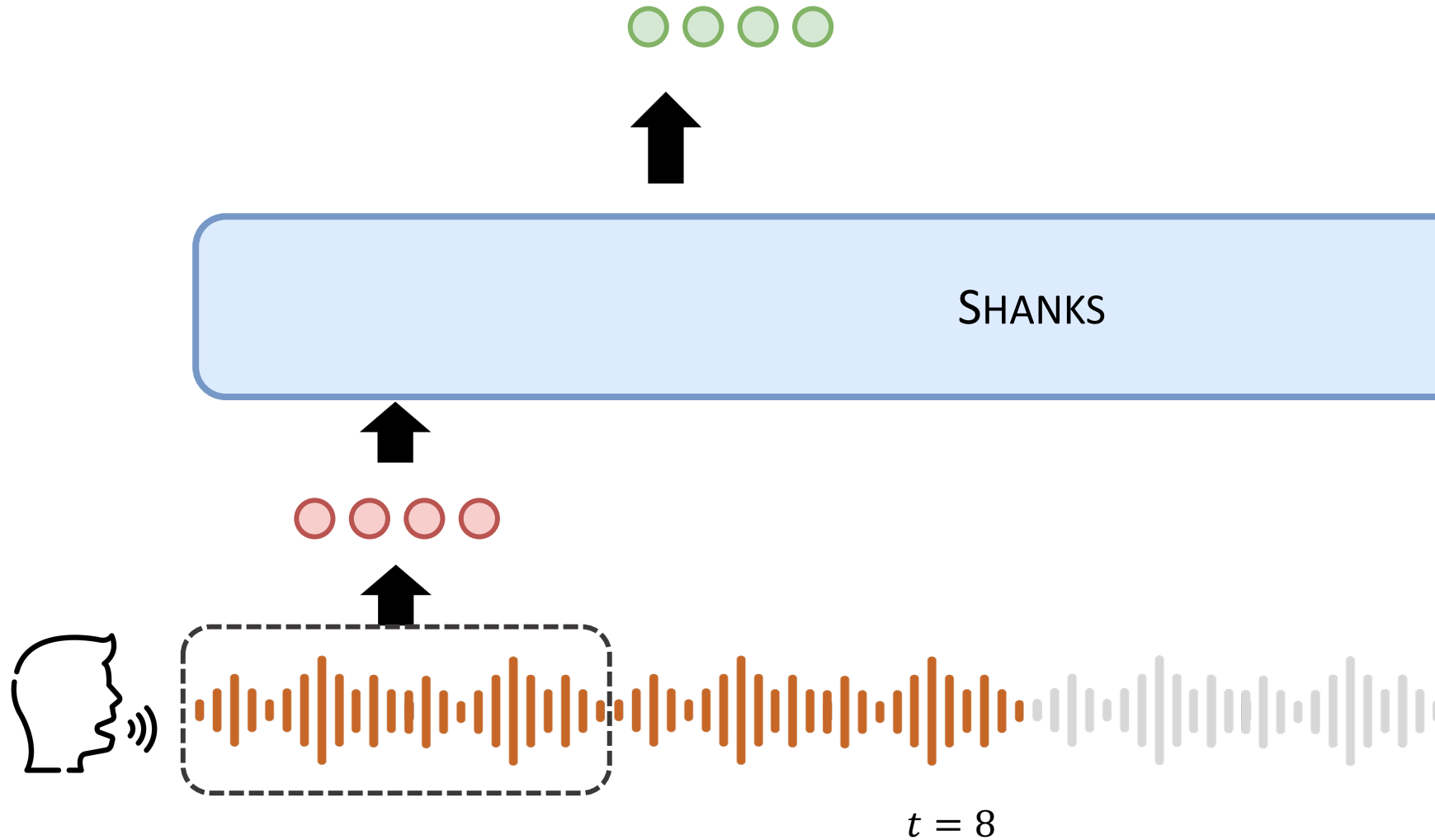
Model continue thinking based on the first chunk ($t = 0$ to $t = 4$)
The user keeps talking



Legends

- Speech token
- Think token

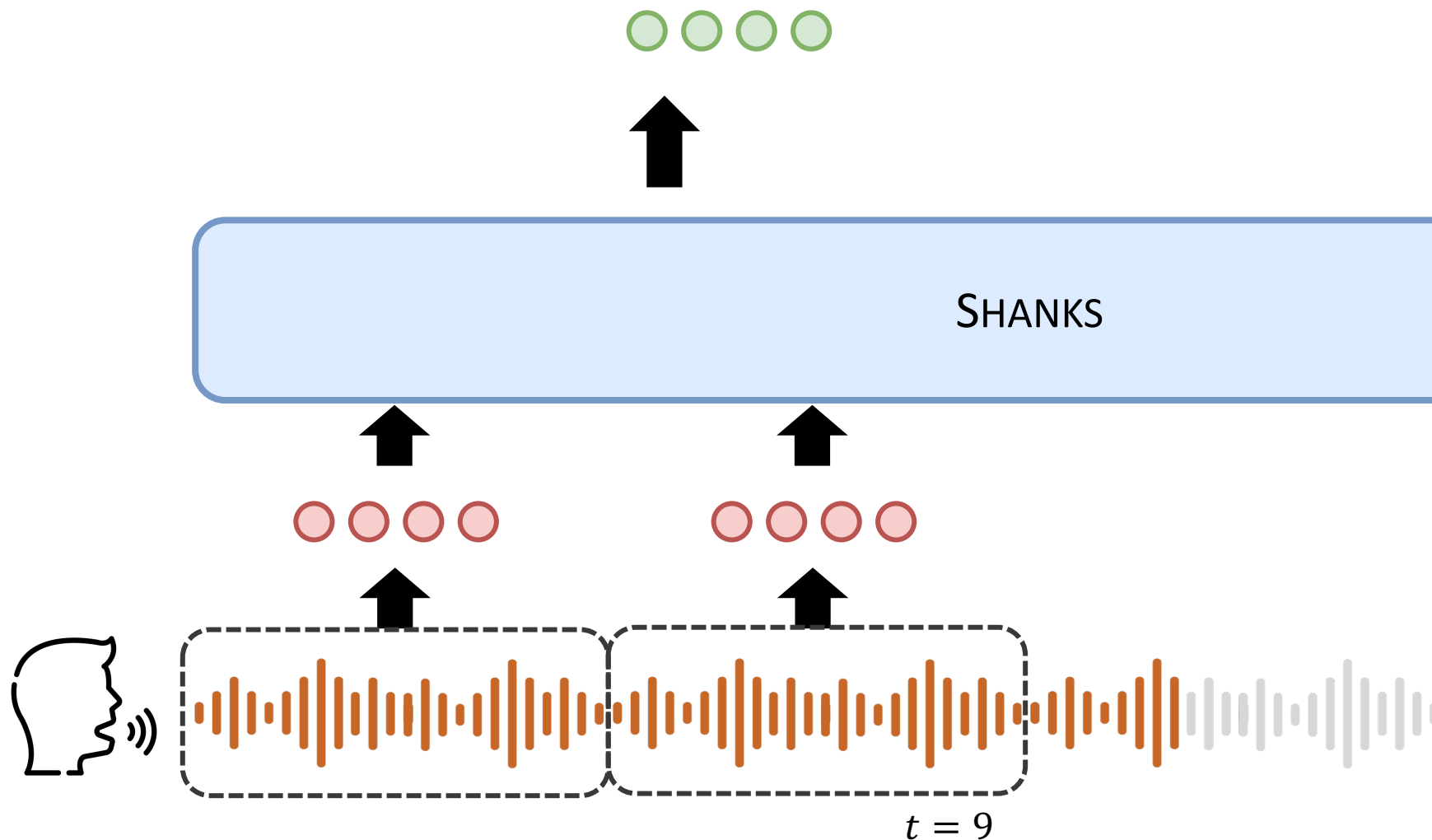
Model continue thinking based on the first chunk ($t = 0$ to $t = 4$)
The user keeps talking
Second chunk ($t = 4$ to $t = 8$) completed



Legends

- Speech token
- Think token

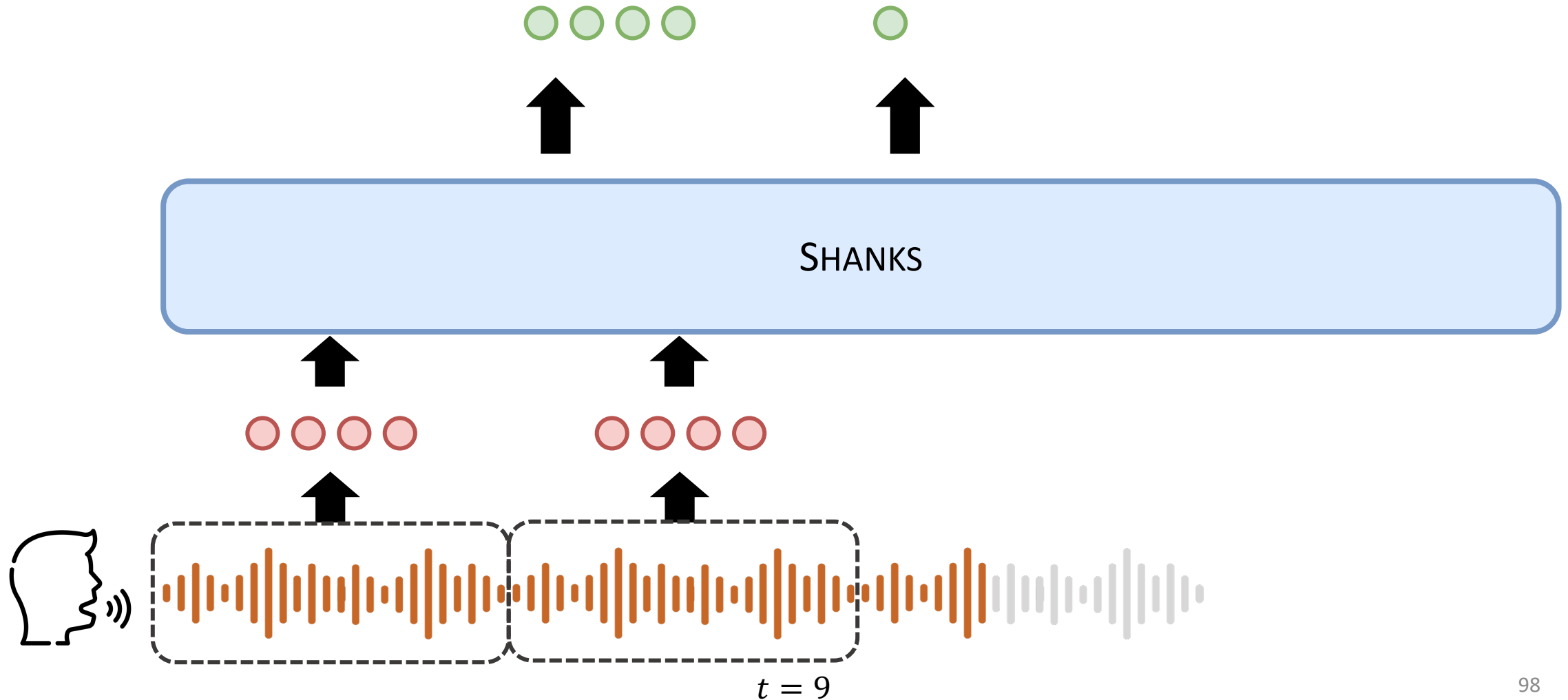
Model continue thinking based on the first chunk ($t = 0$ to $t = 4$)
The user keeps talking
Second chunk ($t = 4$ to $t = 8$) completed



Legends

- Speech token
- Think token

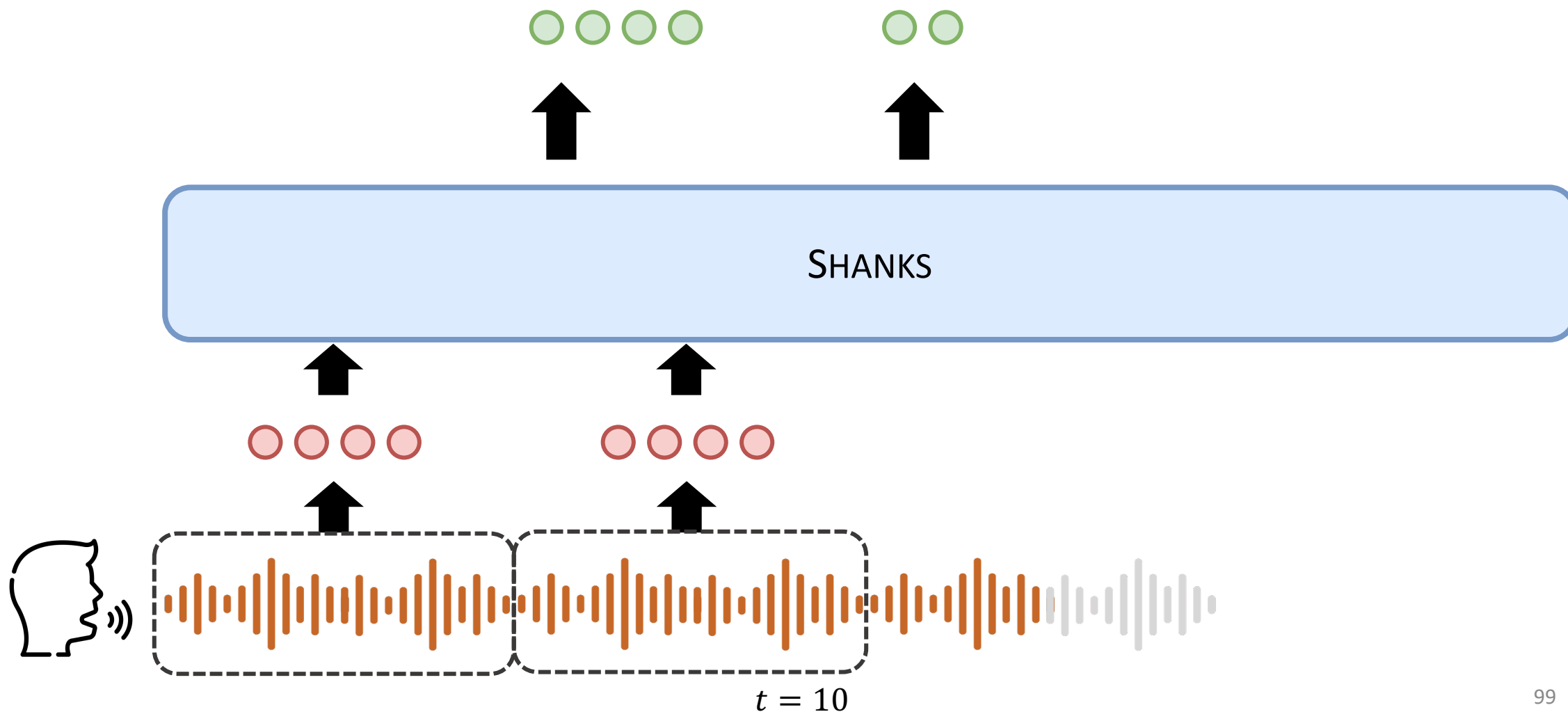
Model continue thinking based on previous chunks ($t = 0$ to $t = 8$) and previous thinking (**not shown**). The user keeps talking



Legends

- Speech token
- Think token

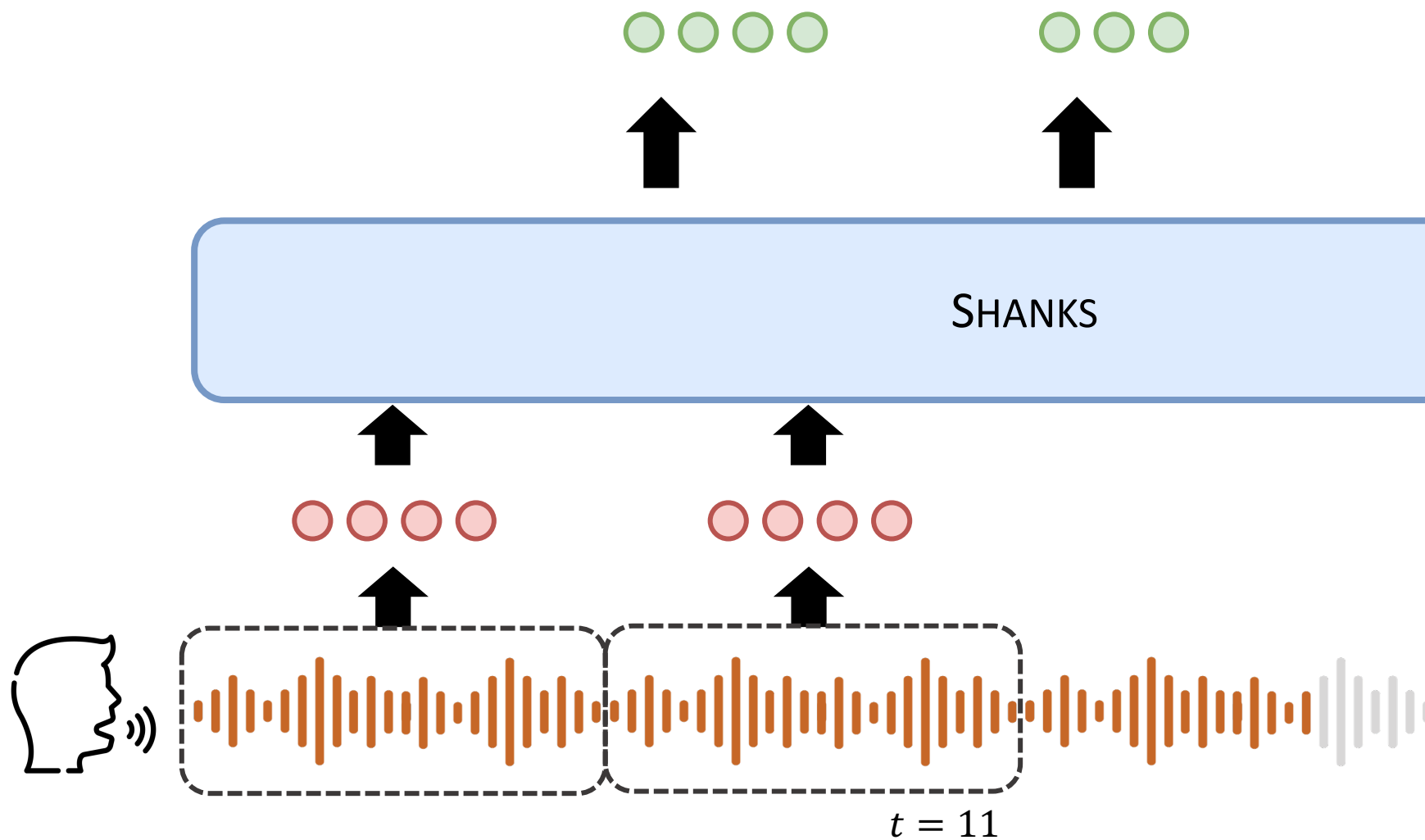
Model continue thinking based on previous chunks ($t = 0$ to $t = 8$) and previous thinking (**not shown**). The user keeps talking



Legends

- Speech token
- Think token

Model continue thinking
The user keeps talking

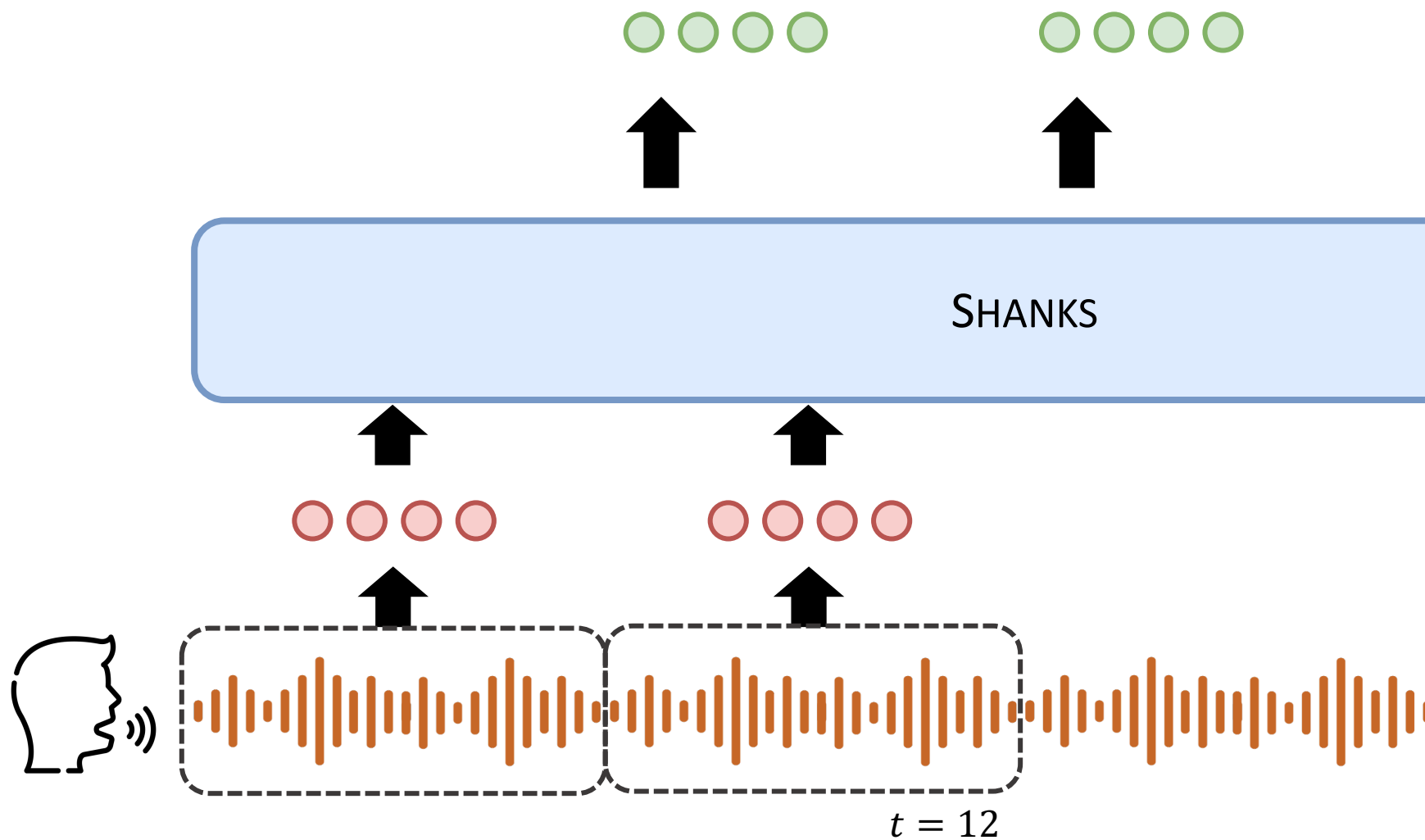


Legends

Speech token

Think token

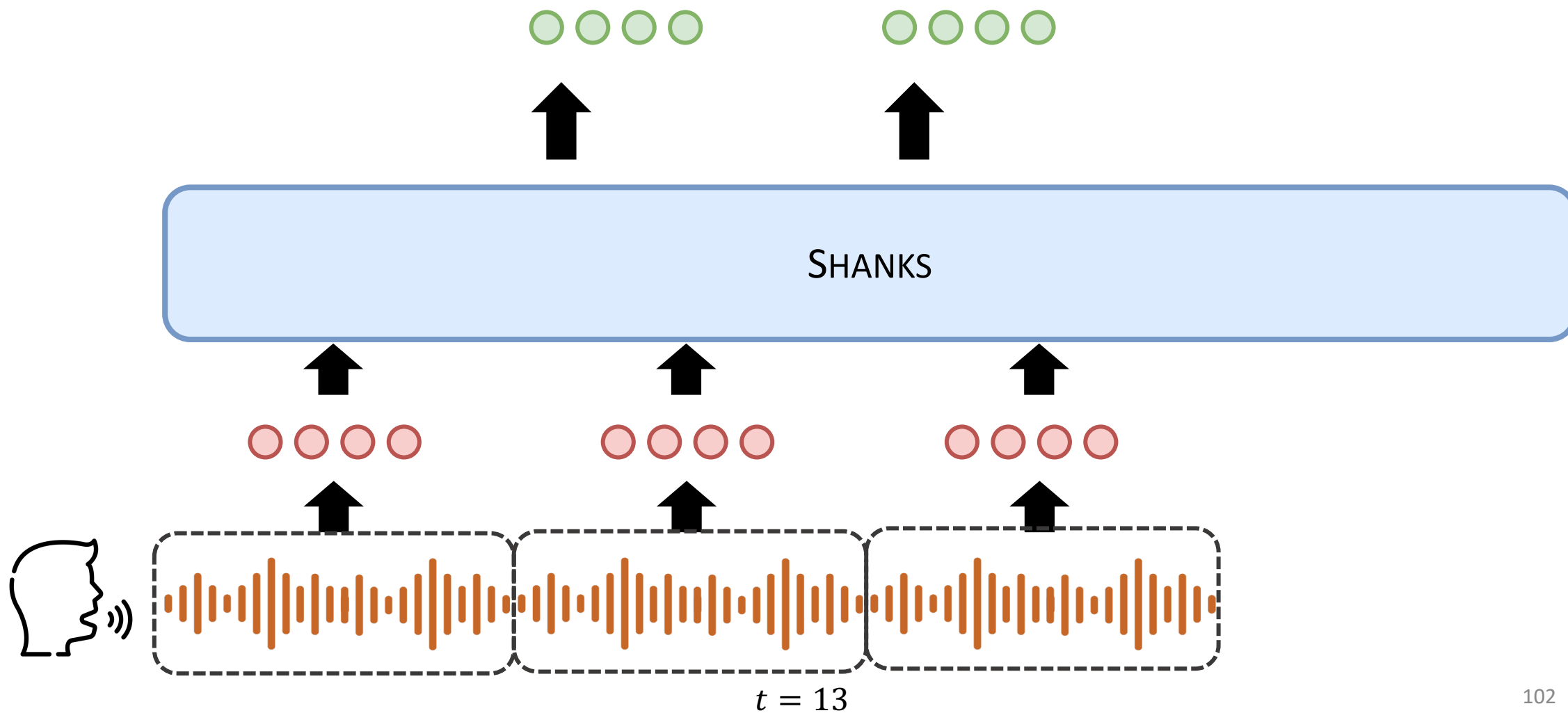
Model continue thinking
The user keeps talking



Legends

- Speech token
- Think token

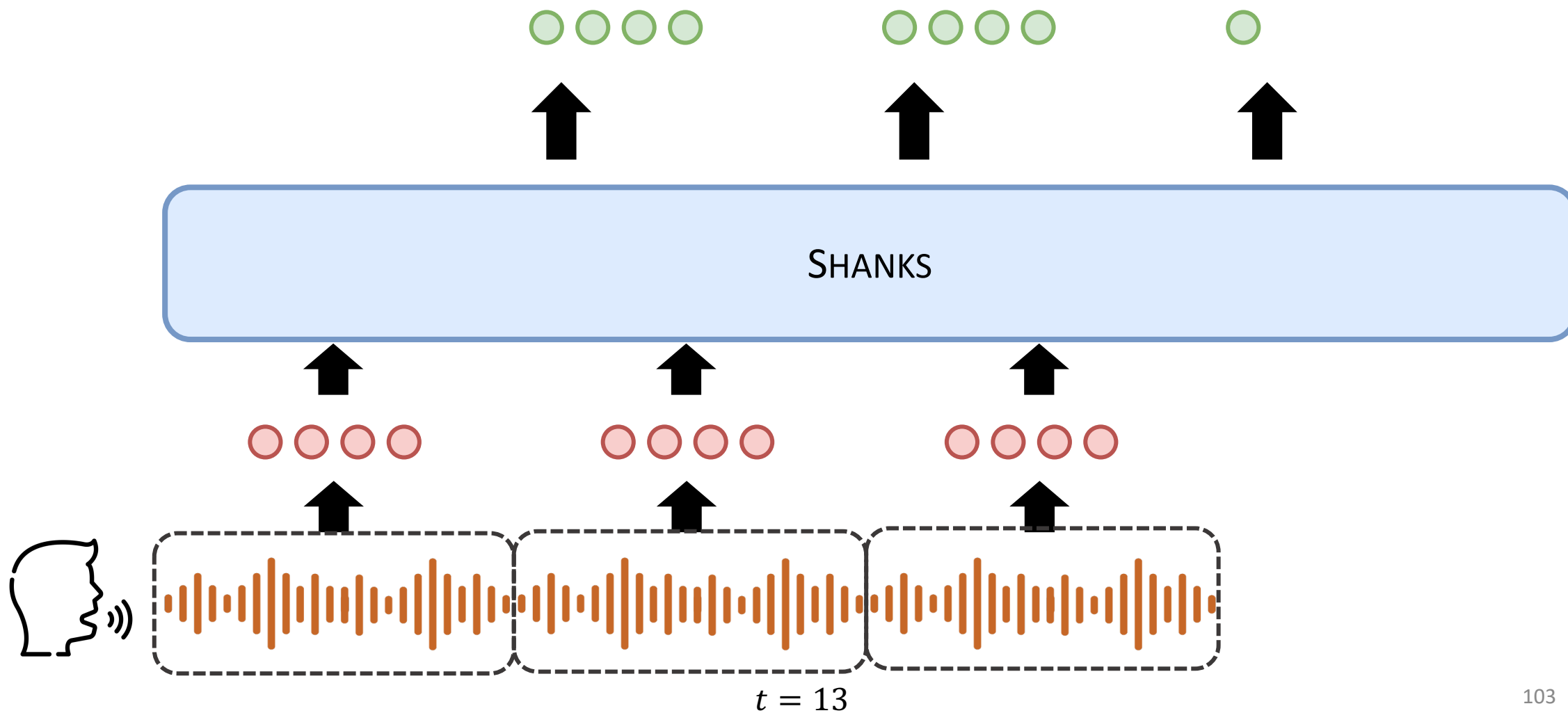
Third chunk ($t = 8$ to $t = 12$) completed. User turn finishes



Legends

- Speech token
- Think token

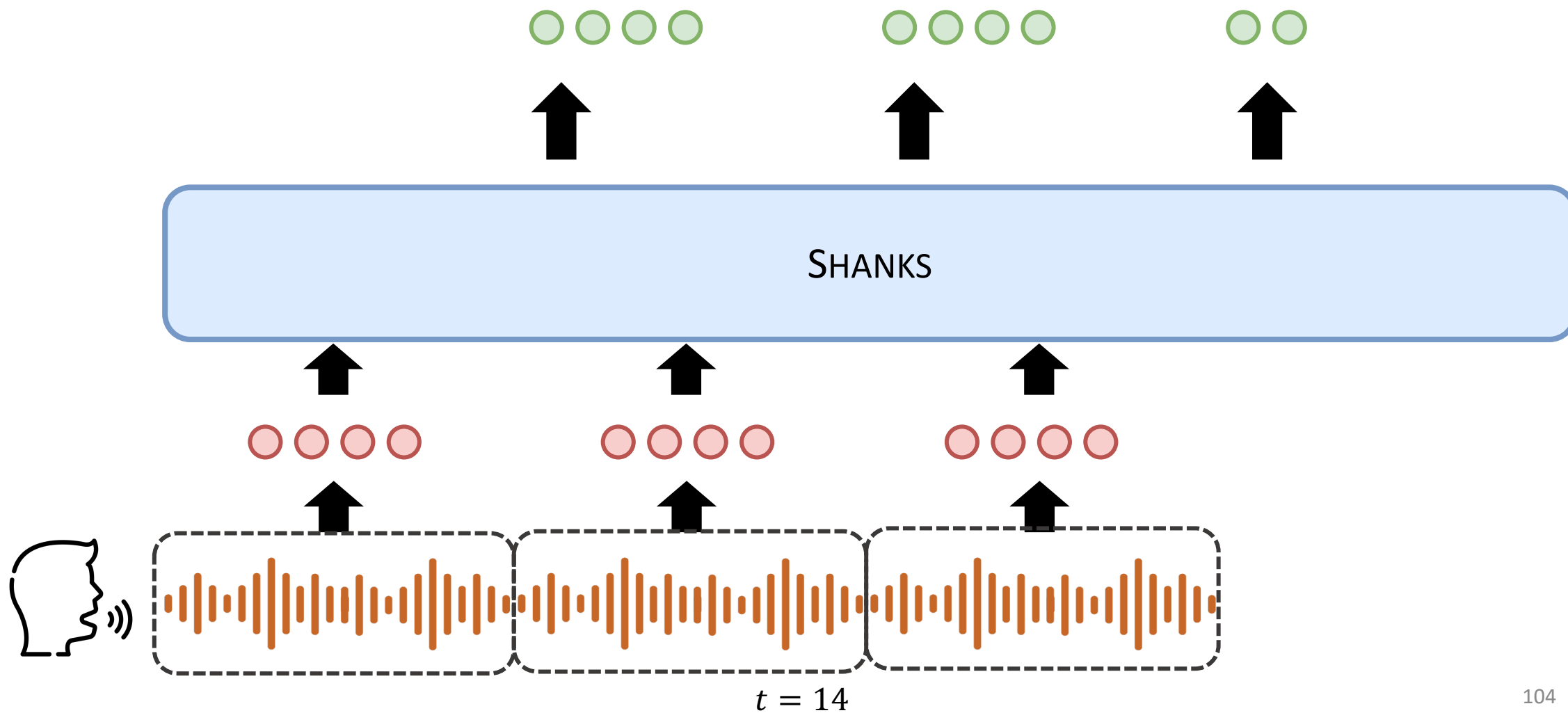
Model keeps thinking



Legends

- Speech token
- Think token

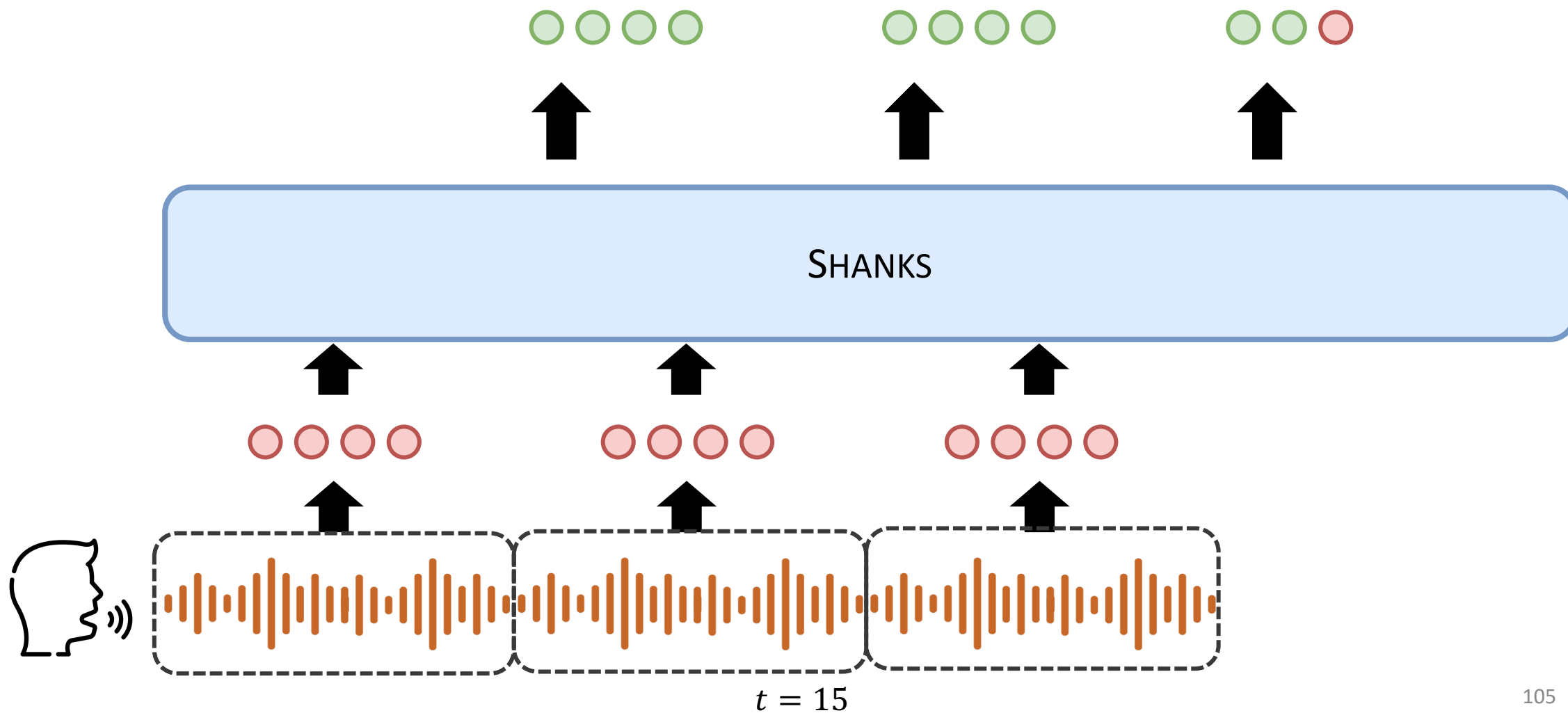
Model keeps thinking



Legends

- Speech token
- Think token

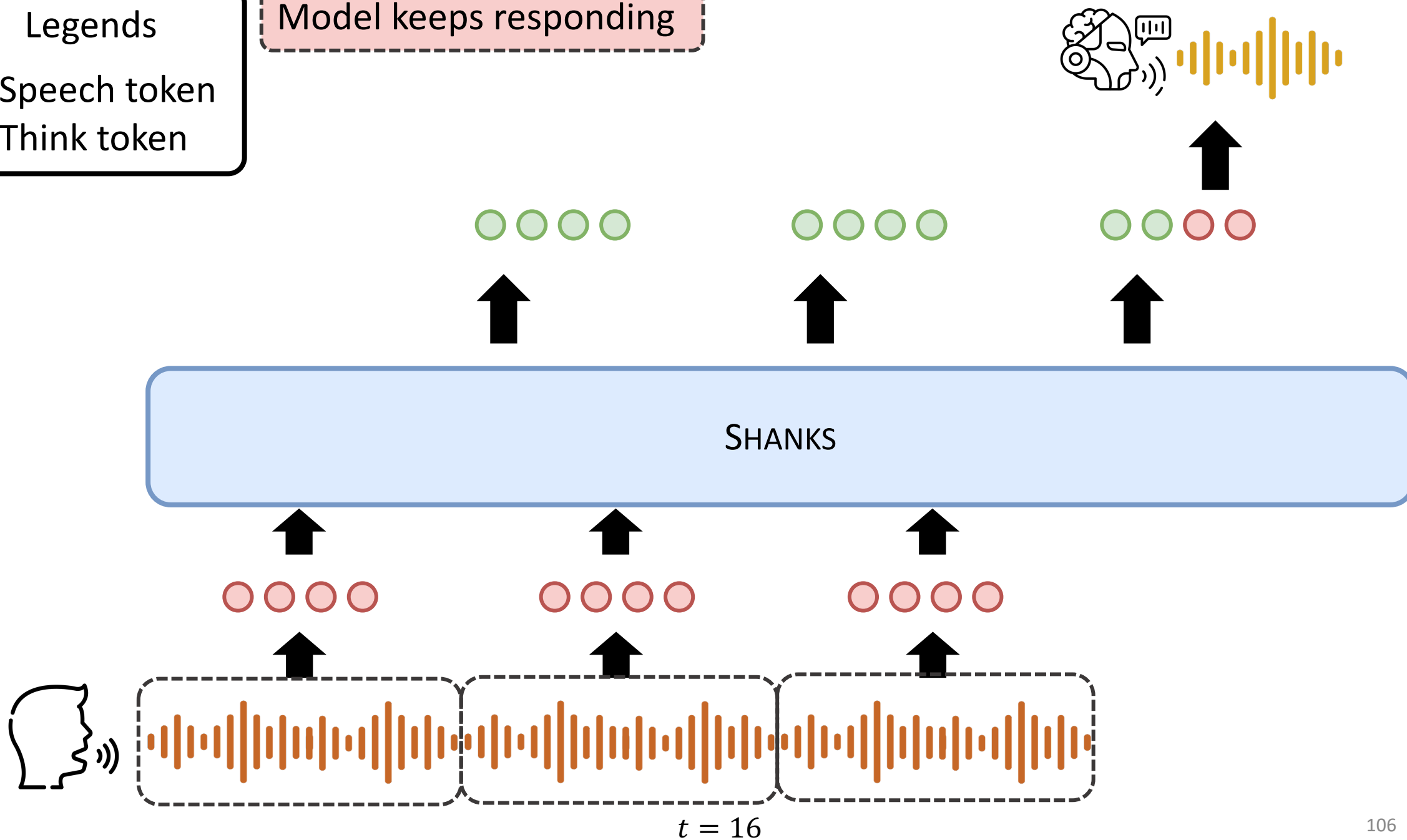
Model finishes thinking and starts responding



Legends

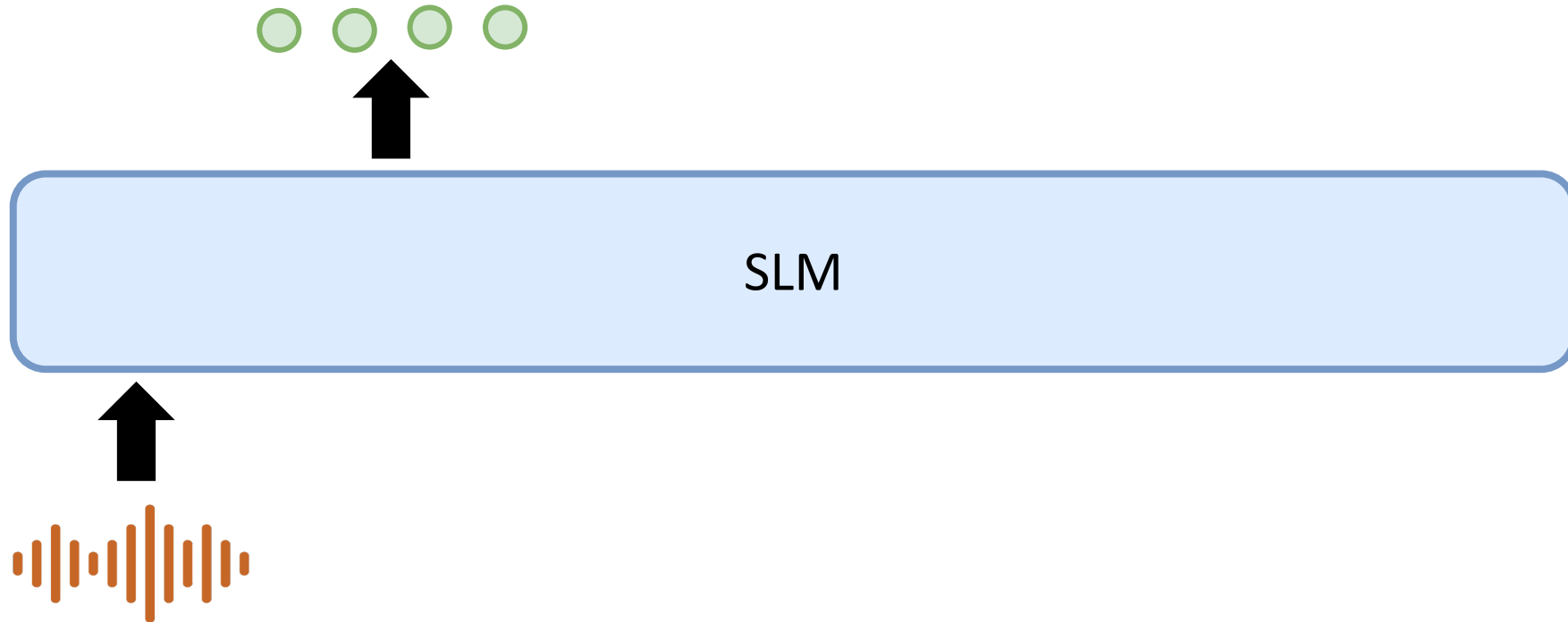
- Speech token
- Think token

Model keeps responding



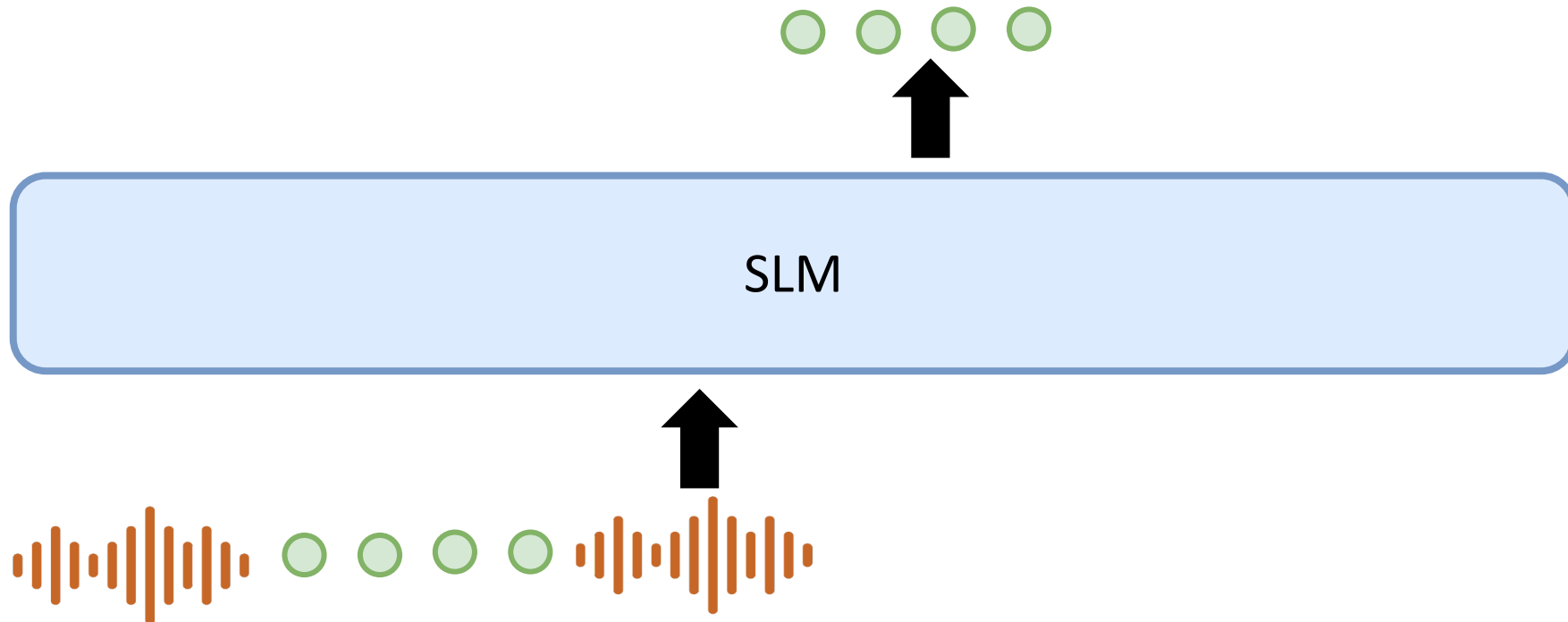
SHANKS: Training

- Given a partial input speech, generate a partial thinking



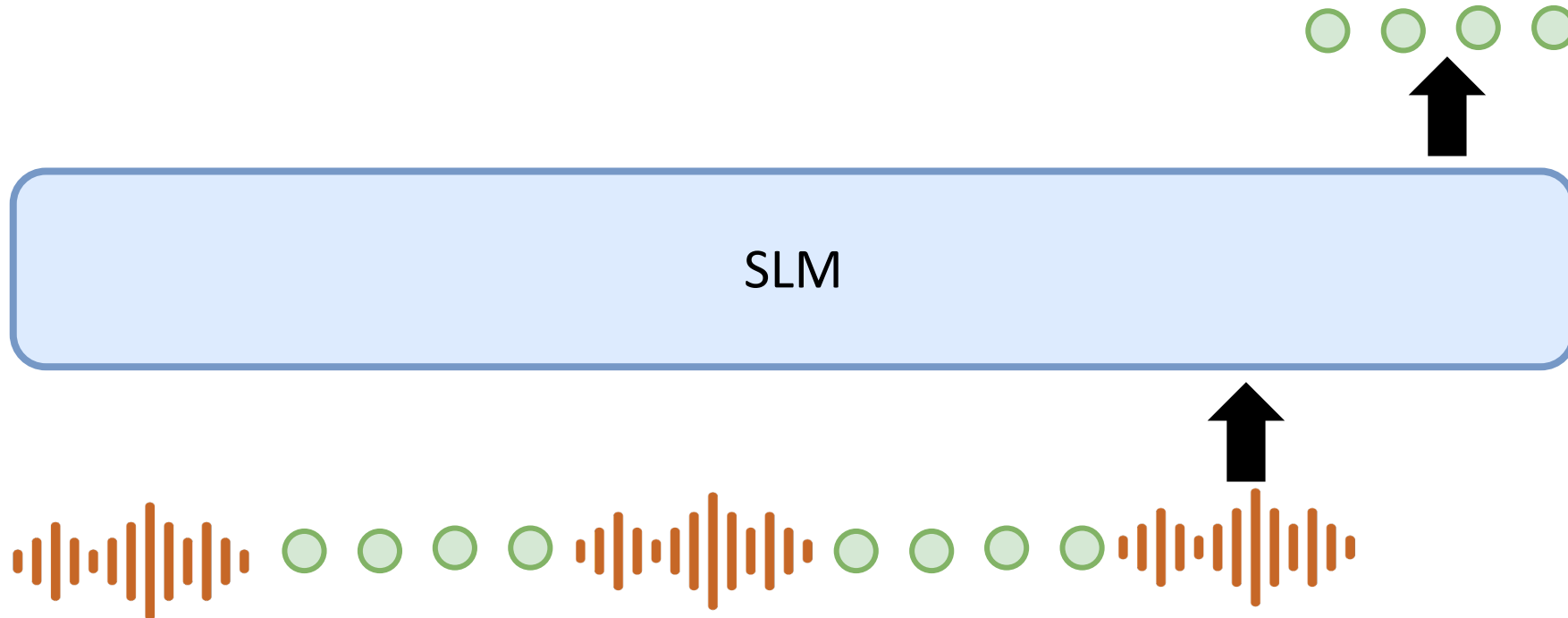
SHANKS: Training

- Given a new partial input speech and the previous thinking, generate a new partial thinking



SHANKS: Training

- Given a new partial input speech and the previous thinking, generate a new partial thinking



Application: Task-Oriented Dialogue + Tool Call

- The customer will specify a list of requests about a travel plan
 - *“Help me find the cheapest flight from Washington to New York on the day before Christmas Eve 2024. I want to know the flight's baggage allowances and also arrange a taxi to Times Square 1 hour after landing.”*



Application: Tool Call during Listening

- When the customer is speaking, the customer service agent is already searching for the hotels and flights when the customer is speaking



Application 2: Tool Call during Listening

- Task: The model acts as a customer service agent
- Goal: Make API calls to search for hotels and flights and fulfill the user's request
 - The model uses the thinking process to call APIs



Legends

 Speech token

 Think token

SHANKS



Check the details of the flight from Hangzhou to Seoul on 12/10, 2024. Please also search the

Legends

 Speech token

 Think token

SHANKS



Check the details of the flight from Hangzhou to Seoul on 12/10, 2024. Please also search the

Legends

Speech token

Think token



SHANKS

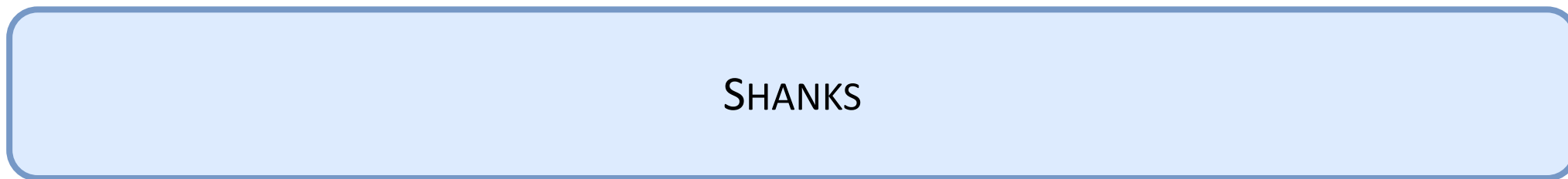
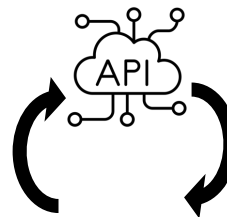


Check the details of the flight from Hangzhou to Seoul on 12/10, 2024. Please also search the

Legends

- Speech token
- Think token

Search for airport at 杭州



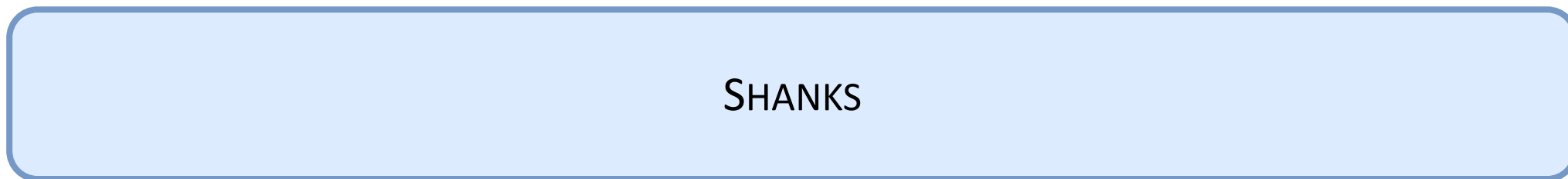
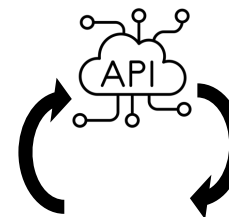
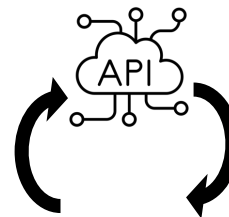
Check the details of the flight from Hangzhou to Seoul on 12/10, 2024. Please also search the

Legends

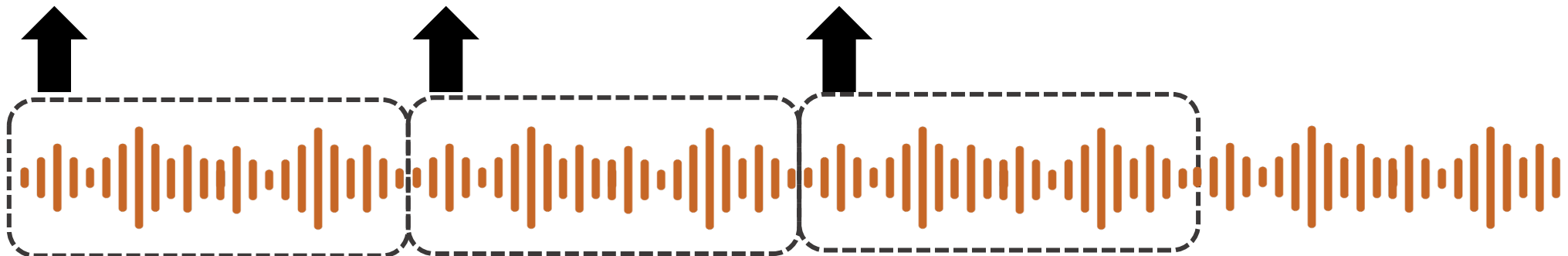
- Speech token
- Think token

Search for airport at 杭州

Search for airport at 首爾



SHANKS



Check the details of the flight from Hangzhou to Seoul on 12/10, 2024. Please also search the

Experiment Setting

- Model input
 - User's request about a travel plan (in speech)
 - Descriptions of the APIs that need to be used to meet the user's request (in text)
 - Each request requires 5.1 API calls on average
- The goal of the model
 - Make API calls (search flights, hotel, car rental information) and then respond the the user
 - Note that it is impossible for the model to respond without calling APIs

Experiment Setting: Metrics

- Evaluation metrics
 - Task success rate

Number of user requests that fully successful

Number of total user requests

A user request is successful if all the required API calls are made. The dataset we use contains the required API calls for each request.



Experiment Setting: Metrics

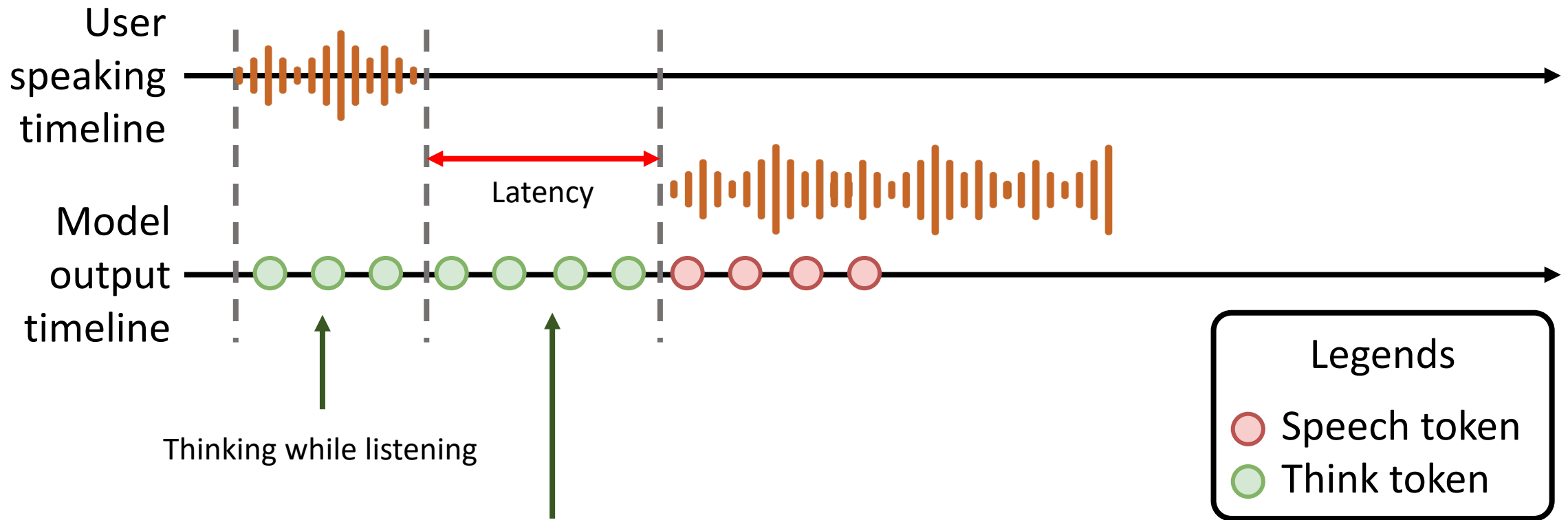
- Evaluation metrics
 - Early API calls: Number of API calls made when the user hasn't finished (including failed API calls)

Experiment Setting: Metrics

- Evaluation metrics
 - Latency: The time after the user finishes and before the model starts to respond

Experiment Setting: Metrics

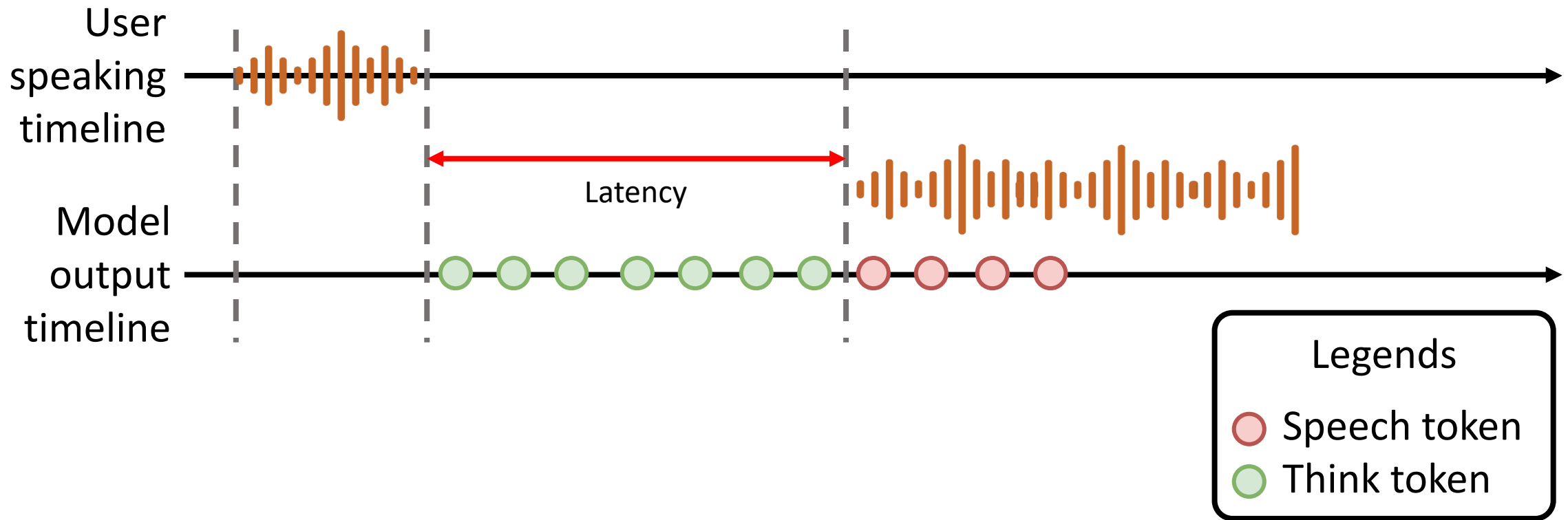
- Evaluation metrics
 - Latency: The time after the user finishes and before the model starts to respond



The model may have some remaining thinking tokens generated after the user finishes

Experiment Setting: Baseline

- Baseline: Call after listening
 - The model starts to call the API after the user finishes



Experiment Results

- SHANKS and the call-after-listening have a similar success rate

Methods	Task Success Rate	Early API Calls	Latency
Call-after-listening	63.2%	0	3.91 s
SHANKS	62.3%	5.49	1.46 s

Experiment Results

- On average, SHANKS can make 5.49 API calls when the user is still speaking

Methods	Task Success Rate	Early API Calls	Latency
Call-after-listening	63.2%	0	3.91 s
SHANKS	62.3%	5.49	1.46 s

Experiment Results

- SHANKS reduce the latency by 62%

Methods	Task Success Rate	Early API Calls	Latency
Call-after-listening	63.2%	0	3.91 s
SHANKS	62.3%	5.49	1.46 s

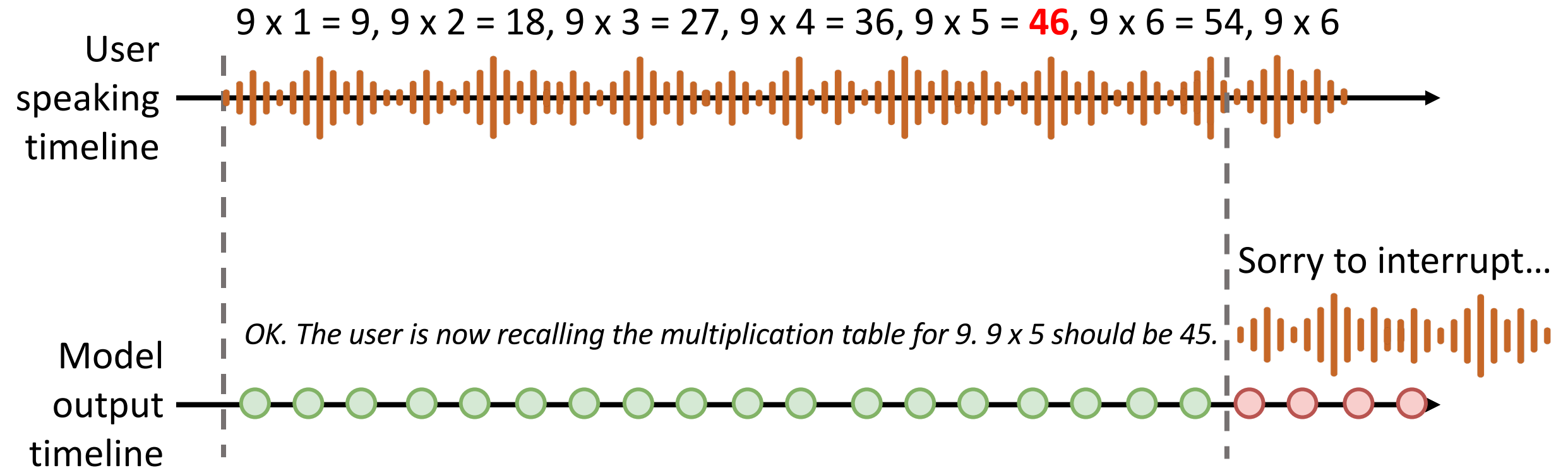
SHANKS: Application 2: Model Interrupts User

- Using SHANKS, we can interrupt the user when the user is saying something wrong
- This application has potential in educational scenarios



SHANKS: Application 2: Model Interrupts User

- SHANKS listen to the user's speech, perform internal reasoning, and then interrupt the user when the user is wrong



SHANKS: Application 2: Model Interrupts User

- Please refer to the paper for this application

Outline

1. Background: Spoken Language Models
2. STITCH: Thinking while speaking
3. SHANKS: Thinking while listening
4. Conclusion

Conclusion

- Introducing reasoning to SLMs require special considerations due to the **streaming** nature and **low-latency** requirement for S2S setting
- We propose STITCH, thinking while speaking, which elegantly uses the time of speaking to generate unspoken text reasoning
- We propose SHANKS, thinking while listening, which elegantly uses the user speaking to perform unspoken text reasoning

Acknowledgement

- The above work was done during my internship at Microsoft GenAI
 - My mentor: Xiaofei Wang
 - Other amazing team members: Linjie Li, Chung-Ching Lin, Kevin Lin, Shujie Liu, Zhendong Wang, Zhengyuan Yang, Lijuan Wang
 - My PhD advisor: Hung-yi Lee

Conclusion

- Introducing reasoning to SLMs require special considerations due to the **streaming** nature and **low-latency** requirement for S2S setting
- We propose STITCH, thinking while speaking, which elegantly uses the time of speaking to generate unspoken text reasoning
- We propose SHANKS, thinking while listening, which elegantly uses the user speaking to perform unspoken text reasoning